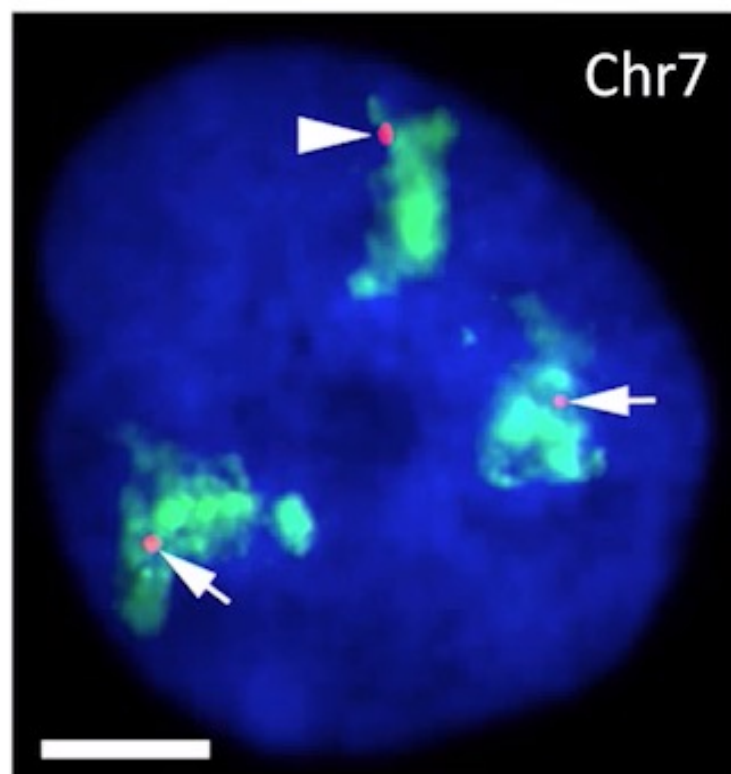


ChIA-PET

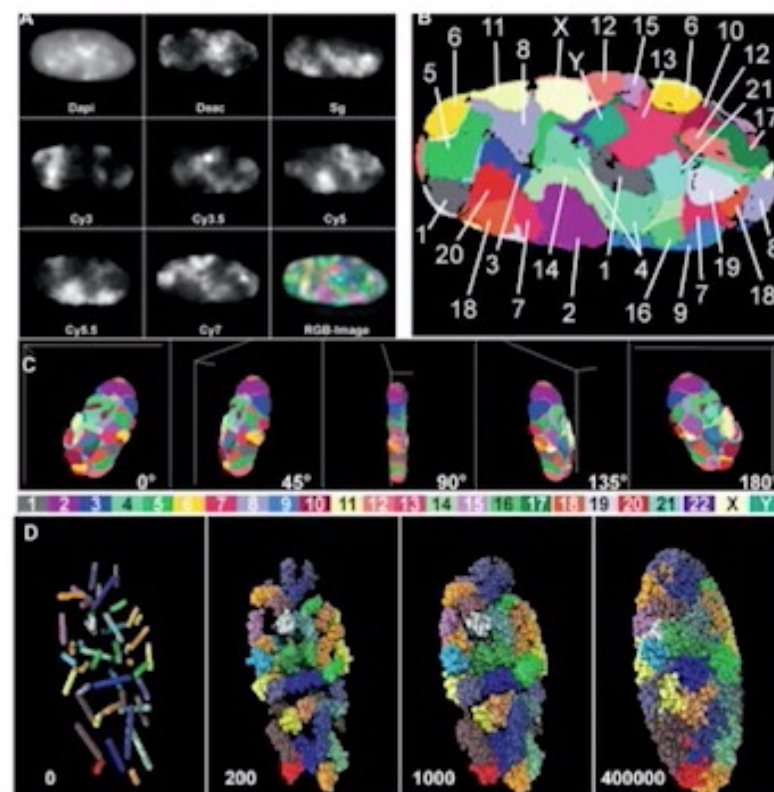
---The interaction techniques for the specific factor mediated long-range interaction for 3D Genome

Chapter 14

Chromosome territories

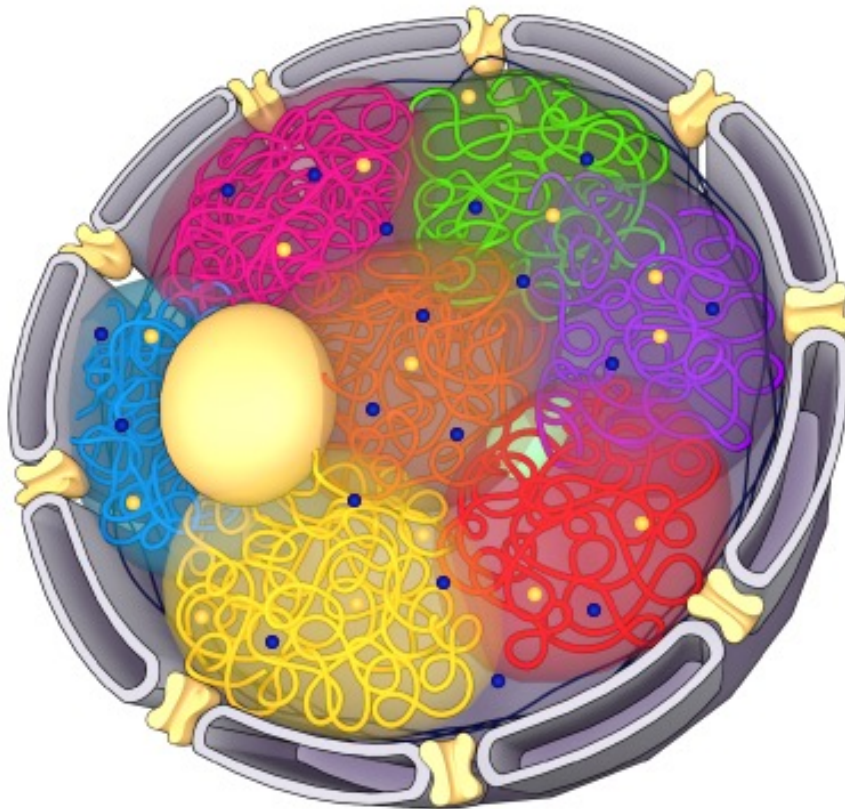


Zink et al 2004

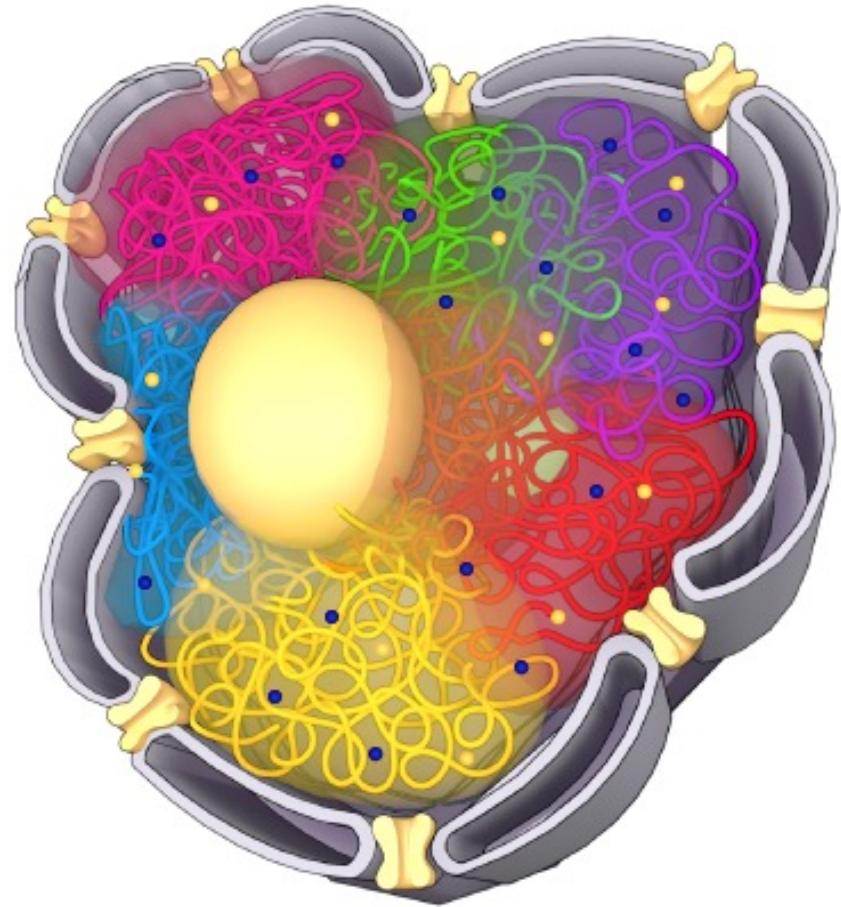


Bolzer et al 2005

Nucleus in healthy cells



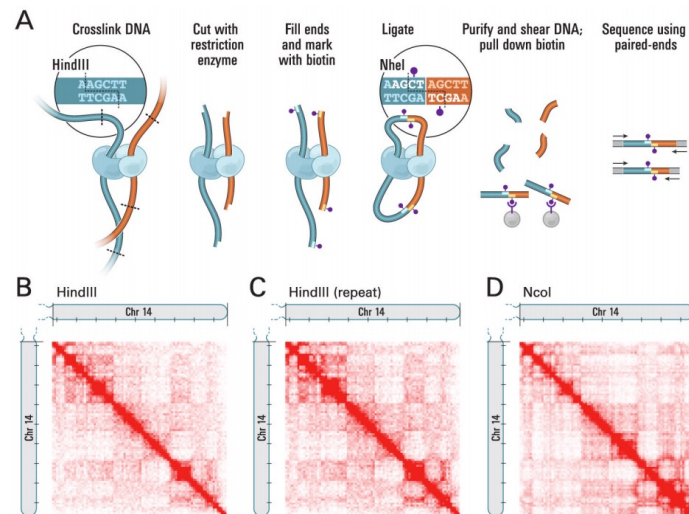
Nucleus in cancer cells



Science 2009

Comprehensive Mapping of Long-Range Interactions Reveals Folding Principles of the Human Genome

Erez Lieberman-Aiden,^{1,2,3,4*} Nynke L. van Berkum,^{5*} Louise Williams,¹ Maxim Imakaev,² Tobias Ragoczy,^{6,7} Agnes Telling,^{6,7} Ido Amit,¹ Bryan R. Lajoie,⁵ Peter J. Sabo,⁸ Michael O. Dorschner,⁸ Richard Sandstrom,⁸ Bradley Bernstein,^{1,9} M. A. Bender,¹⁰ Mark Groudine,^{6,7} Andreas Gnirke,¹ John Stamatoyannopoulos,⁸ Leonid A. Mirny,^{2,11} Eric S. Lander,^{1,12,13†} Job Dekker^{5†}



Hi-C

Nature 2009

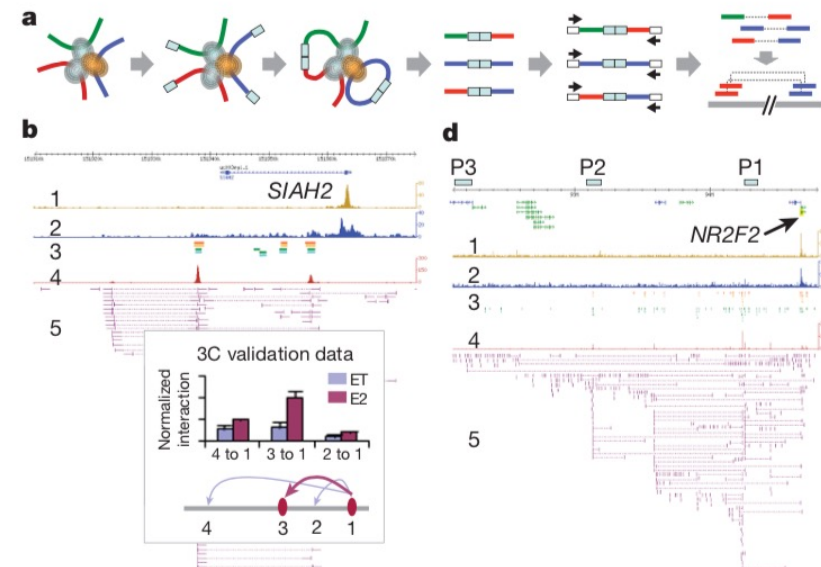
nature

Vol 462 | 5 November 2009 | doi:10.1038/nature08497

ARTICLES

An oestrogen-receptor- α -bound human chromatin interactome

Melissa J. Fullwood¹, Mei Hui Liu¹, You Fu Pan¹, Jun Liu¹, Han Xu¹, Yusoff Bin Mohamed¹, Yuriy L. Orlov¹, Stoyan Velkov¹, Andrea Ho¹, Poh Huay Mei¹, Elaine G. Y. Chew¹, Phillips Yao Hui Huang¹, Willem-Jan Welboren², Yuyuan Han¹, Hong Sain Ooi¹, Pramila N. Ariyaratne¹, Vinsensius B. Vega¹, Yanquan Luo¹, Peck Yean Tan¹, Pei Ye Choy¹, K. D. Senali Abayaratna Wansa¹, Bing Zhao¹, Kar Sian Lim¹, Shi Chi Leow¹, Jit Sin Yow¹, Roy Joseph¹, Haixia Li¹, Kartiki V. Desai¹, Jane S. Thomsen¹, Yew Kok Lee¹, R. Krishna Murthy Karuturi¹, Thoreau Herve¹, Guillaume Bourque¹, Hendrik G. Stunnenberg², Xiaolan Ruan¹, Valere Cacheux-Rataboul¹, Wing-Kin Sung^{1,3}, Edison T. Liu¹, Chia-Lin Wei¹, Edwin Cheung^{1,4,5} & Yijun Ruan^{1,4}



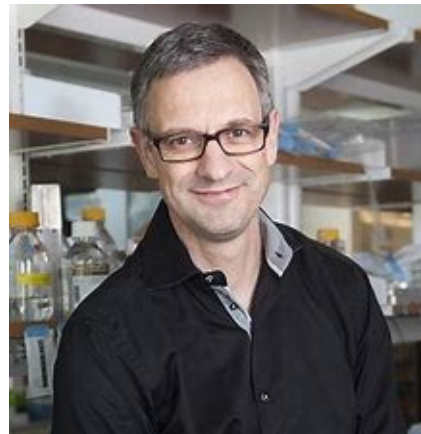
ChIA-PET

Hi-C

Science 2009

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ChIA-PET

Nature 2009

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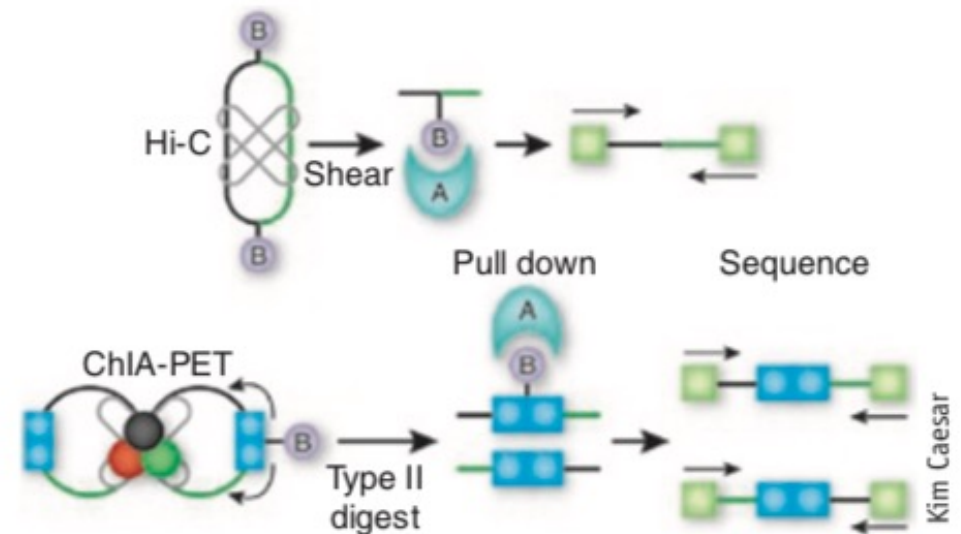
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When ChIA PETs meet Hi-C

Two methods enable the drawing of genome-wide chromatin interaction maps: one looks at protein-independent folding principles, the other at protein-mediated functional interactions.

In its stretched out form, human chromatin adds up to a roughly two meter long polymer that needs to fit into a nucleus approximately ten micrometers in diameter. In addition, genes and their regulatory elements need to communicate, and chromatin must open to allow transcription and replication. Important questions are ‘how does the linear order of genes relate to their spatial arrangement?’ and ‘how are genes regulated by distant elements?’. Two independent groups recently tackled these questions from opposite directions.

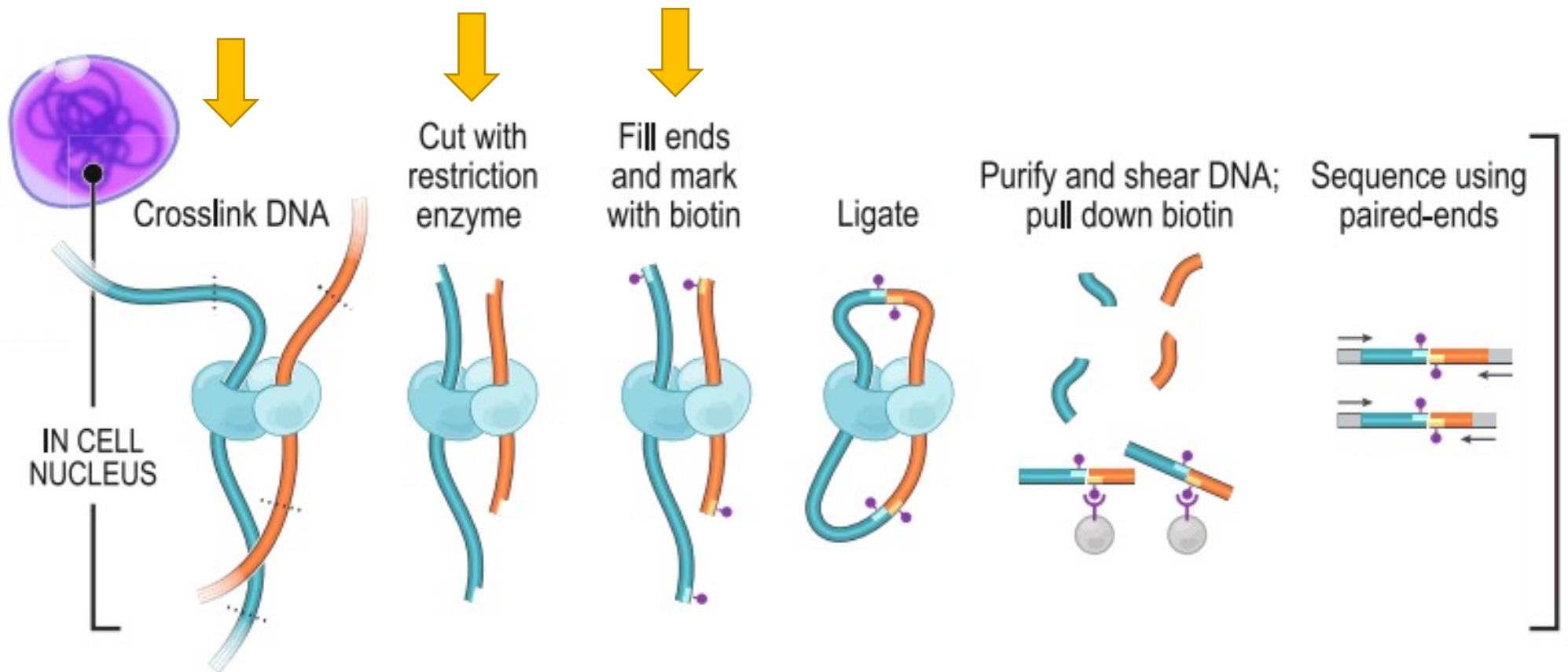


Schematic of Hi-C and ChIA-PETs.

one-kilobase resolution, hundreds of millions of reads will be needed in addition to a new way to model the interaction data.

But in the meantime, the scientists found surprising results with the number of reads

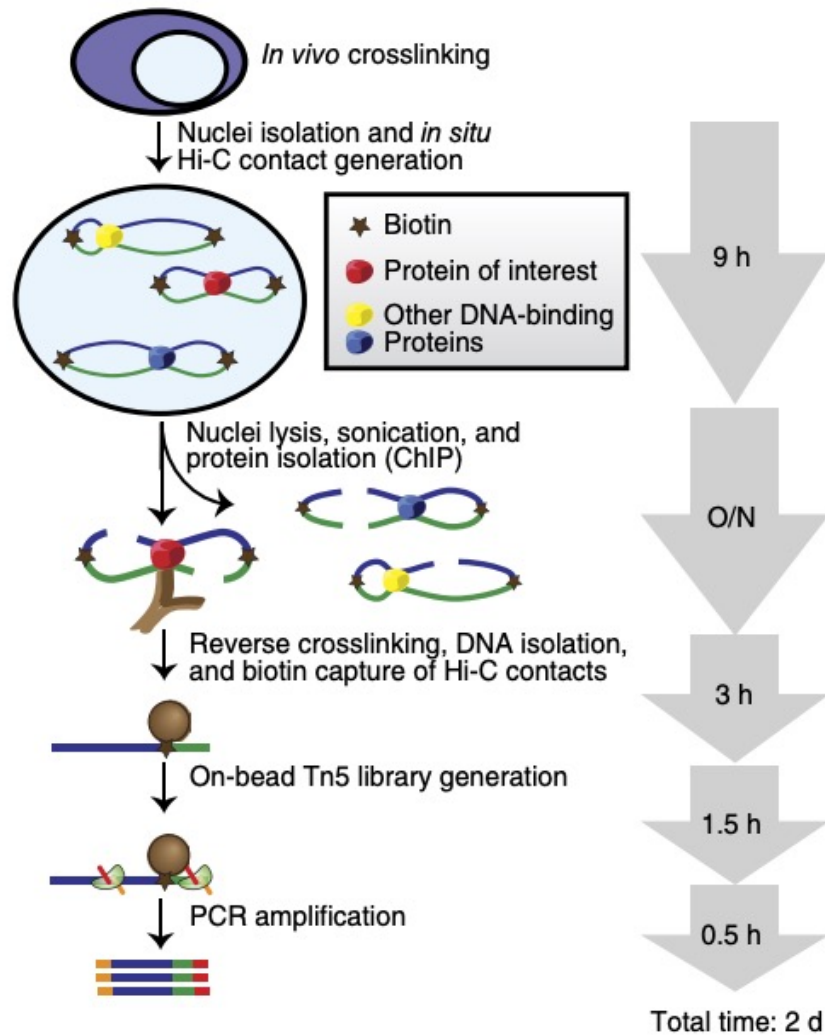
Hi-C method optimization and development



➤ How about ChIA-PET?

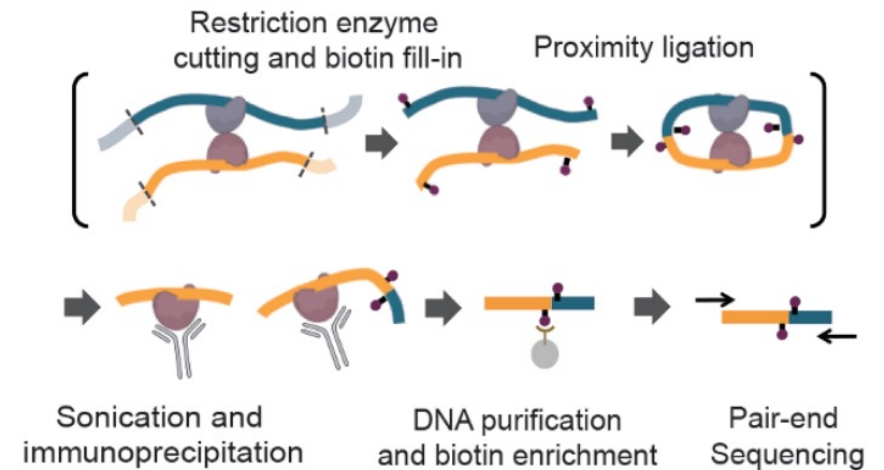
Until to 2016

HiChIP



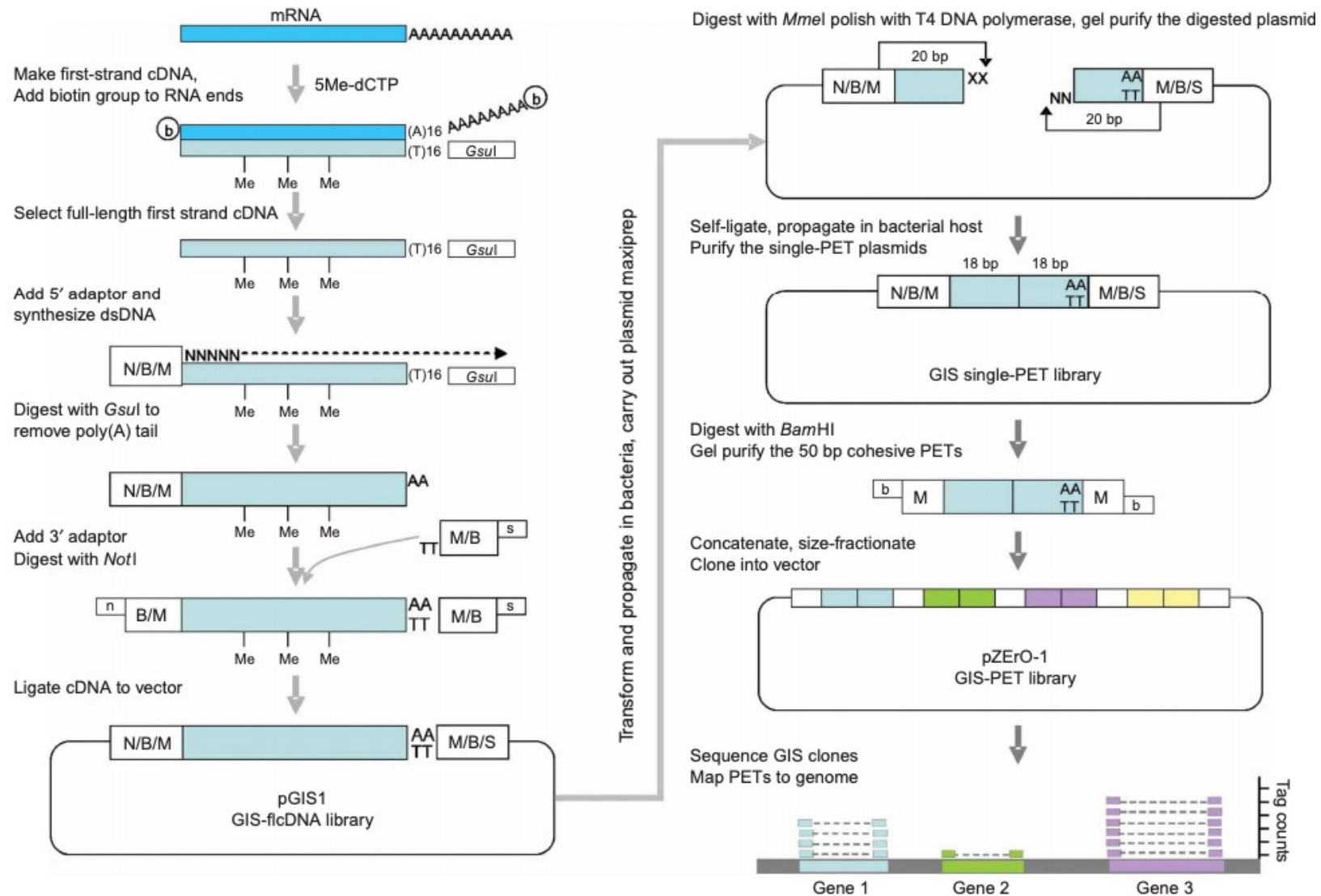
Mumbach et al., Nature Methods 2016

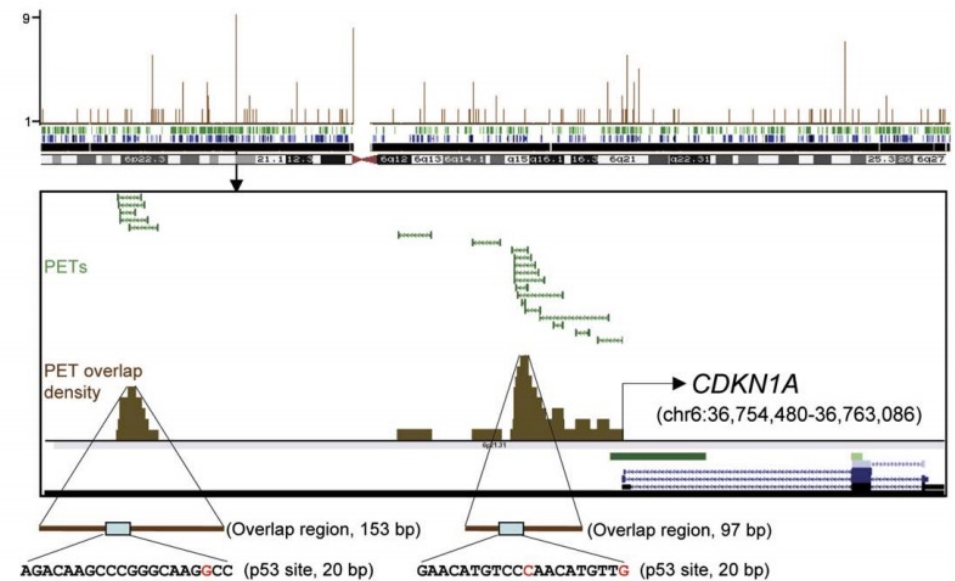
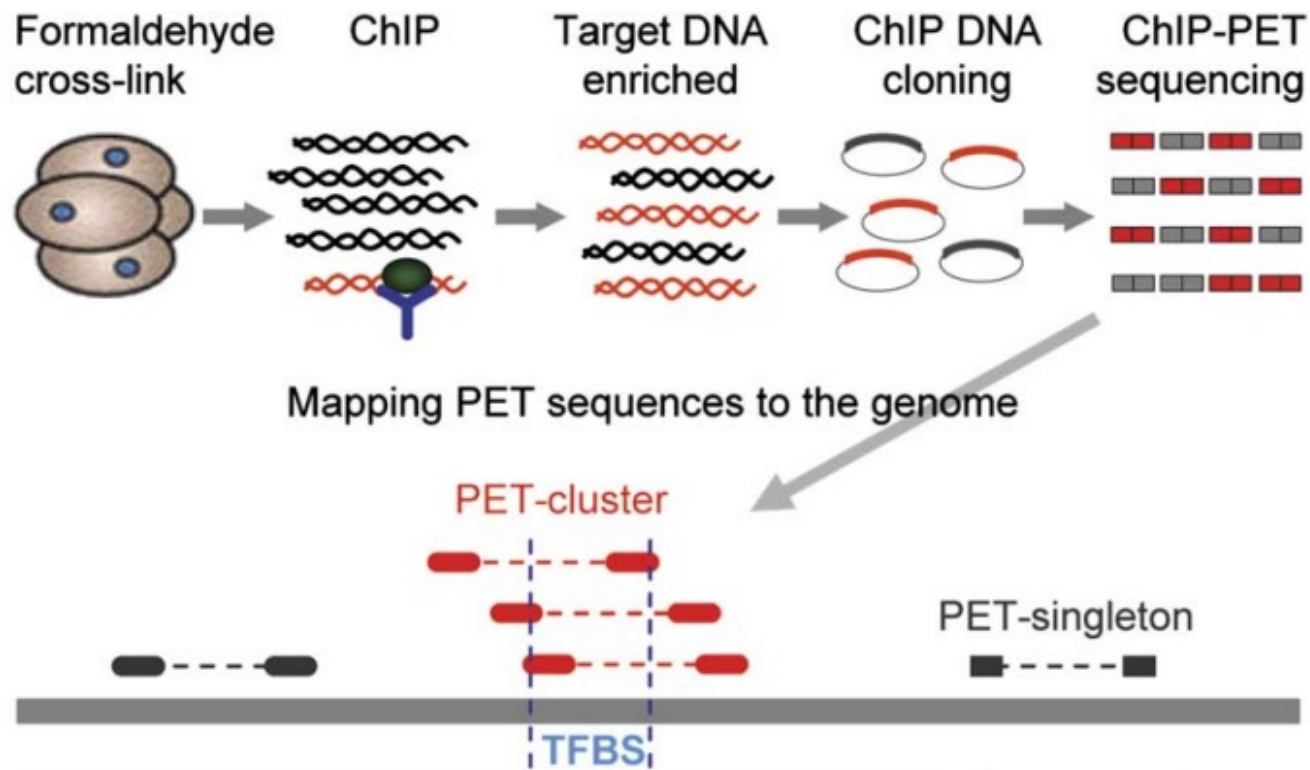
PLAC-seq



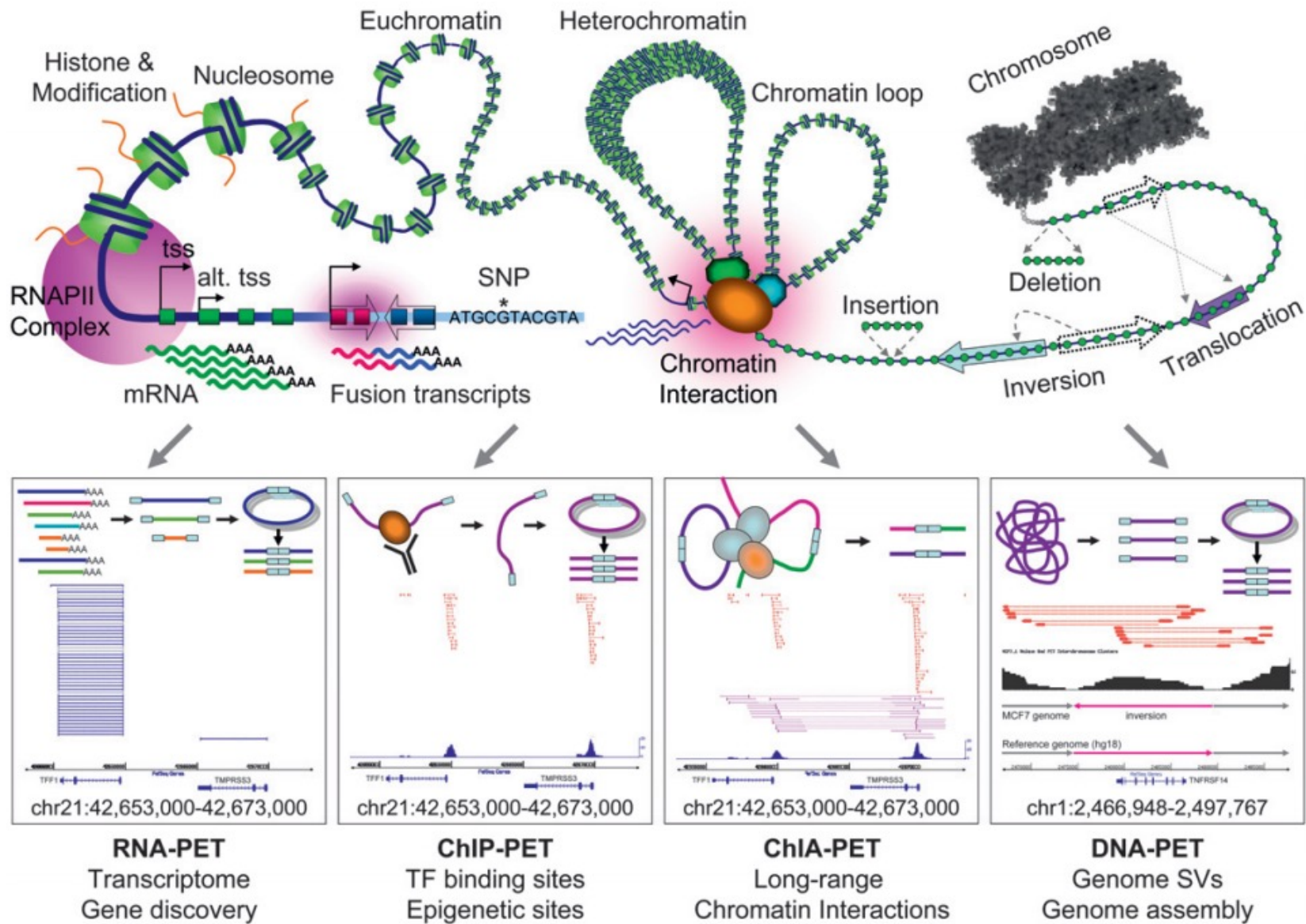
Fang et al., Cell Research 2016

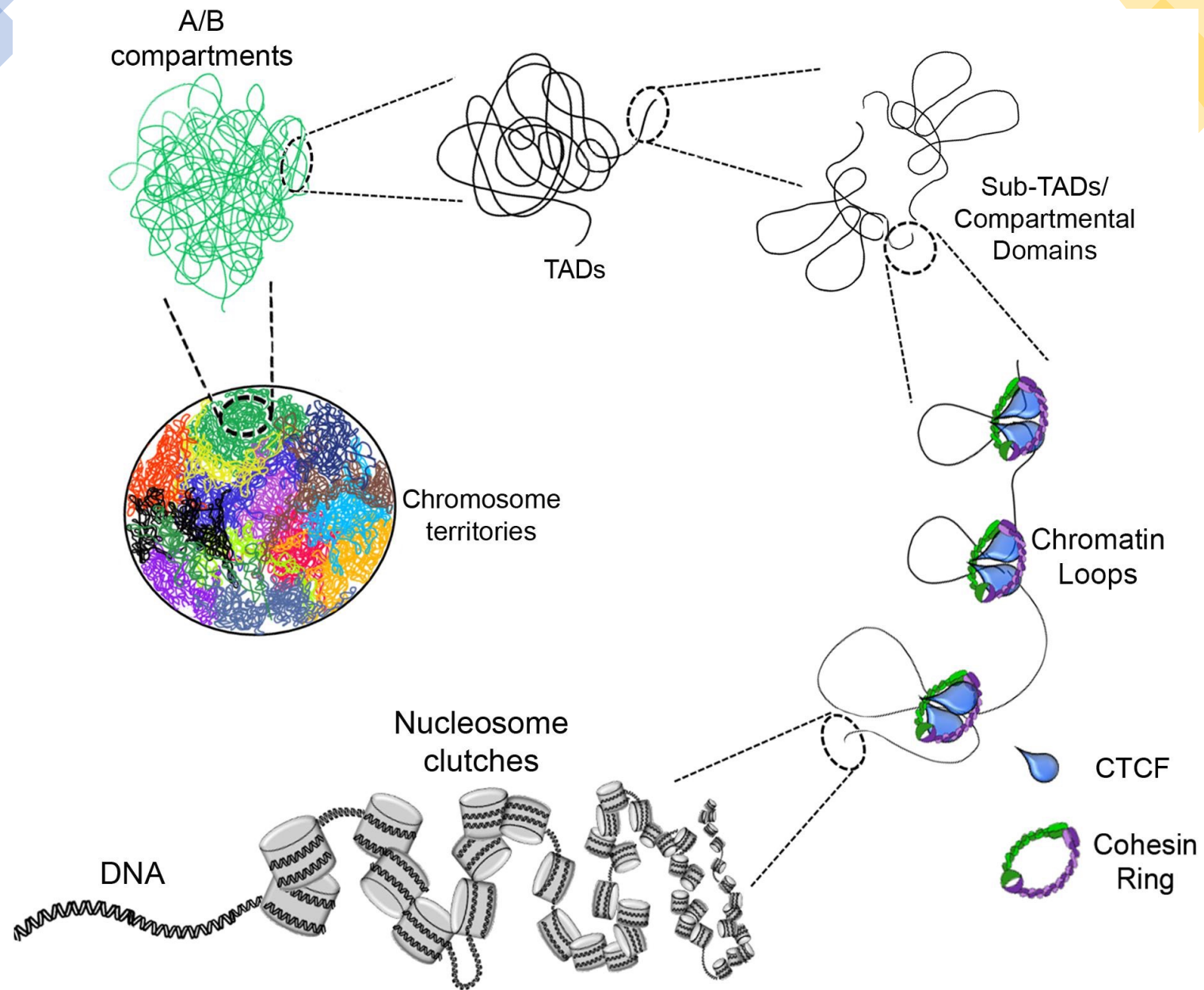
Pair-end Tag

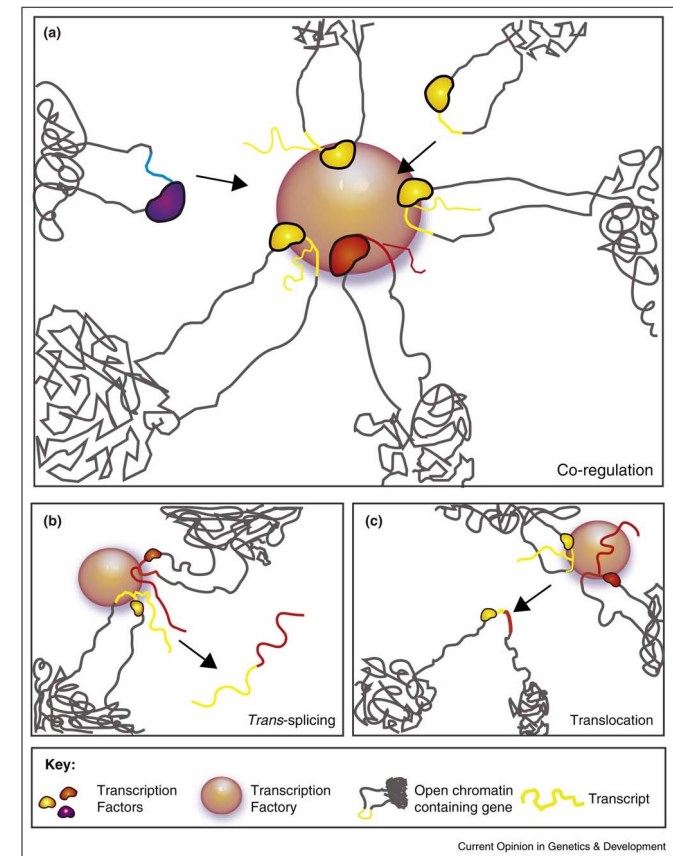
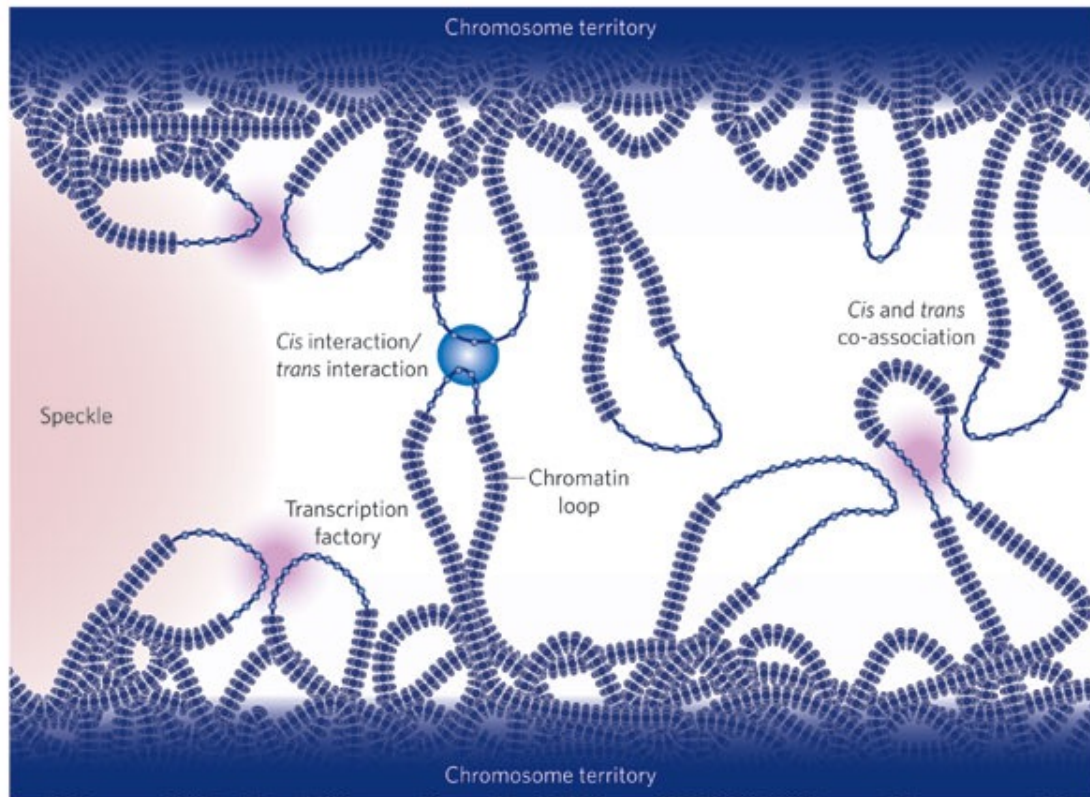


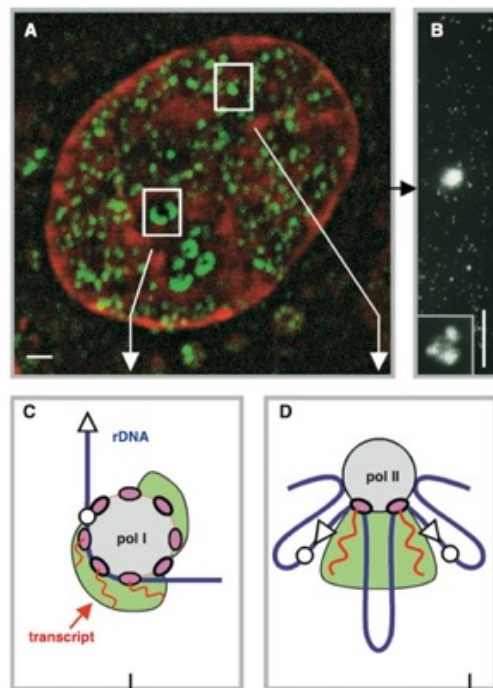
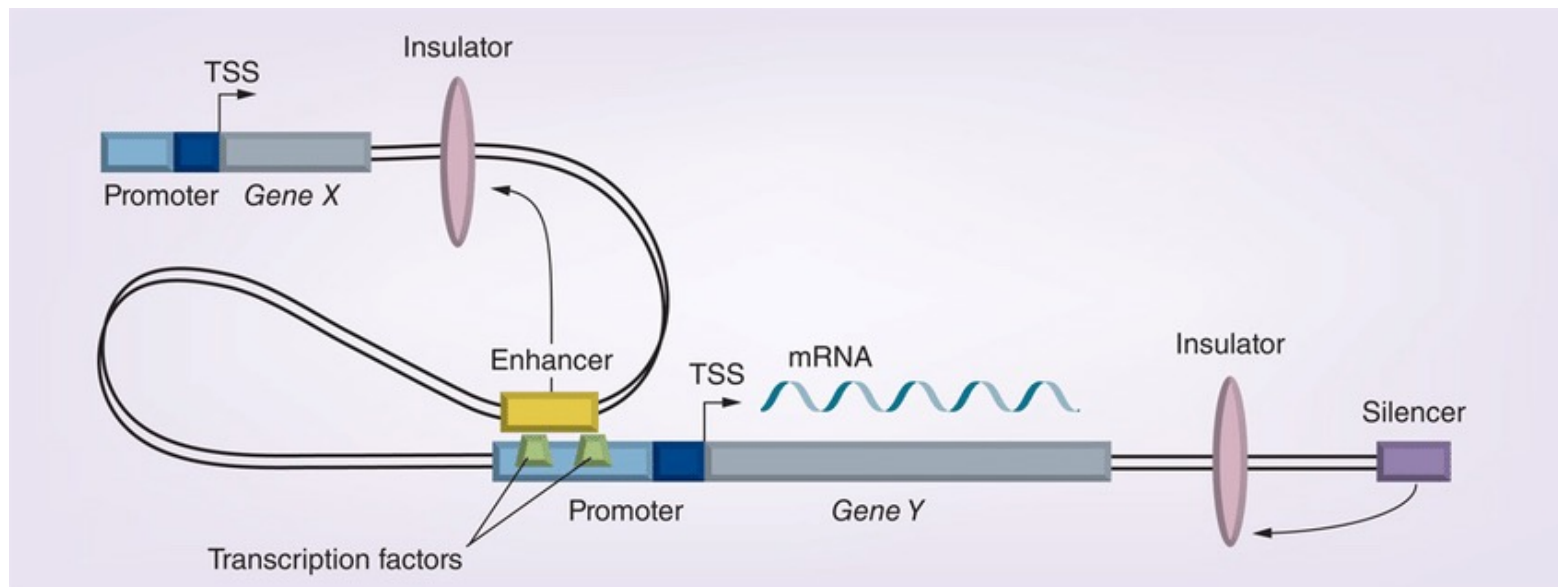


Background

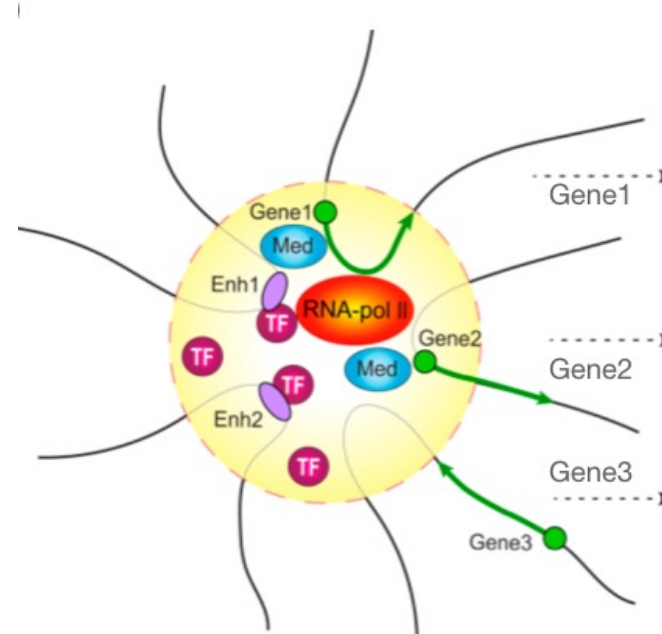






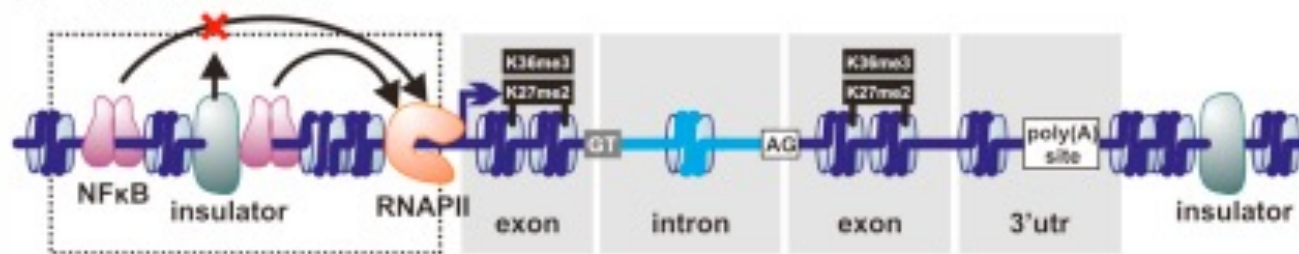


(Cook et al. 1999)



(Ibragimov et al. 2020)

(a) Linear structure

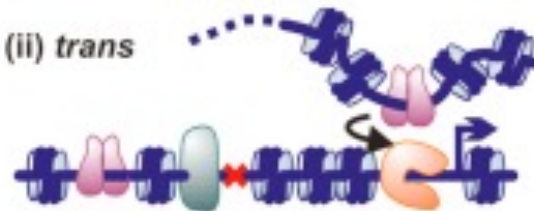


(b) Enhancer-promoter interaction

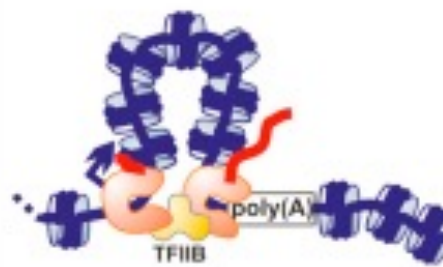
(i) *cis*



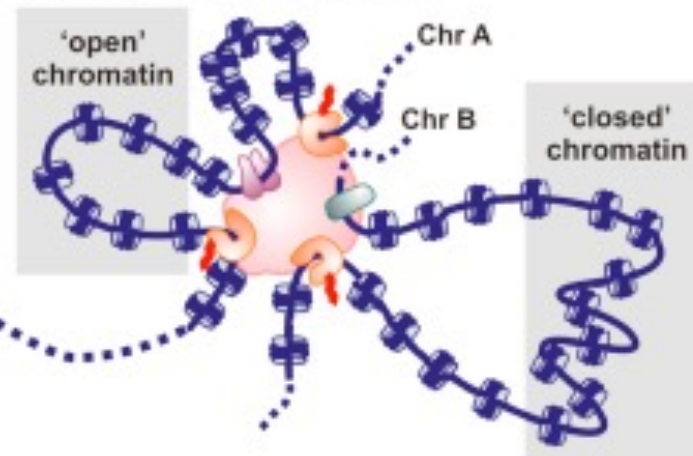
(ii) *trans*



(c) Gene loops

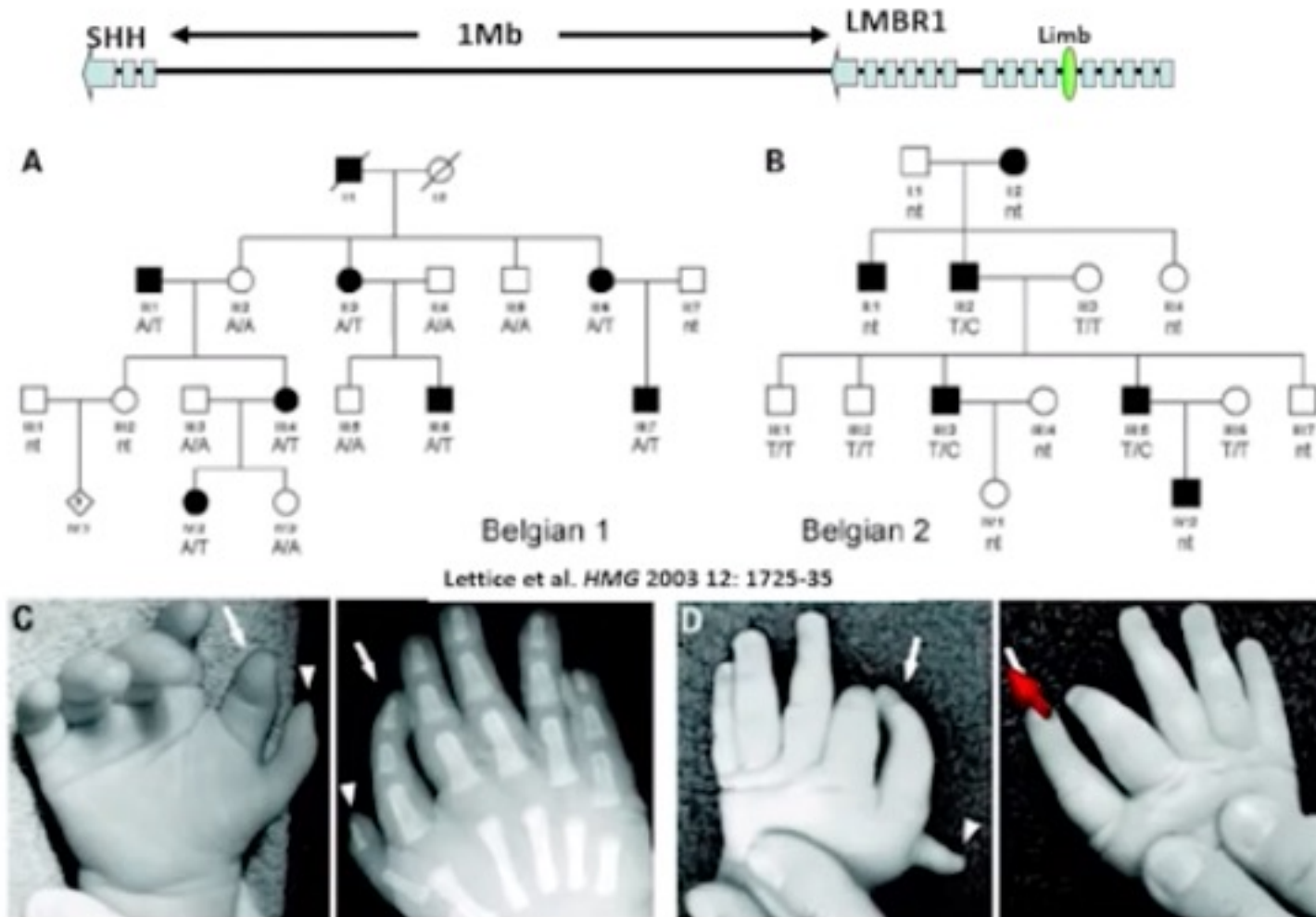


(d) Transcription factories



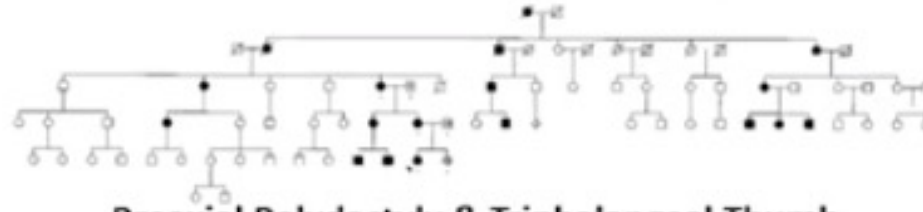
Current Opinion in Cell Biology

Mutations in the SHH limb enhancer (ZRS) lead to limb malformations



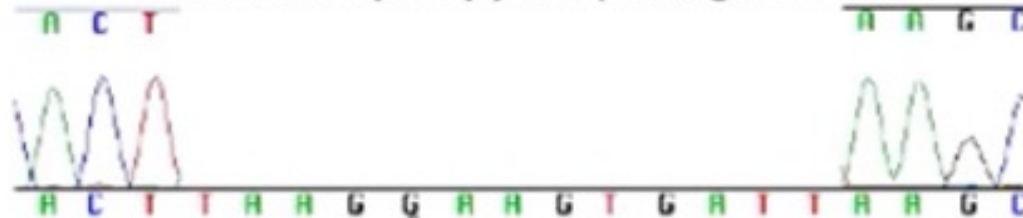


13bp insertion in the *SHH* limb enhancer

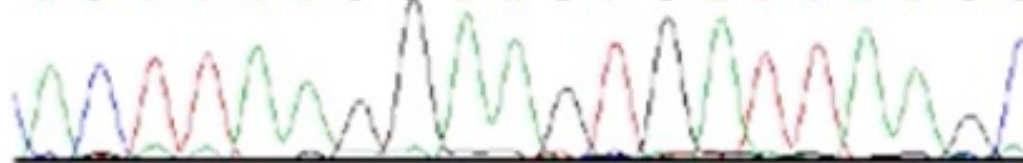


Preaxial Polydactyly & Triphalangeal Thumb

Unaffected



Affected

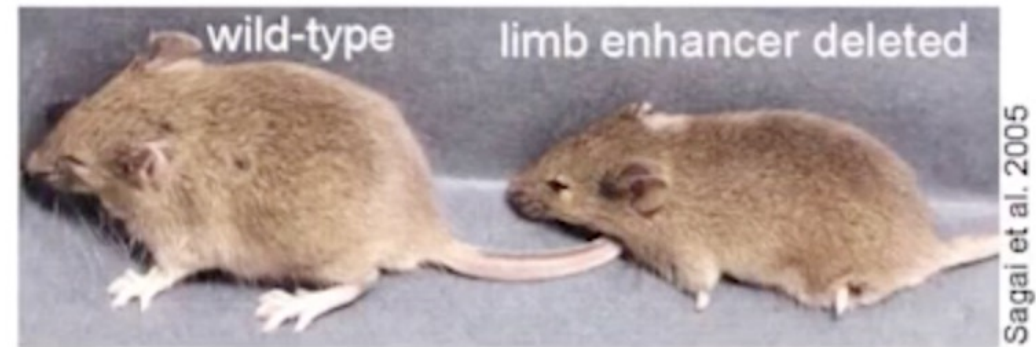
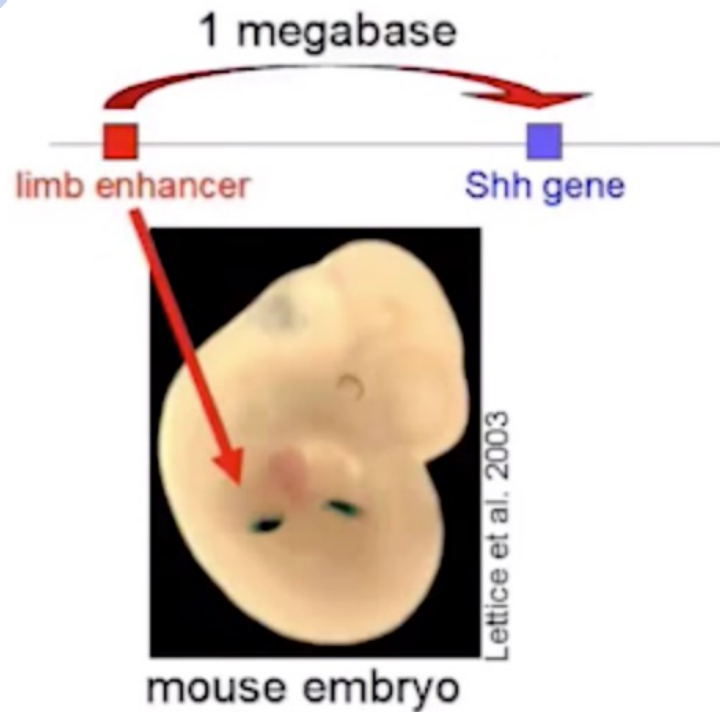


CTAACA CT **TAAGGAAGTGATT** AAGCAGCA

TBX5/6
EN1
CDX2
PITX1/2
PBX1
SHOX2
HOXA9, B8,
C4/5/6/8/9



Laurell et al. *Human Mutation* 2012



human *Shh* enhancer point mutations



Long-range enhancer controls the transcription regulation of
Sonic hedgehog (SHH)

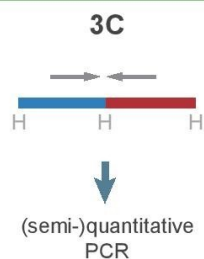
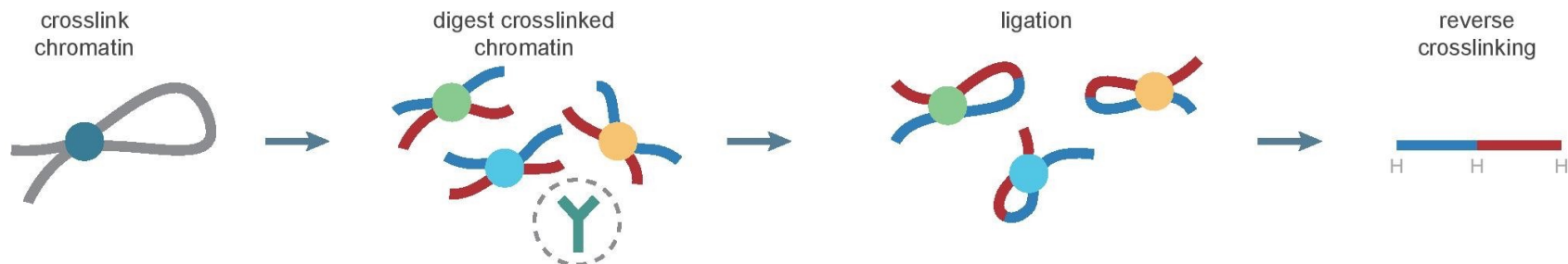
Main ChIA-PET Techniques

ChIA-PET

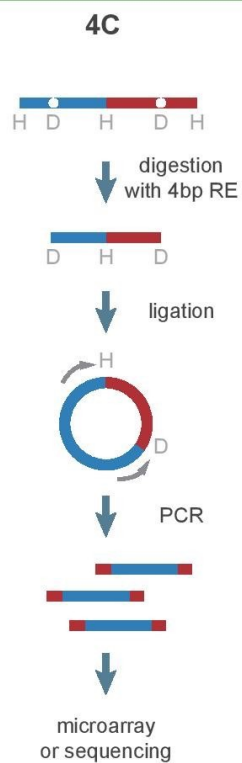
Long-reads
ChIA-PET

In situ ChIA-PET

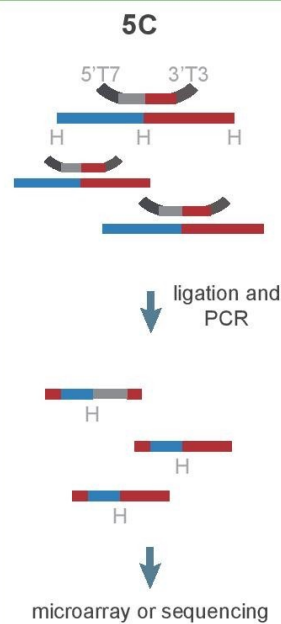
Chromosome Conformation Technologies



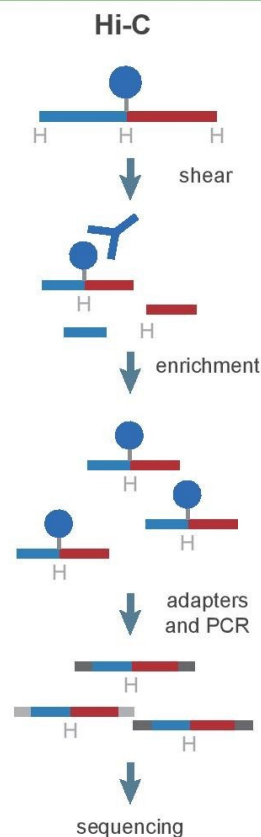
one vs one



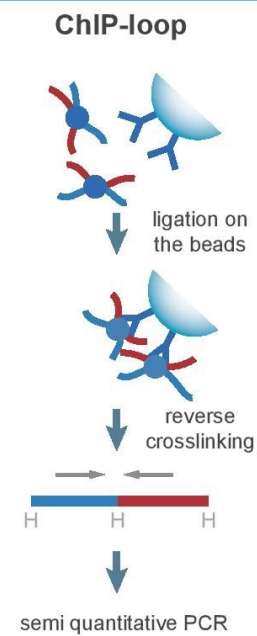
one vs all



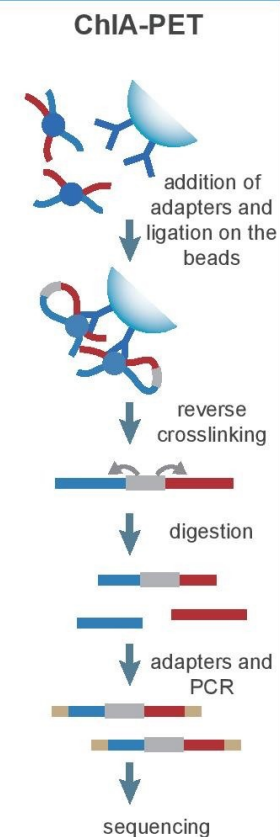
many vs many



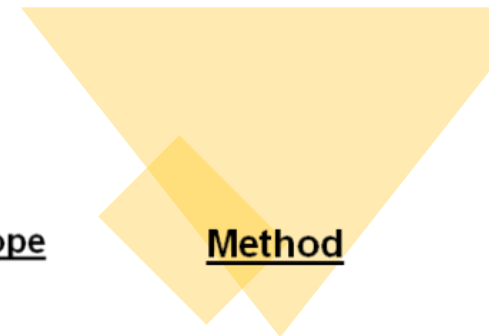
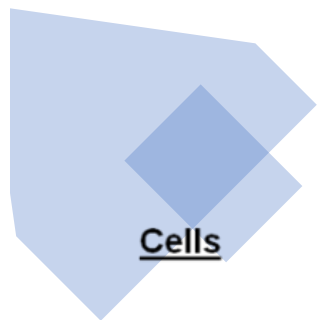
all vs all



one vs one



all vs all



Cells

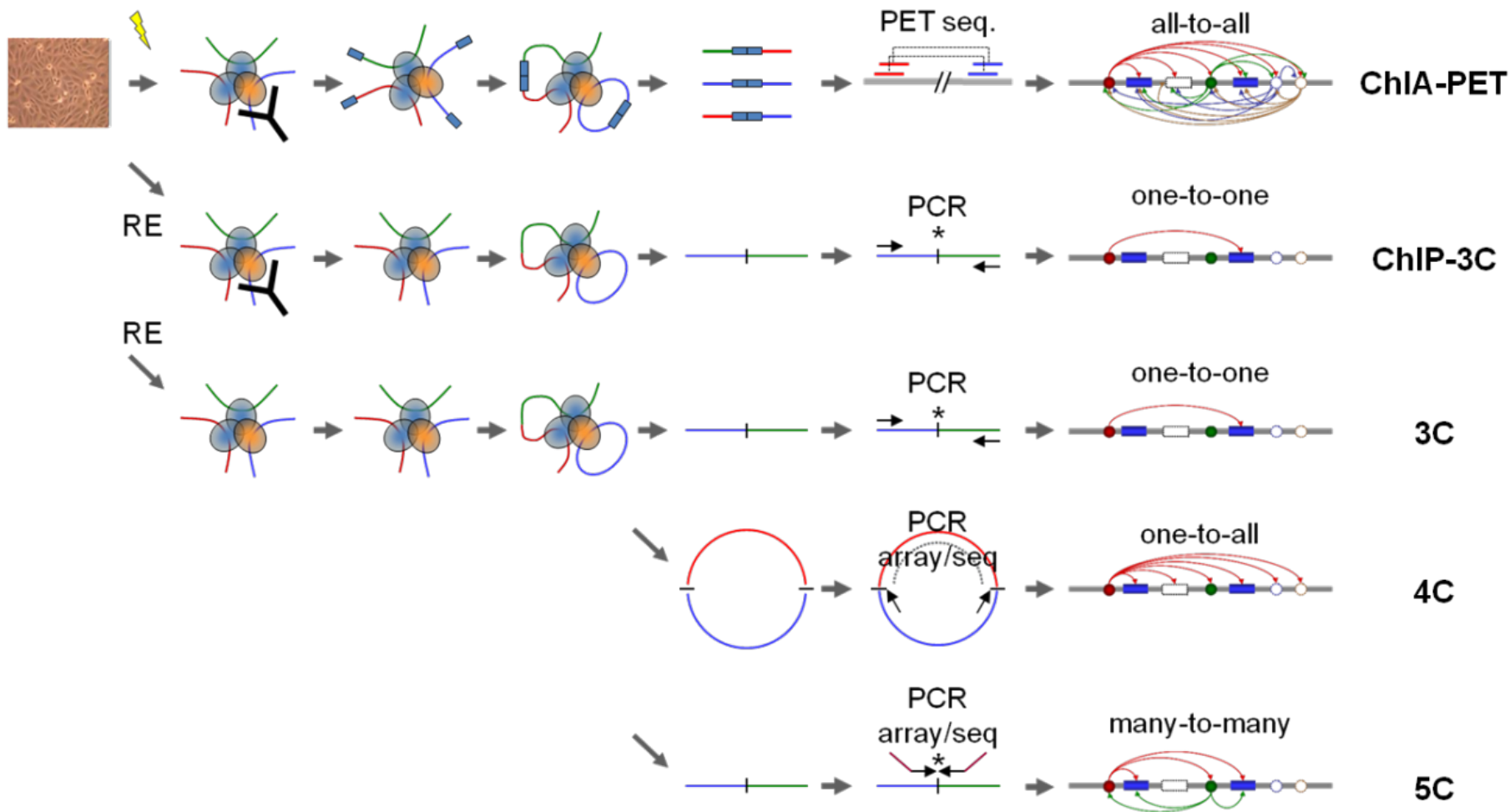
Chromatin preparation

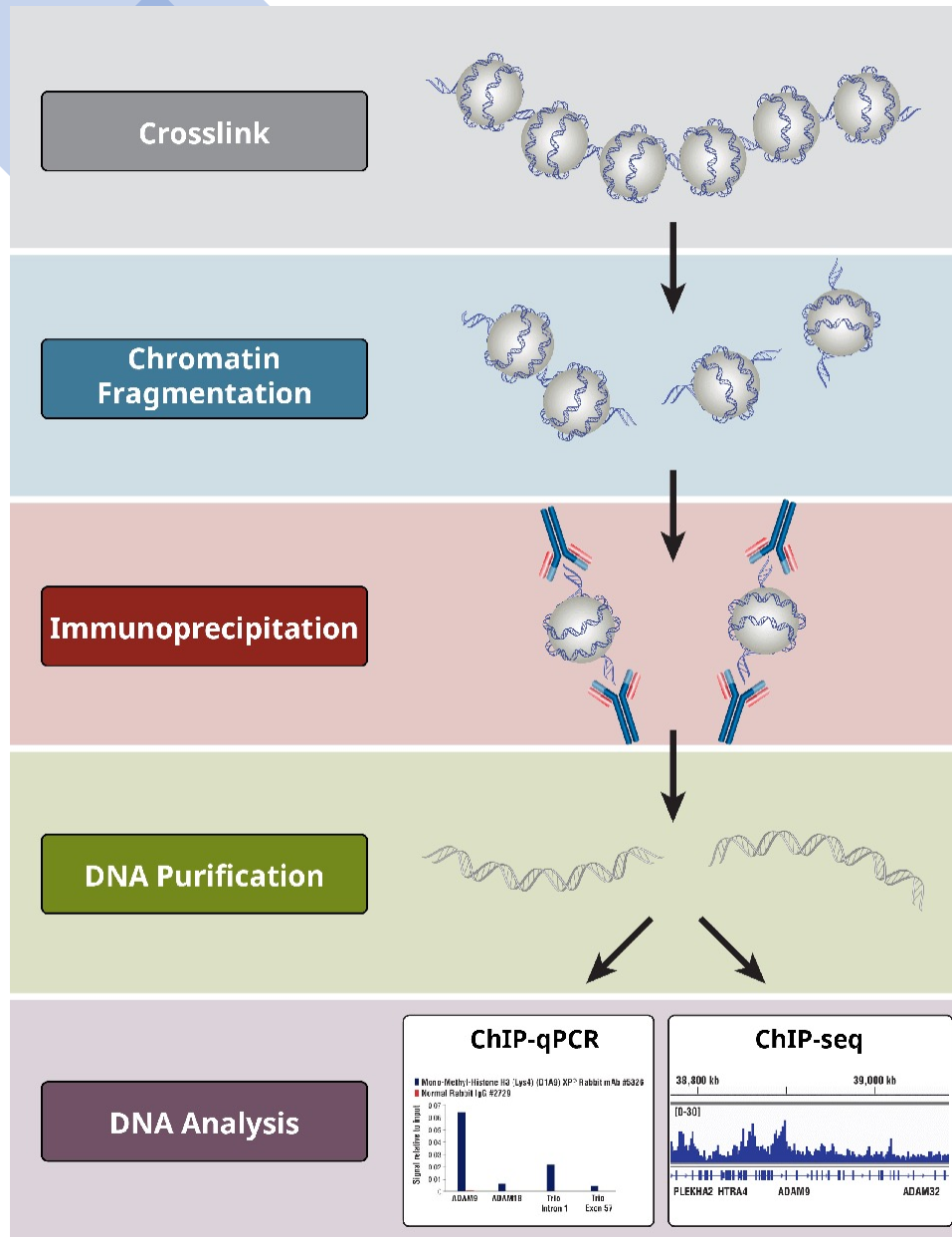
Templates

Detection

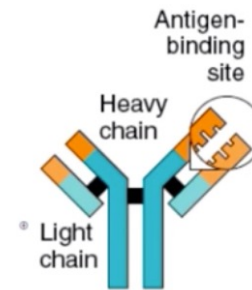
Scope

Method



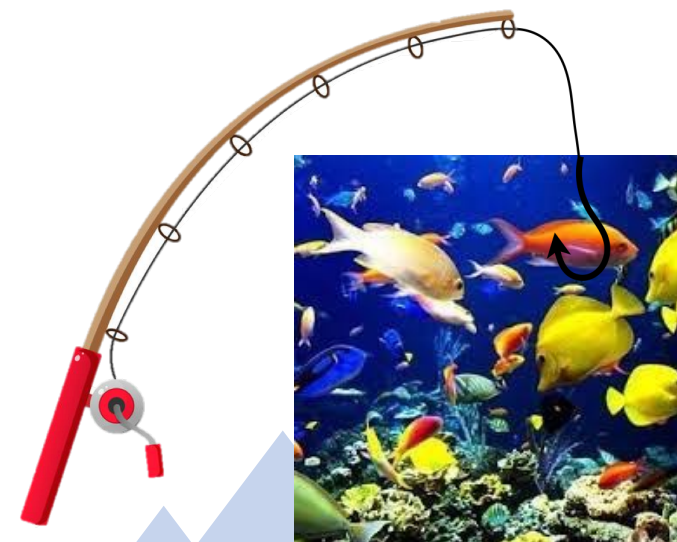
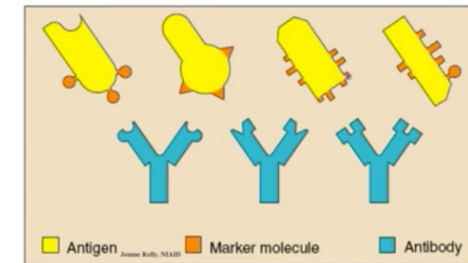


Antibody Structure



- Variable region
- Constant region

Antigen-Antibody Specificity



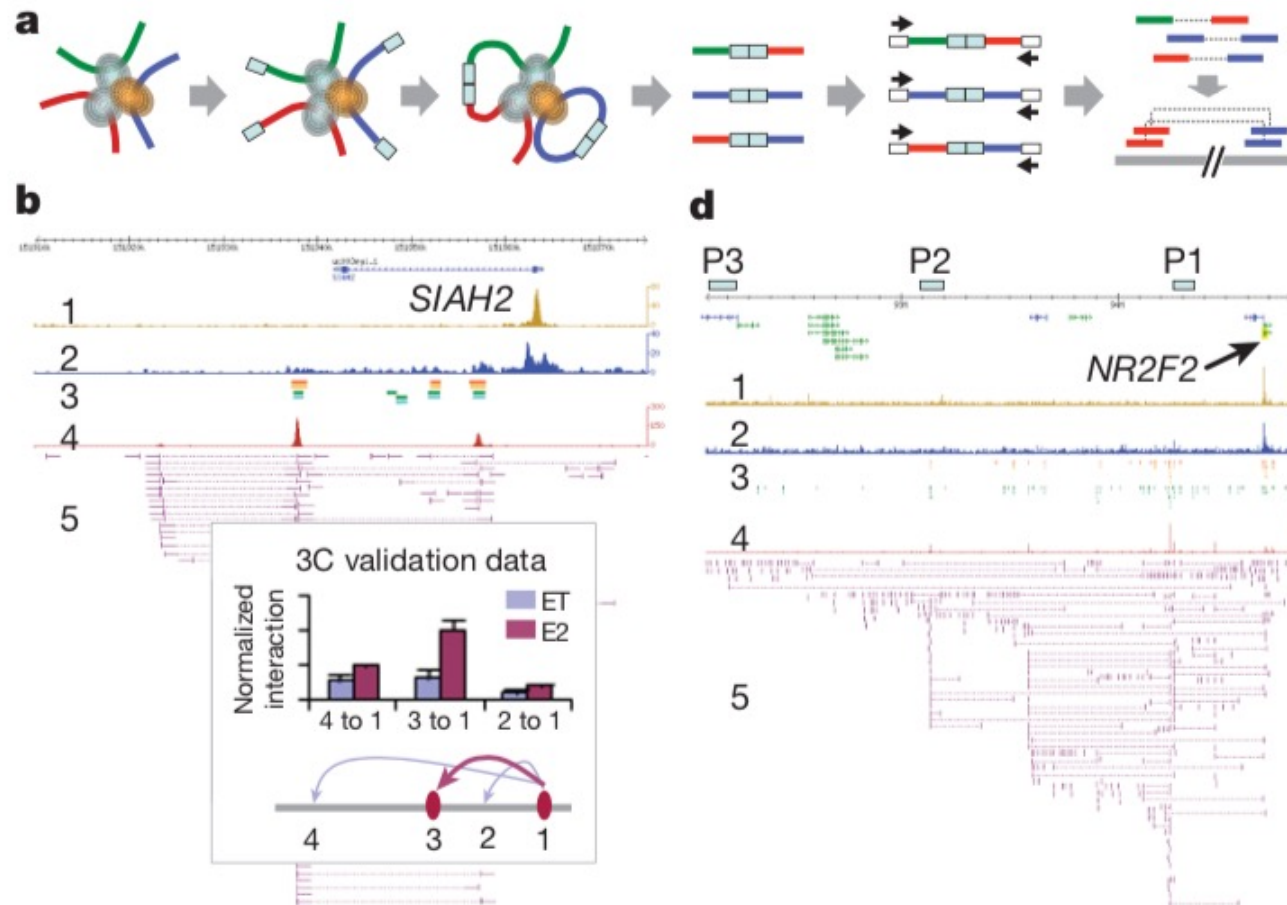
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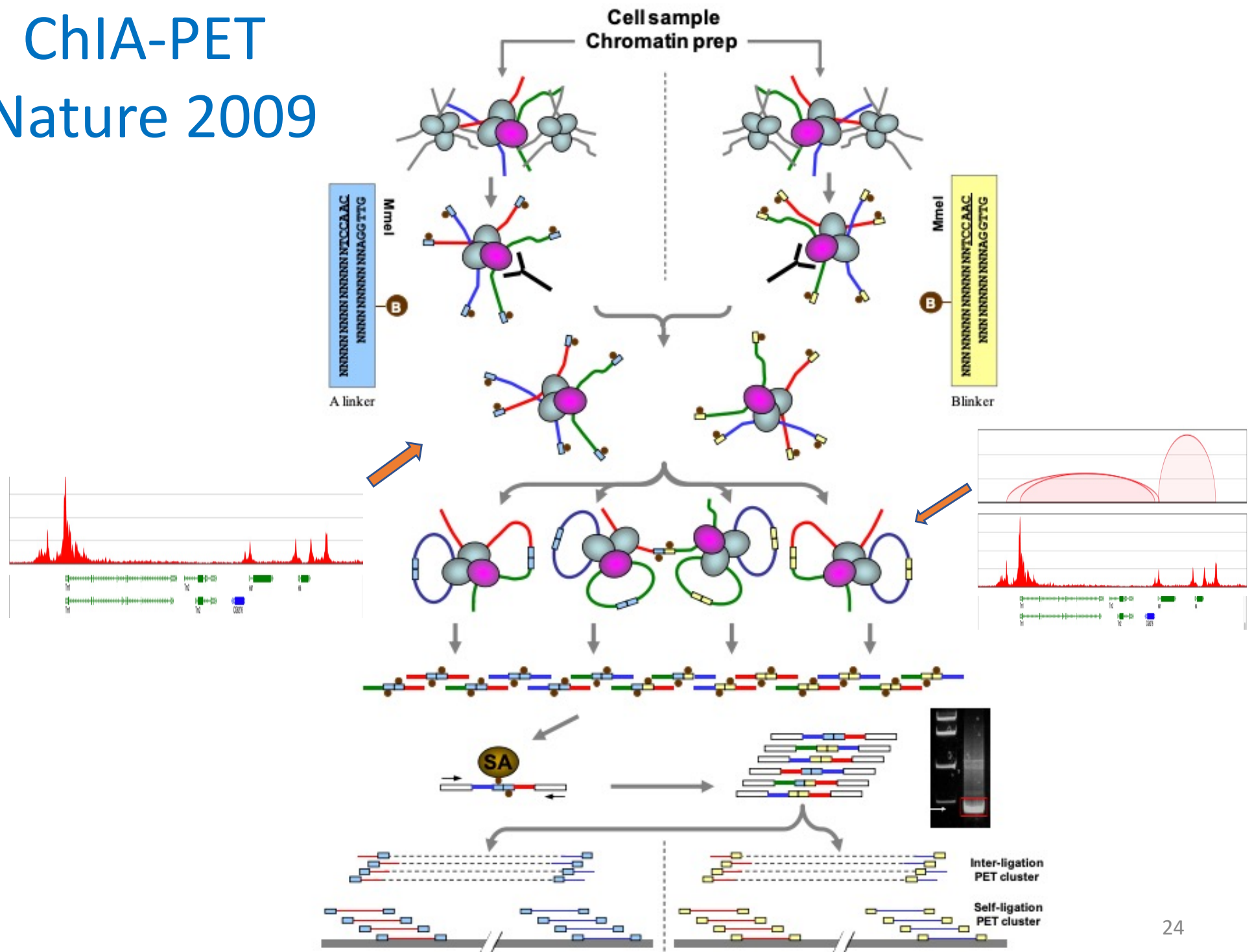
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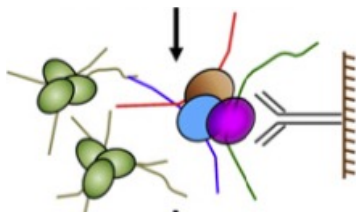
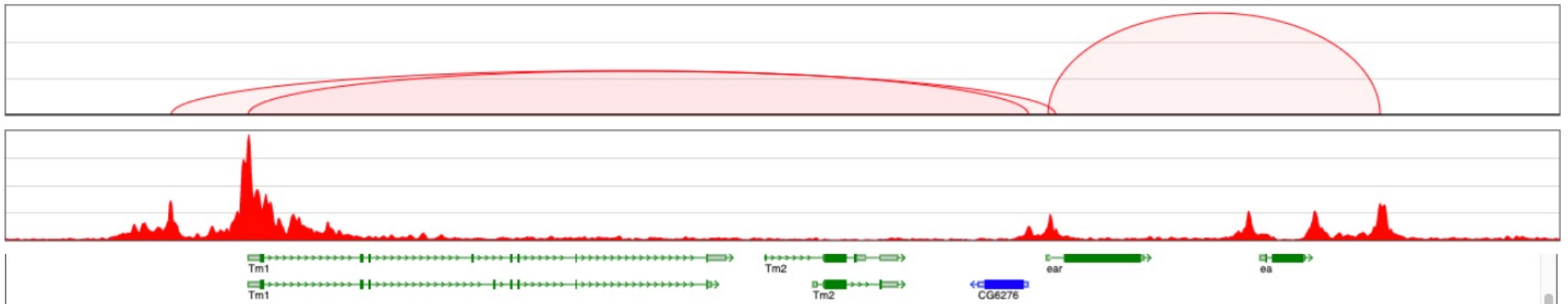
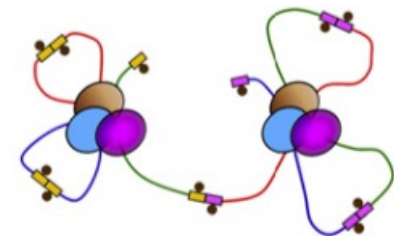
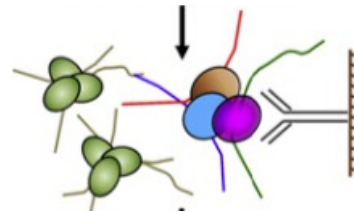
Nature 2009

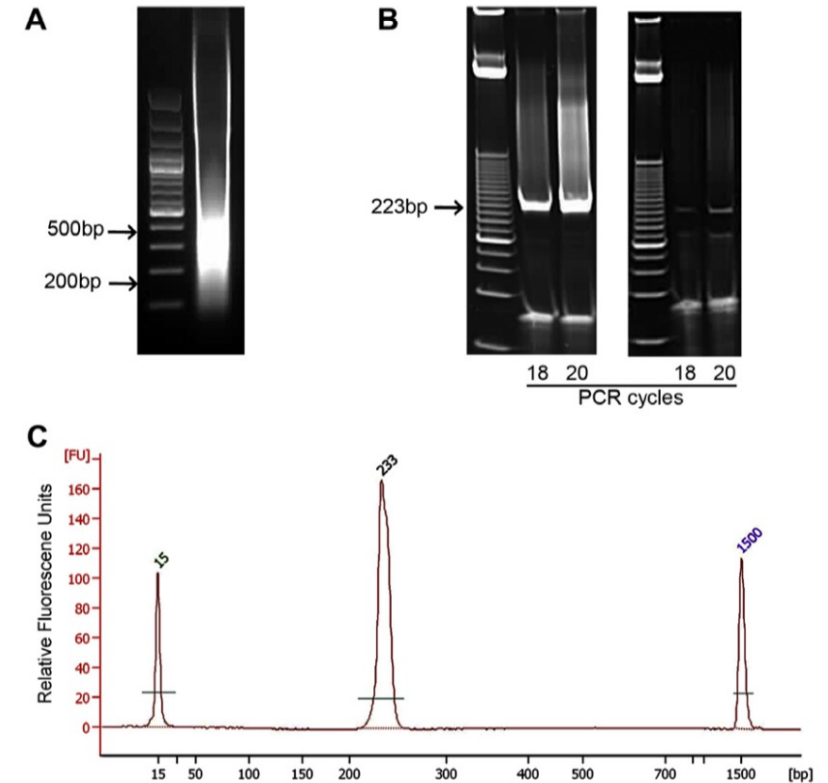
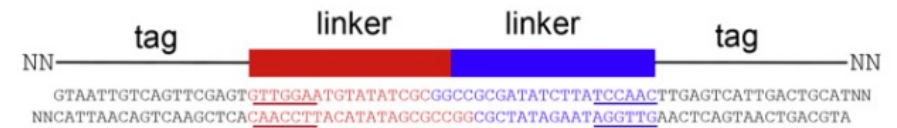
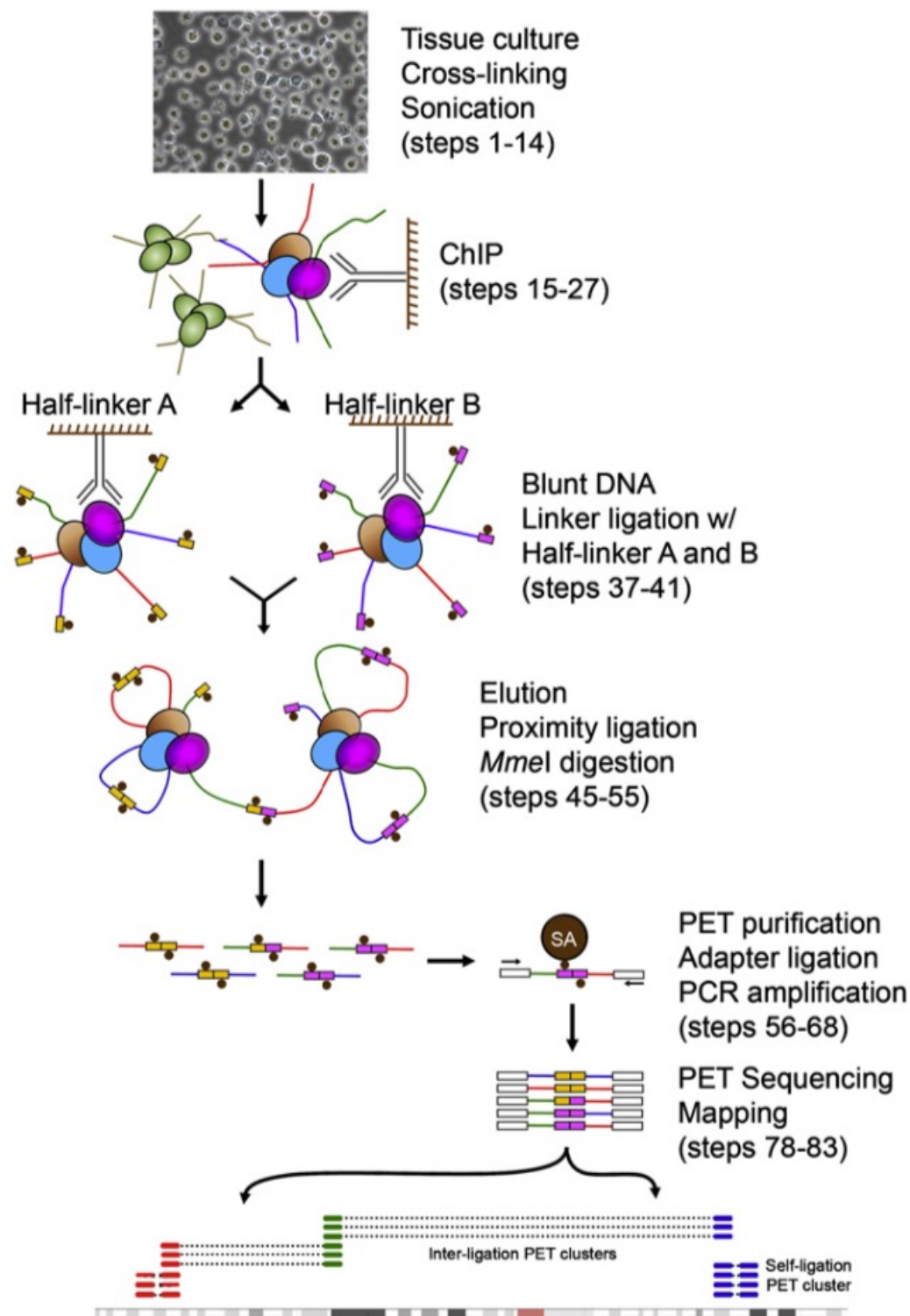


ChIA-PET

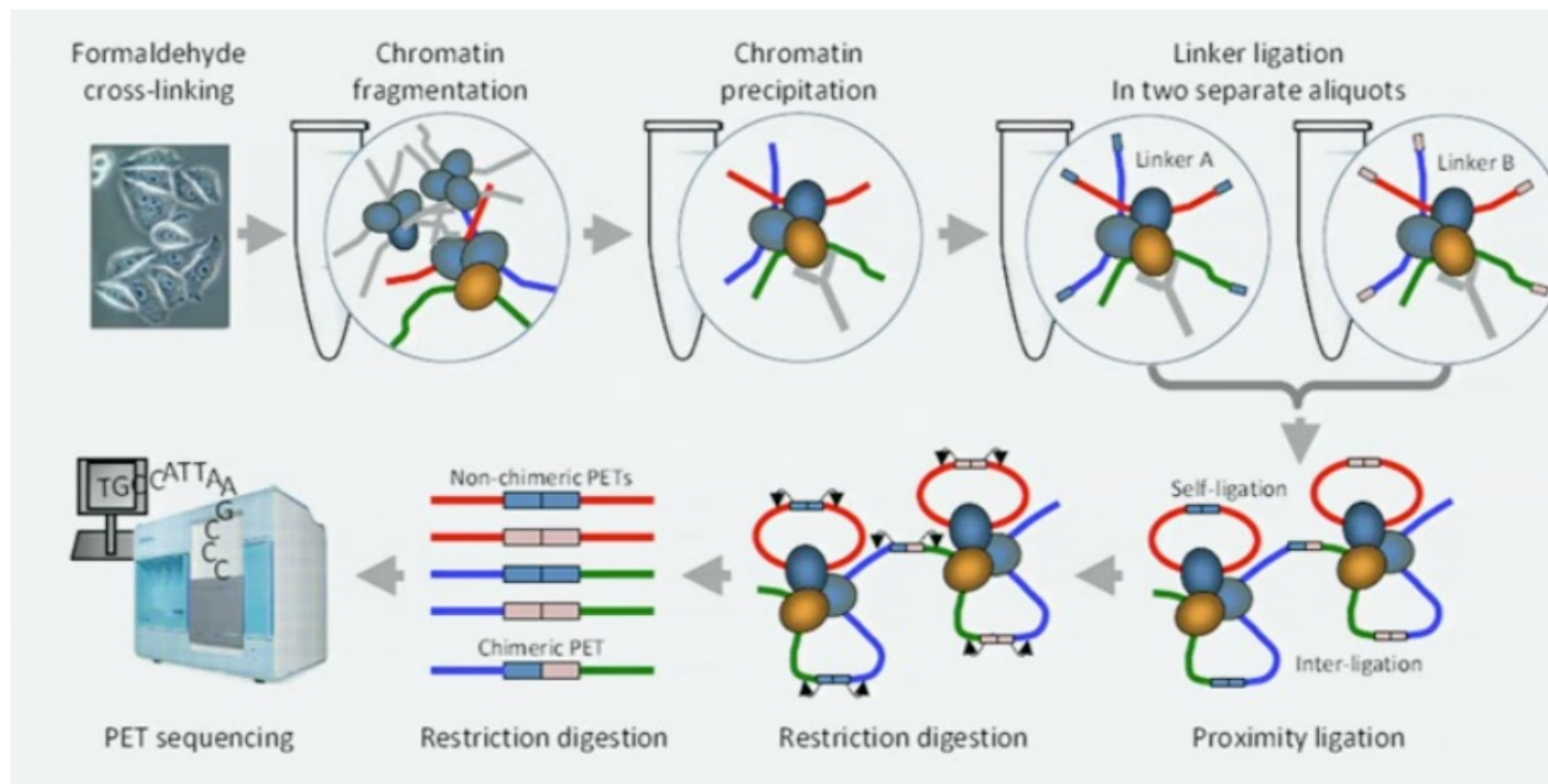
Nature 2009





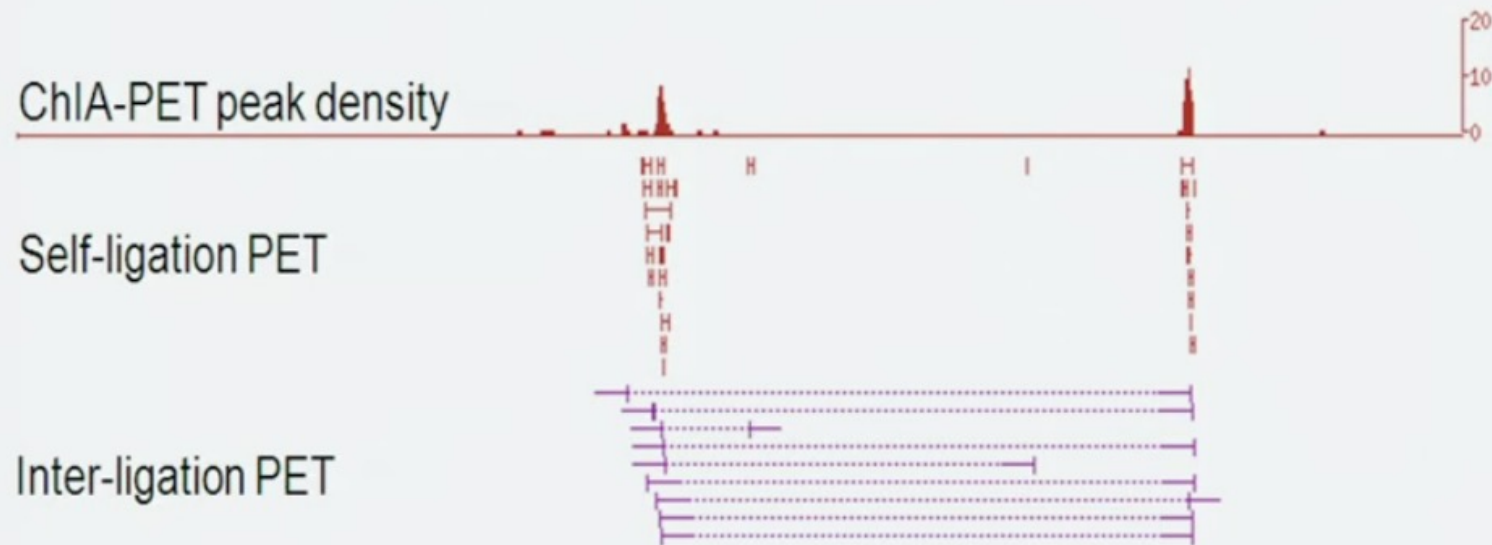


ChIA-PET Procedure



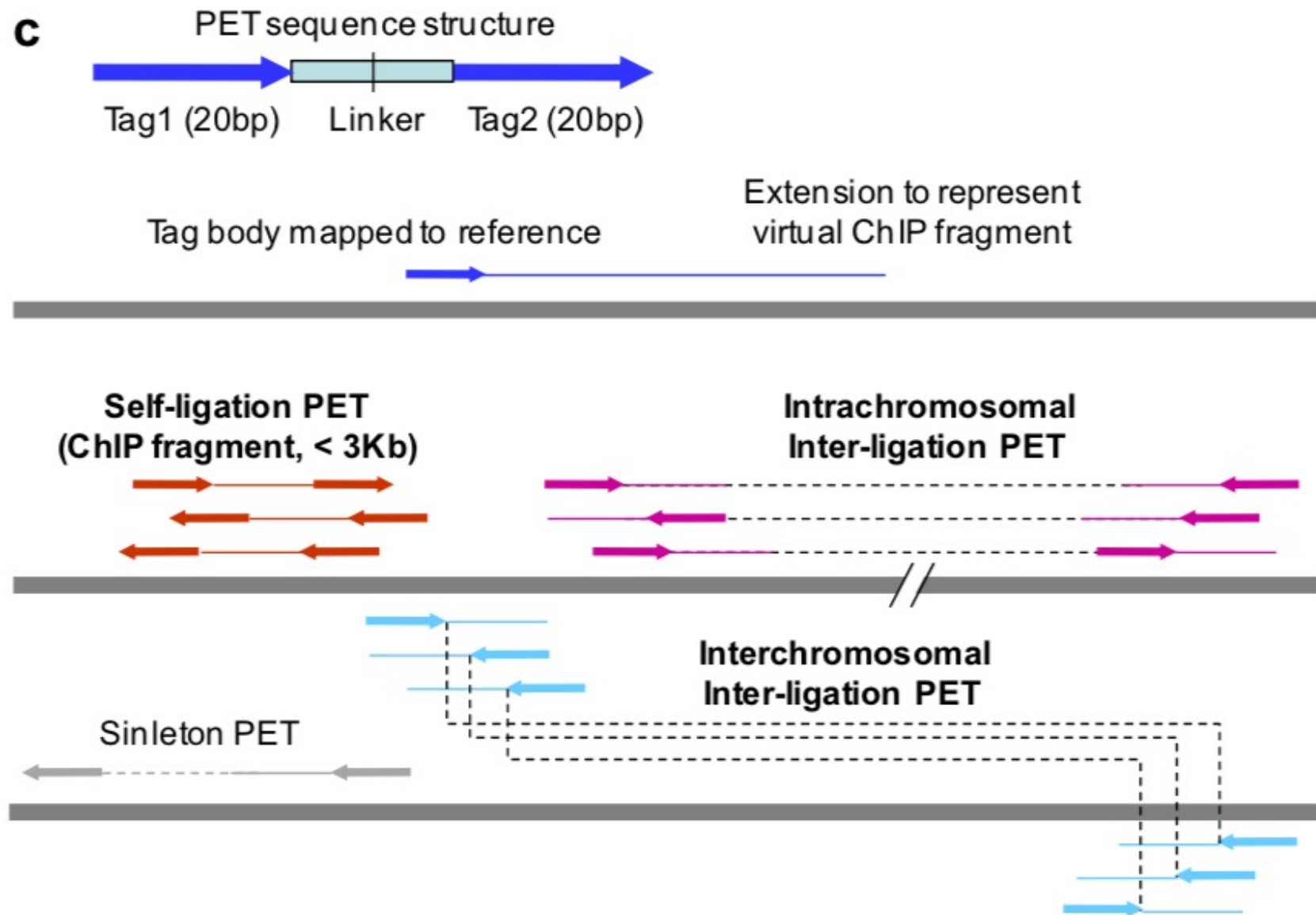
ChIA-PET Browser View

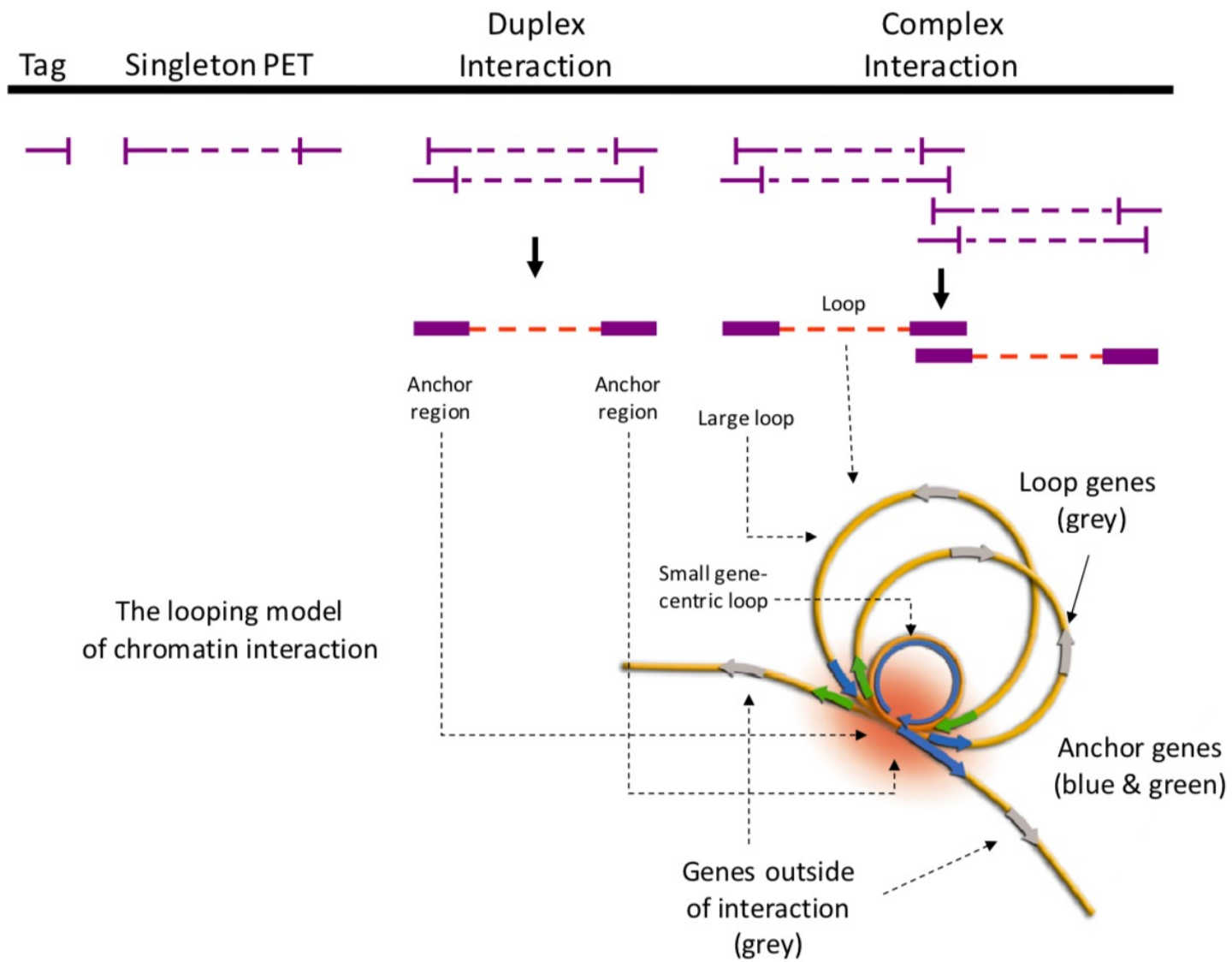
Computational analysis pipeline for ChIA-PET
(Li *et al.*, Genome Biology, 2010)

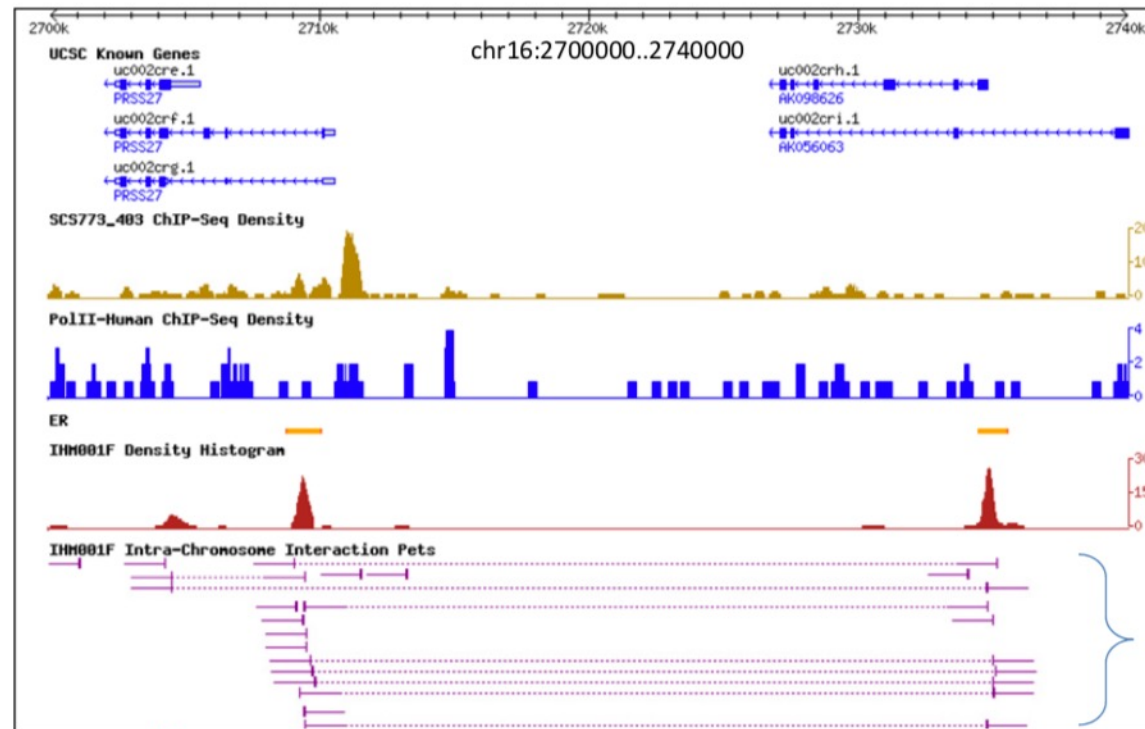


Is this a significant interaction?

c







Anchors

Weak
Anchorpoint

Duplex interaction

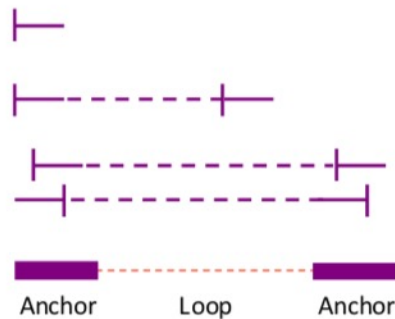
Strong
Anchorpoint

Tag defined ChIP fragment

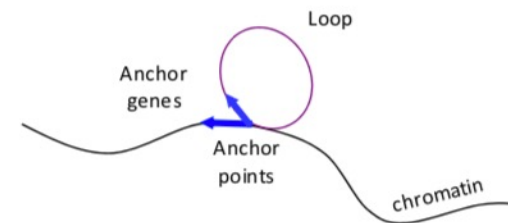
Connection of 2 ChIP fragments
in a inter-ligation PET

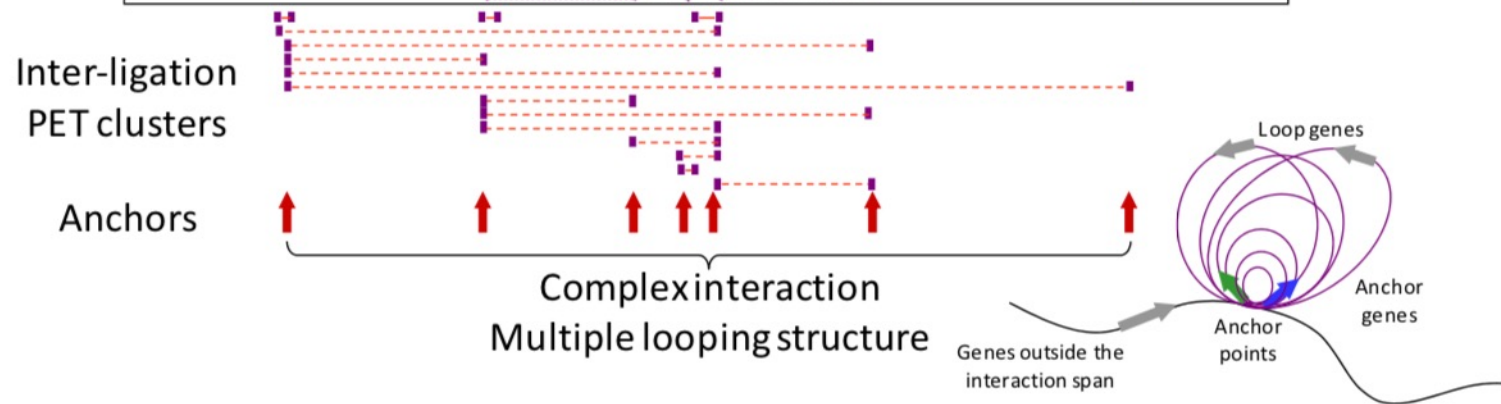
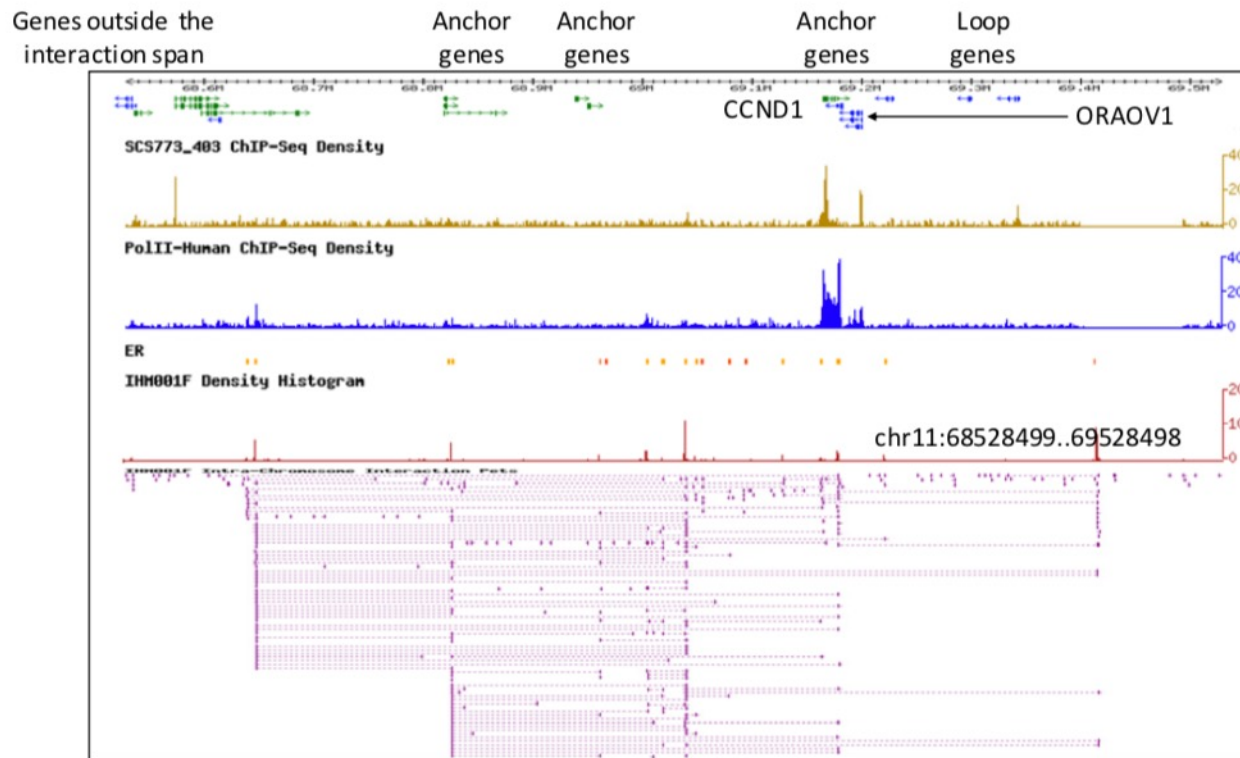
Cluster of overlapping
Inter-ligation PETs

Duplex interaction defined
by inter-ligation PET cluster

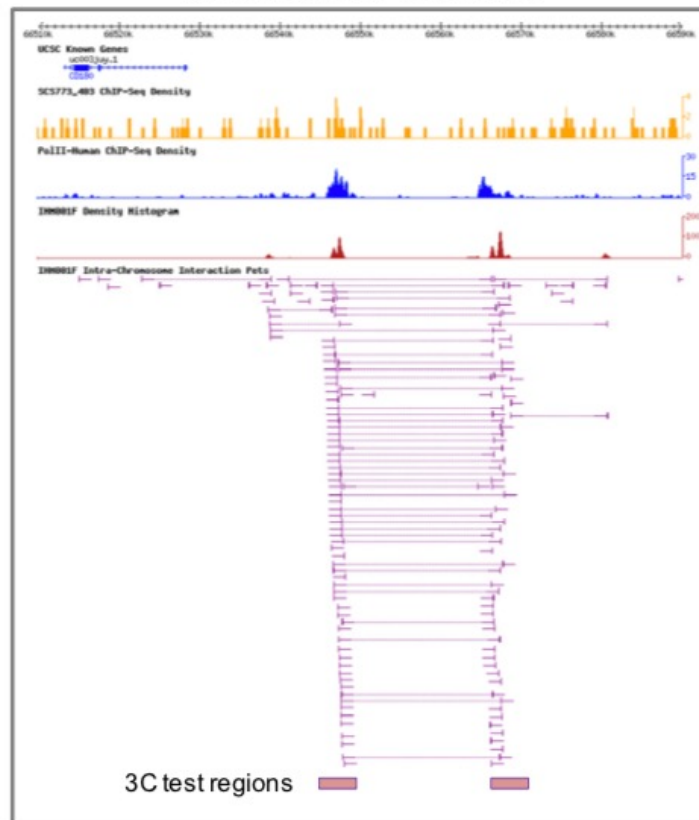


Suggested looping structure

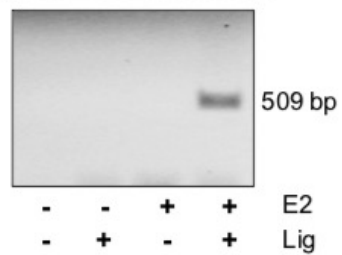




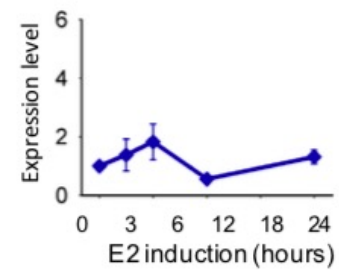
chr5:66510000-66590000

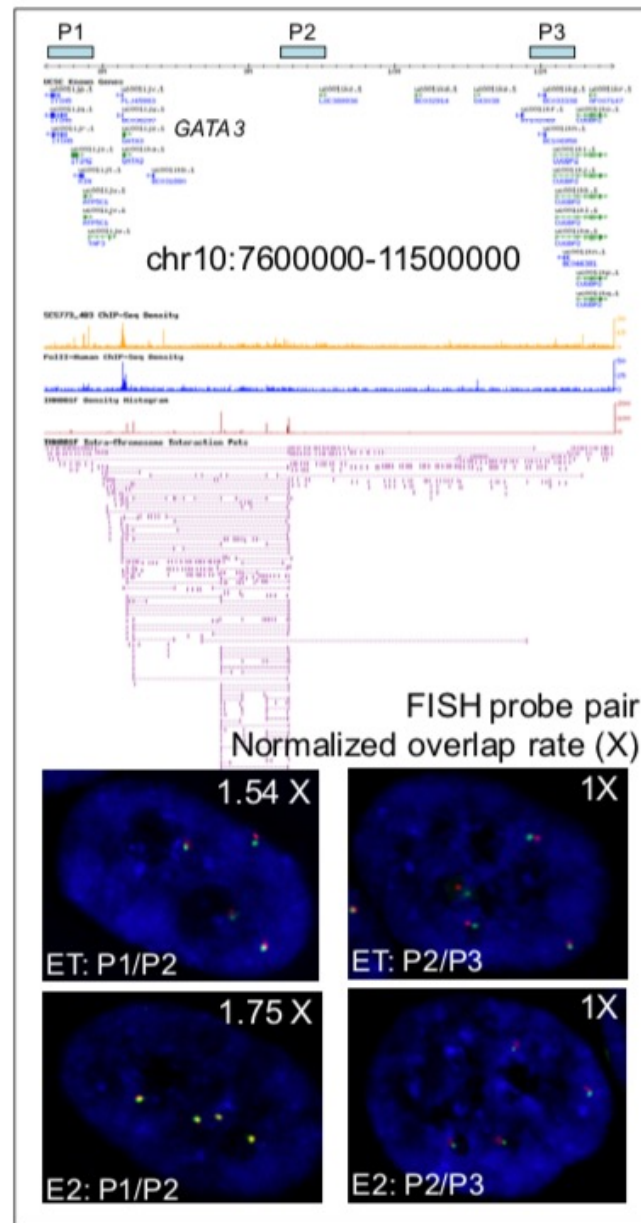


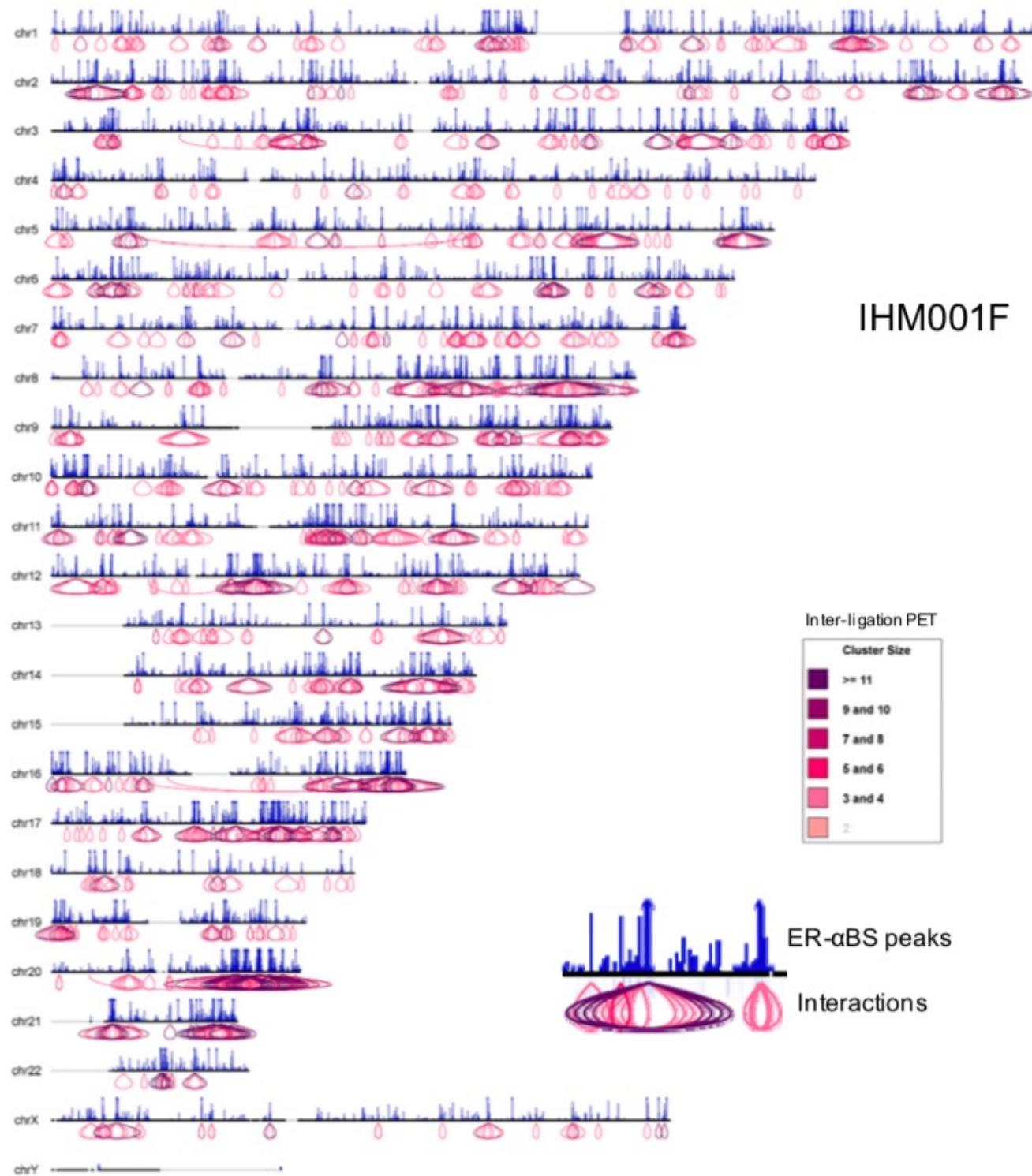
ChIP-3C validation

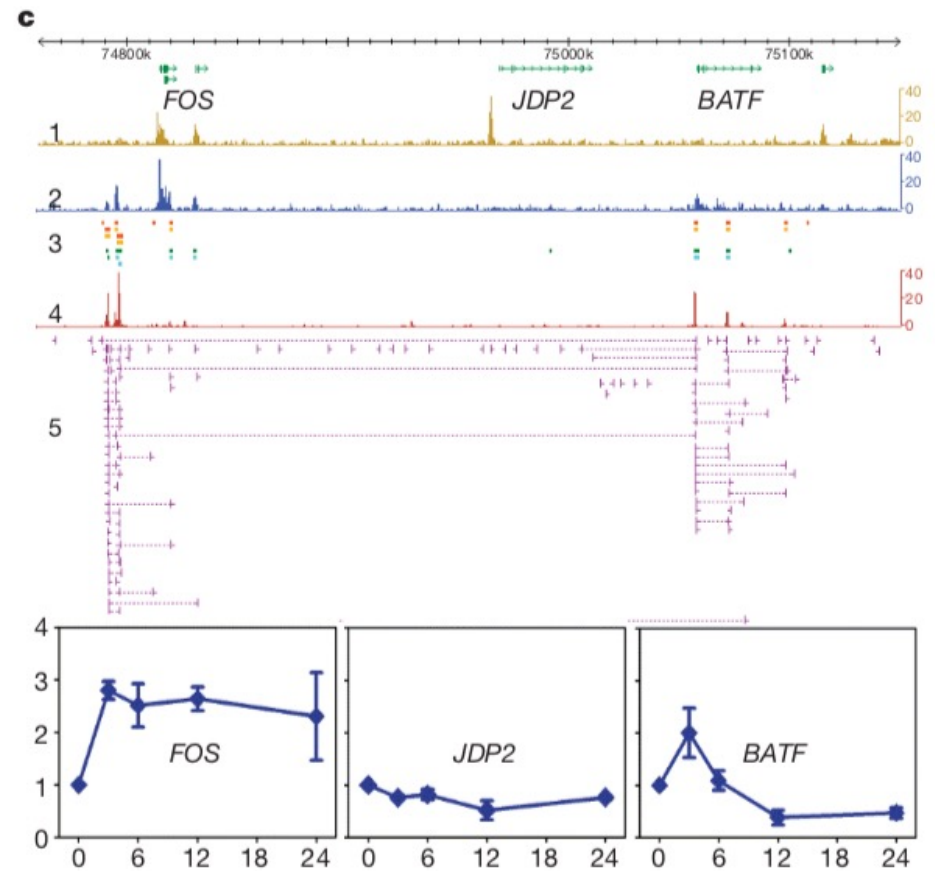
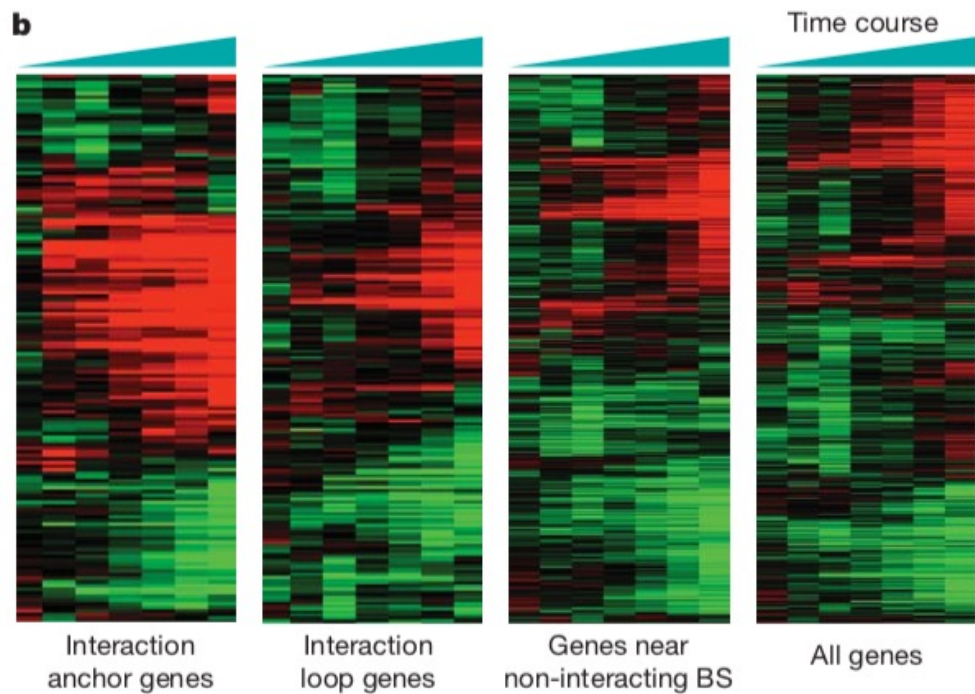
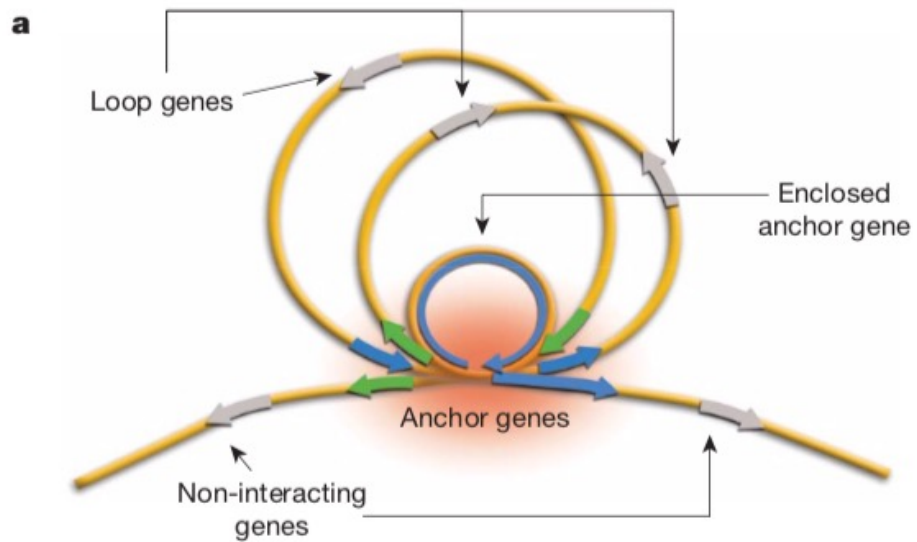


RT-qPCR

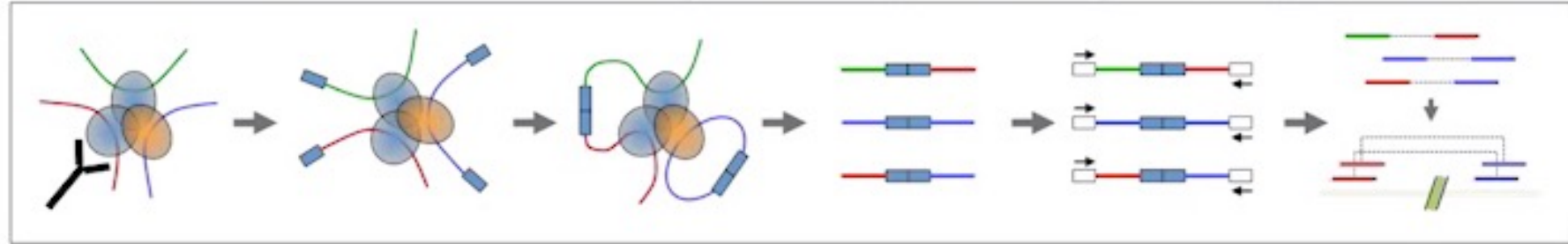




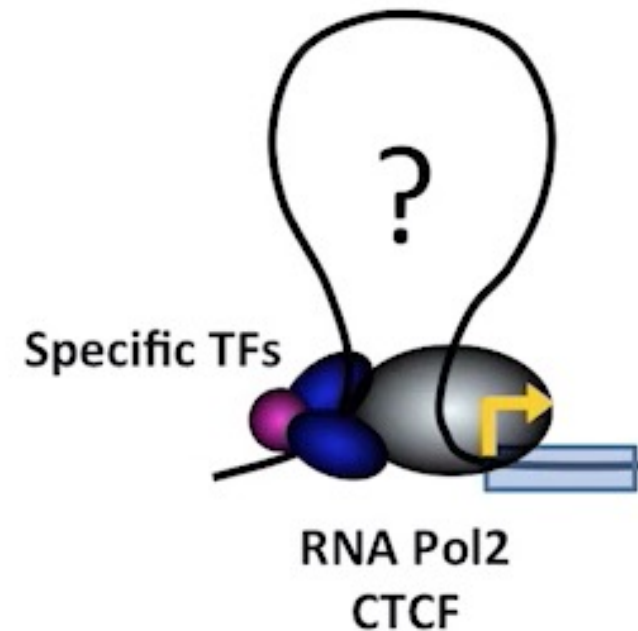
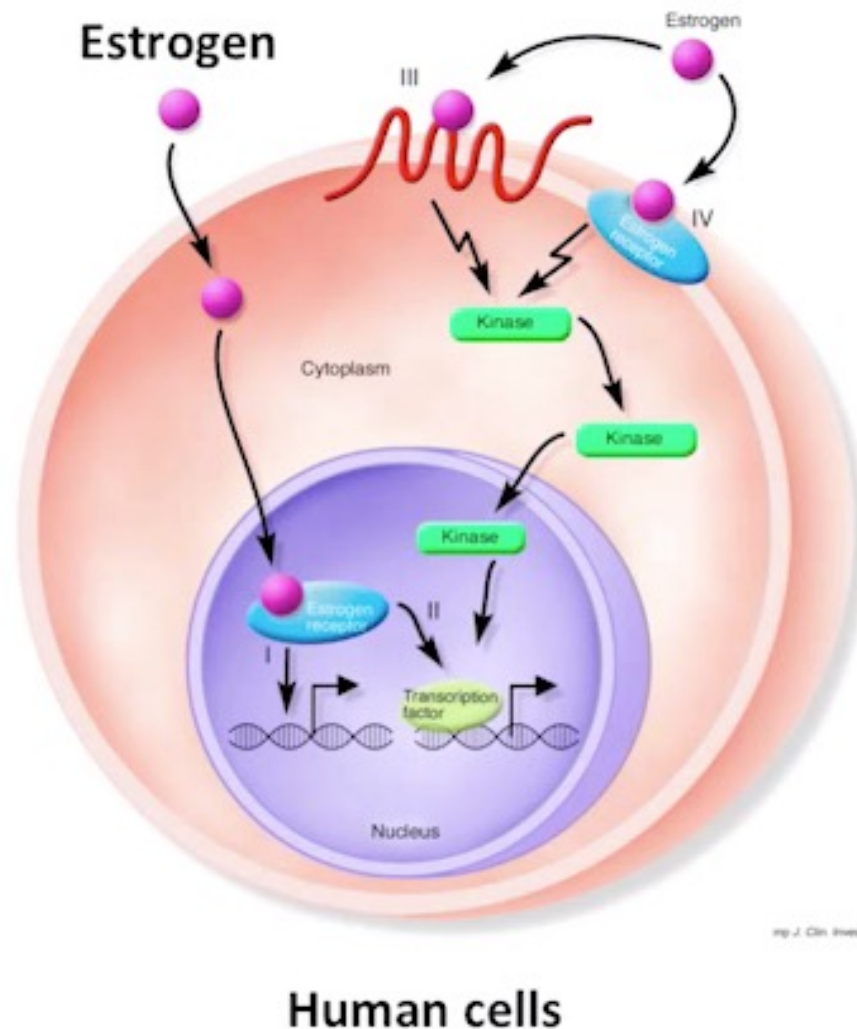




Chromatin Interaction Analysis using PET sequencing (ChIA-PET)



Fullwood et al. (2009) Nature



Mechanisms of transcription regulation

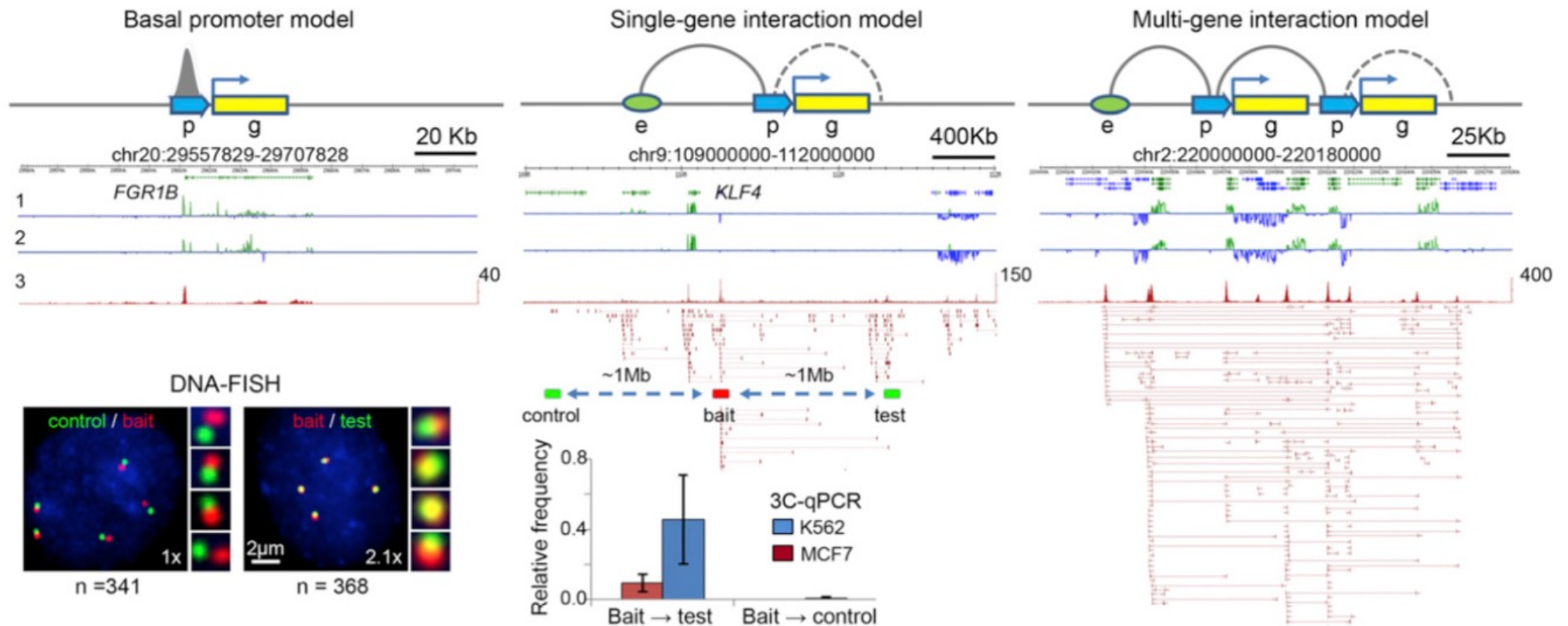
Specific TFs: ER, p53, RARA, ...

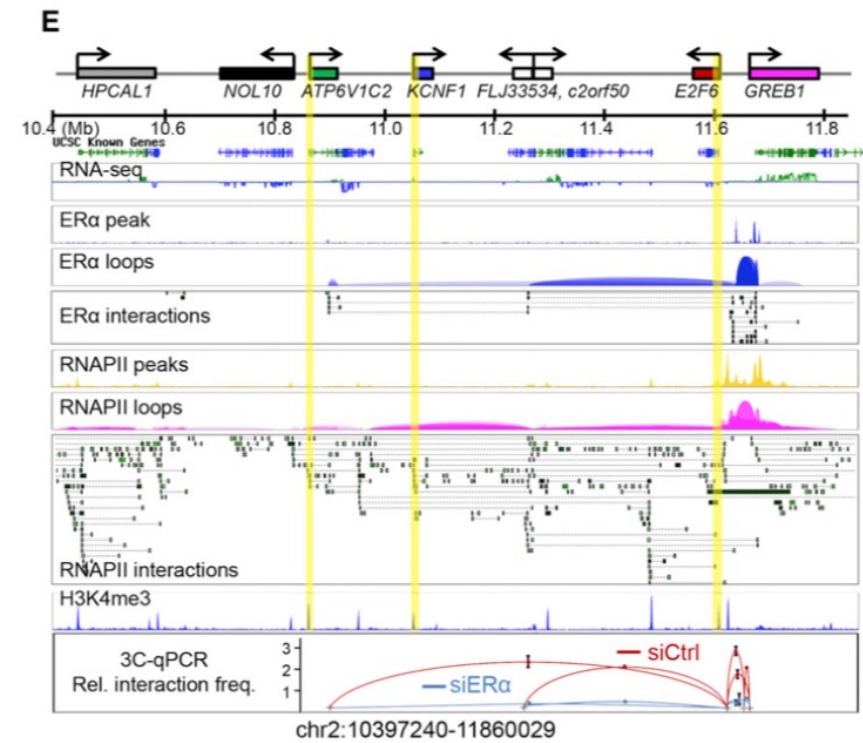
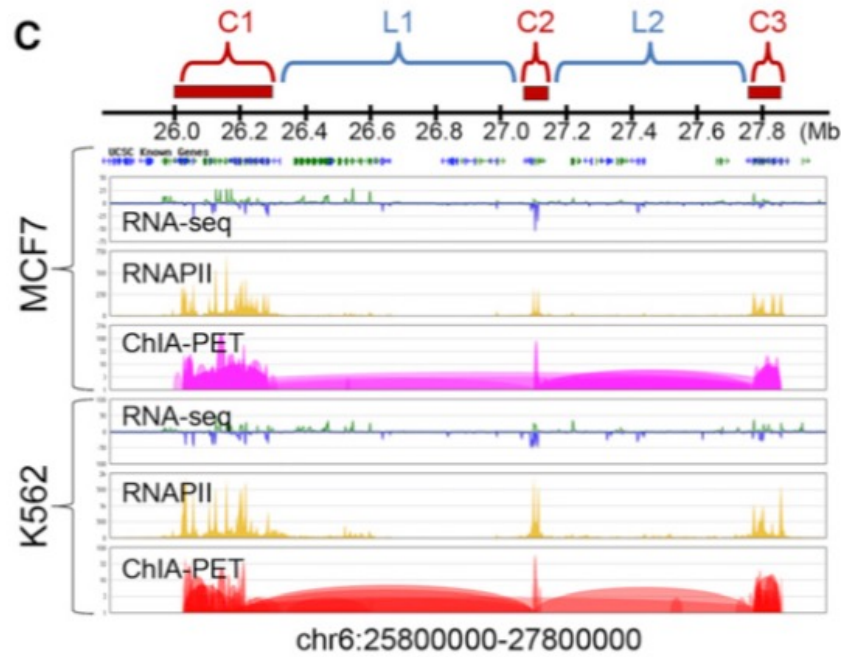
General TFs: RNA Pol 2, TAF1, ...

Chromatin Fs: CTCF, cohesin, ...

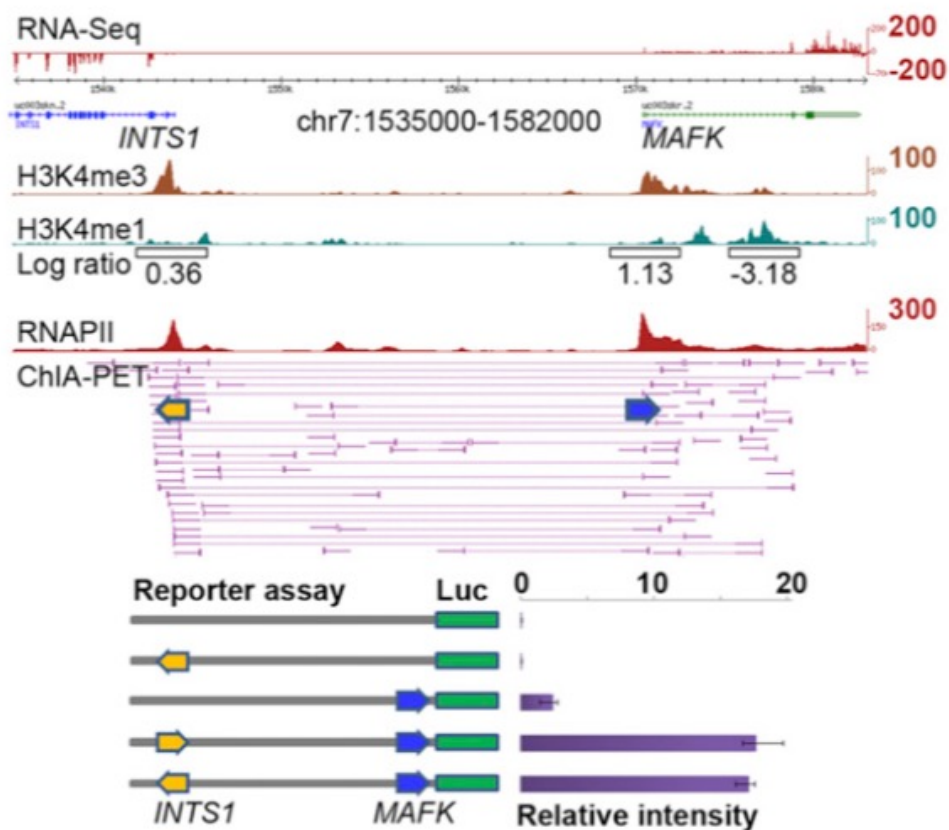
Extensive Promoter-Centered Chromatin Interactions Provide a Topological Basis for Transcription Regulation

Guoliang Li,^{1,10} Xiaolan Ruan,^{1,10} Raymond K. Auerbach,^{2,10} Kuljeet Singh Sandhu,^{1,10} Meizhen Zheng,¹ Ping Wang,¹ Huay Mei Poh,¹ Yufen Goh,¹ Joanne Lim,¹ Jingyao Zhang,¹ Hui Shan Sim,¹ Su Qin Peh,¹ Fabianus Hendriyan Mulawadi,¹ Chin Thing Ong,¹ Yuriy L. Orlov,¹ Shuzhen Hong,¹ Zhizhuo Zhang,³ Steve Landt,⁴ Debasish Raha,⁴ Ghia Euskirchen,⁴ Chia-Lin Wei,¹ Weihong Ge,⁵ Huairen Wang,⁶ Carrie Davis,⁶ Katherine I. Fisher-Aylor,⁷ Ali Mortazavi,⁷ Mark Gerstein,² Thomas Gingeras,⁶ Barbara Wold,⁷ Yi Sun,⁵ Melissa J. Fullwood,¹ Edwin Cheung,^{1,8} Edison Liu,¹ Wing-Kin Sung,^{1,3} Michael Snyder,^{4,*} and Yijun Ruan^{1,9,*}

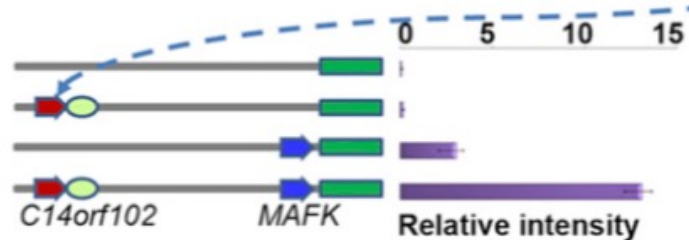




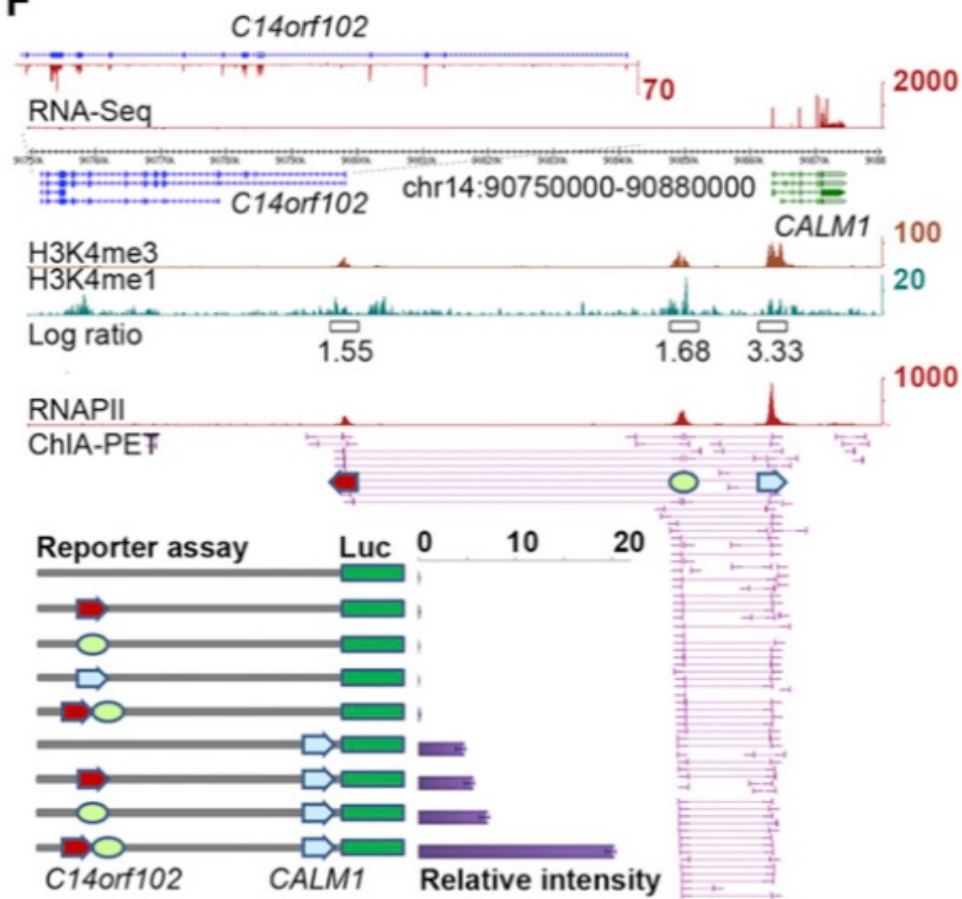
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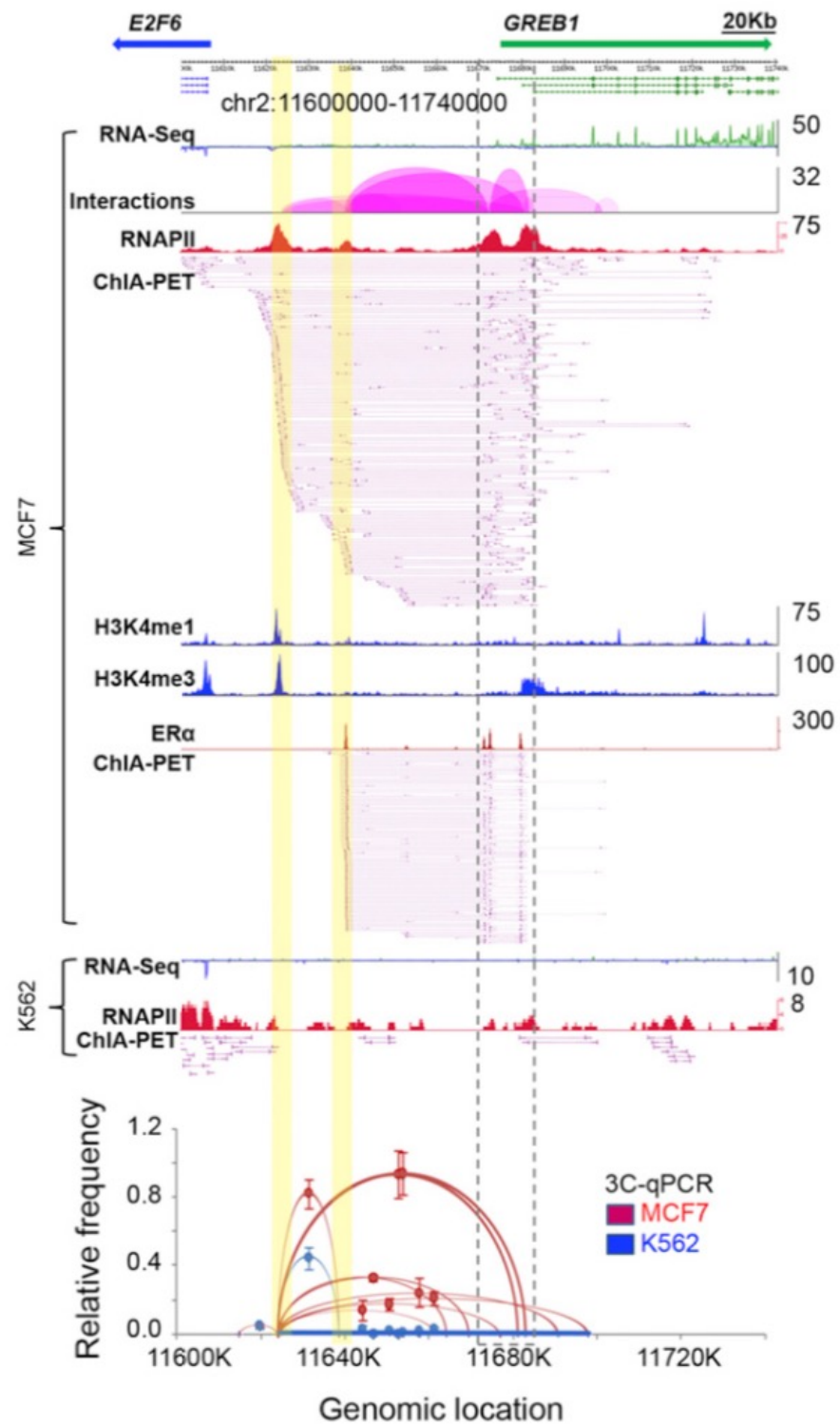


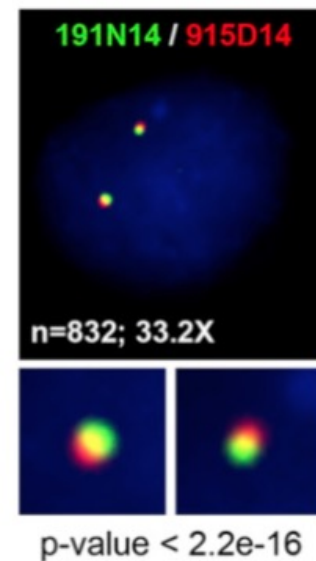
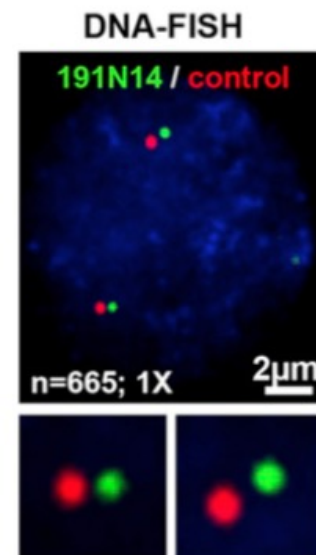
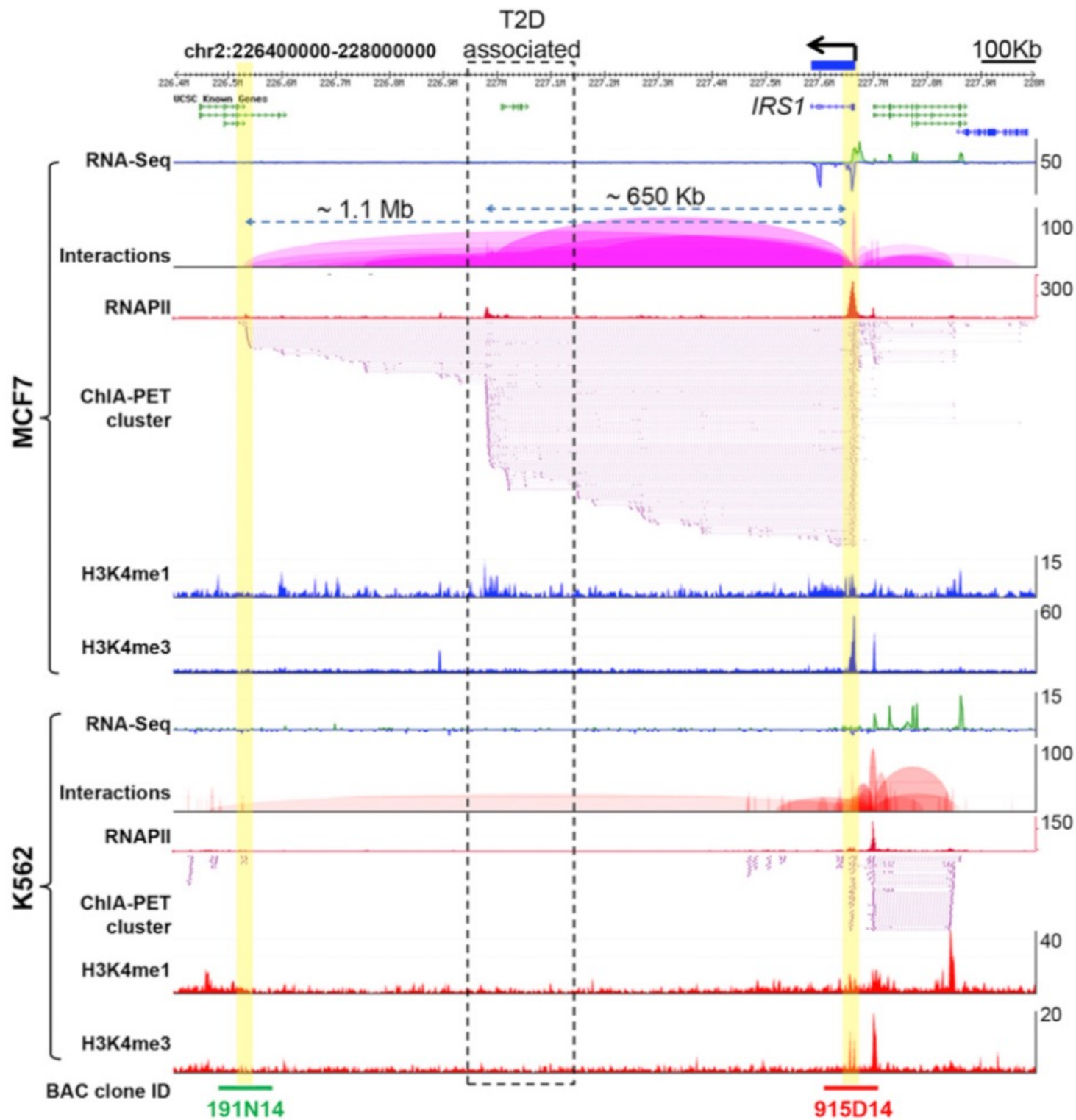
G



F

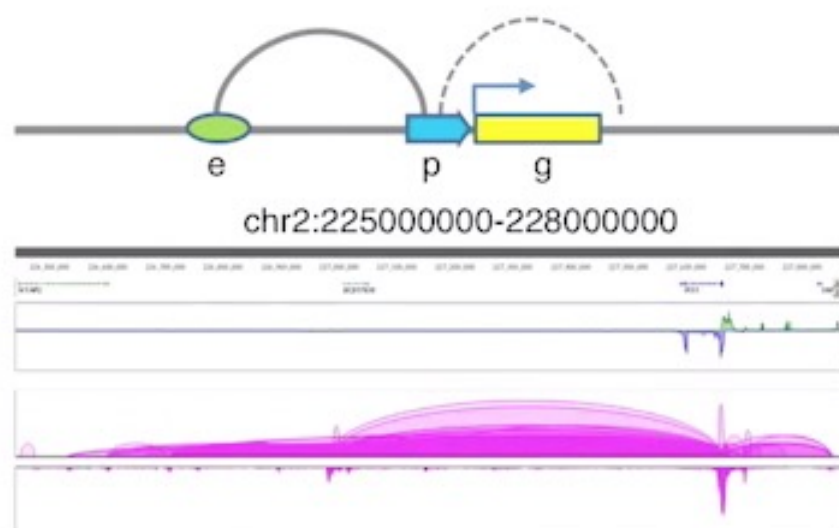




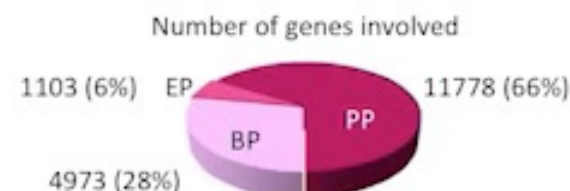
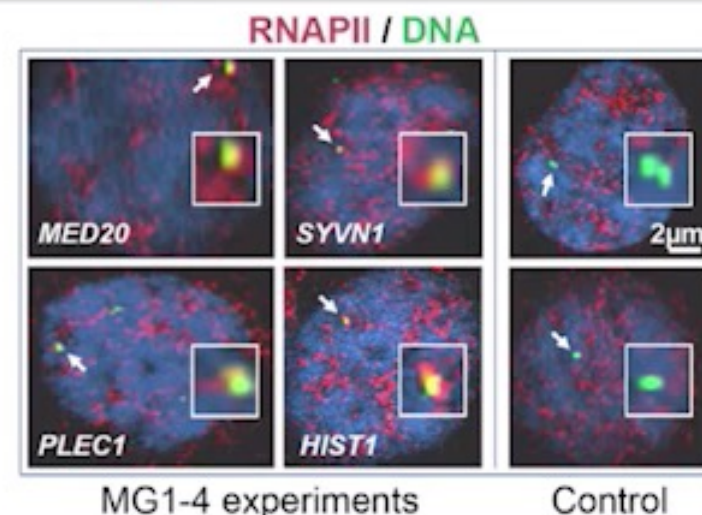
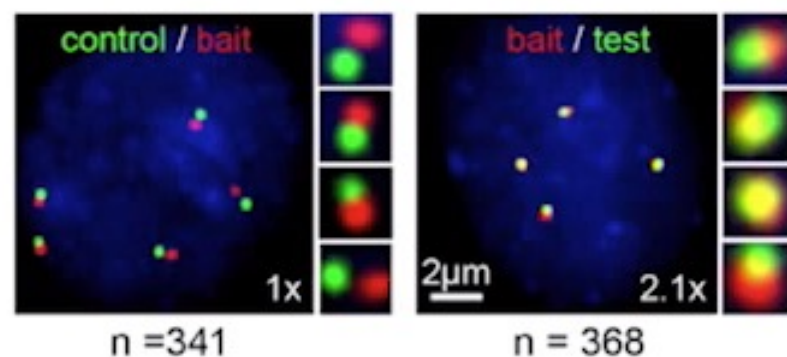
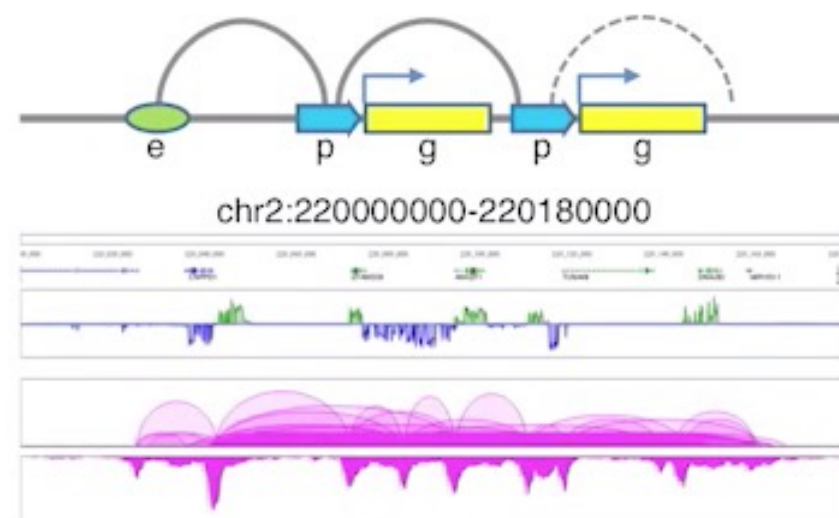


RNAPII associated chromatin interaction models

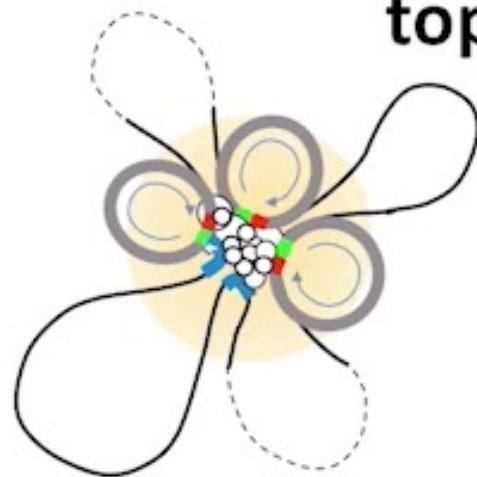
Extra-genic enhancer-promoter interaction (EP)
Single gene complex



Inter-genic promoter-promoter interaction (PP)
Multi-gene complex

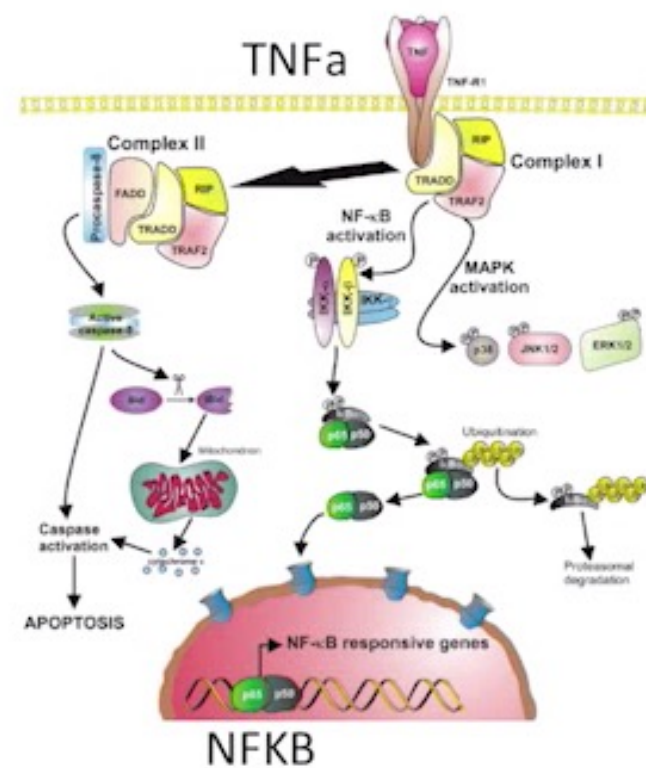
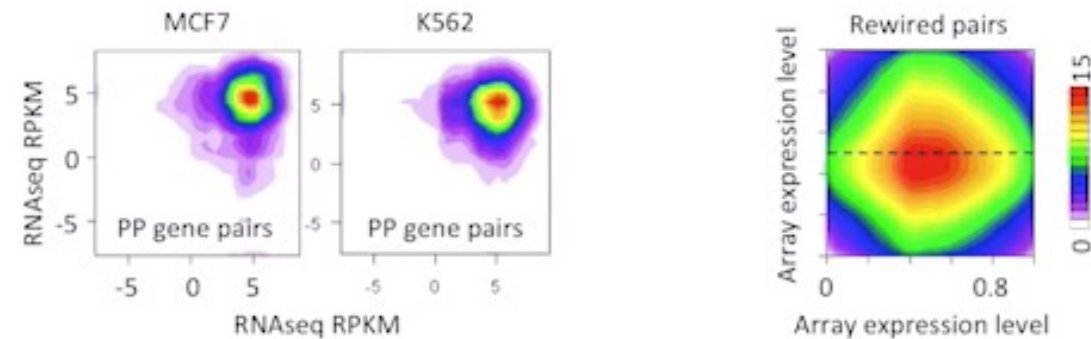


Multi-gene interaction complex provides the topological basis for co-transcription regulation

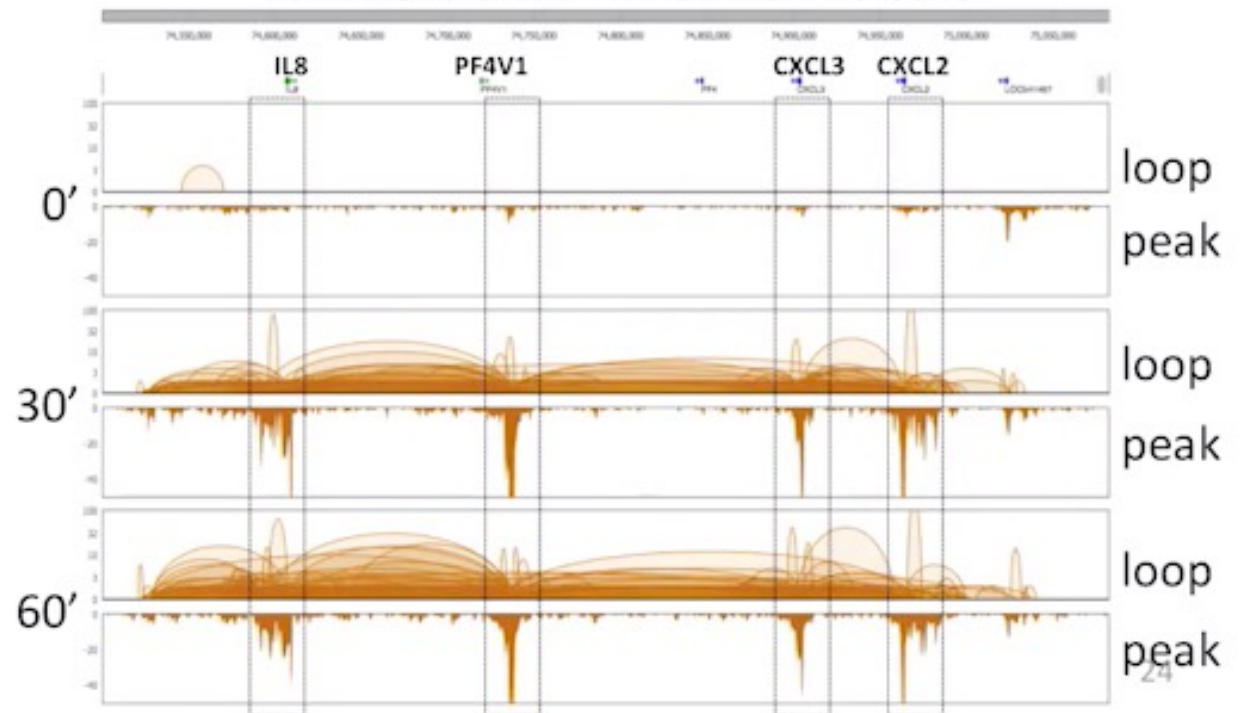


Multi-gene complex

Co-transcription

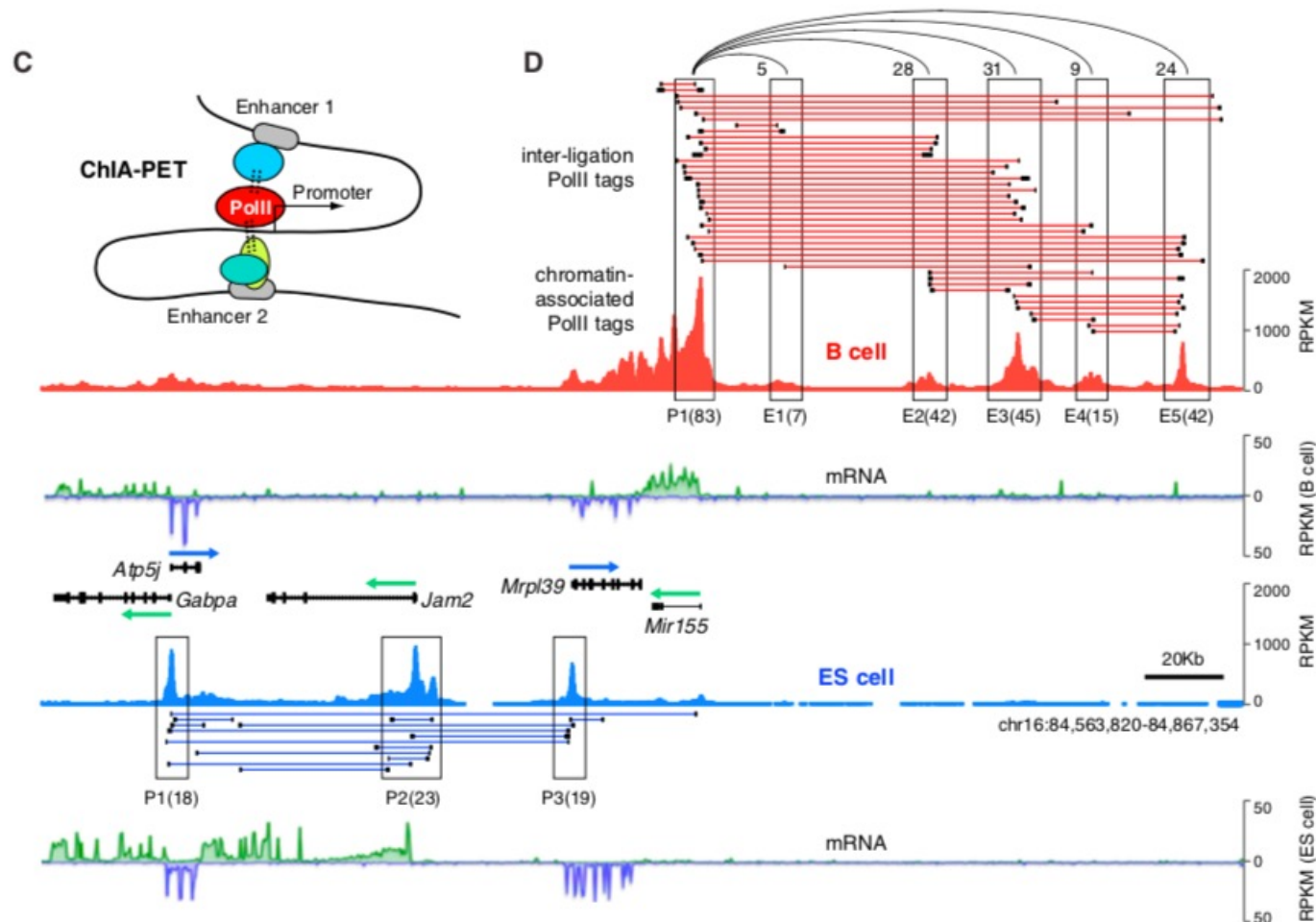


RNA Pol2 ChIA-PET of HUVEC cells



Interactome Maps of Mouse Gene Regulatory Domains Reveal Basic Principles of Transcriptional Regulation

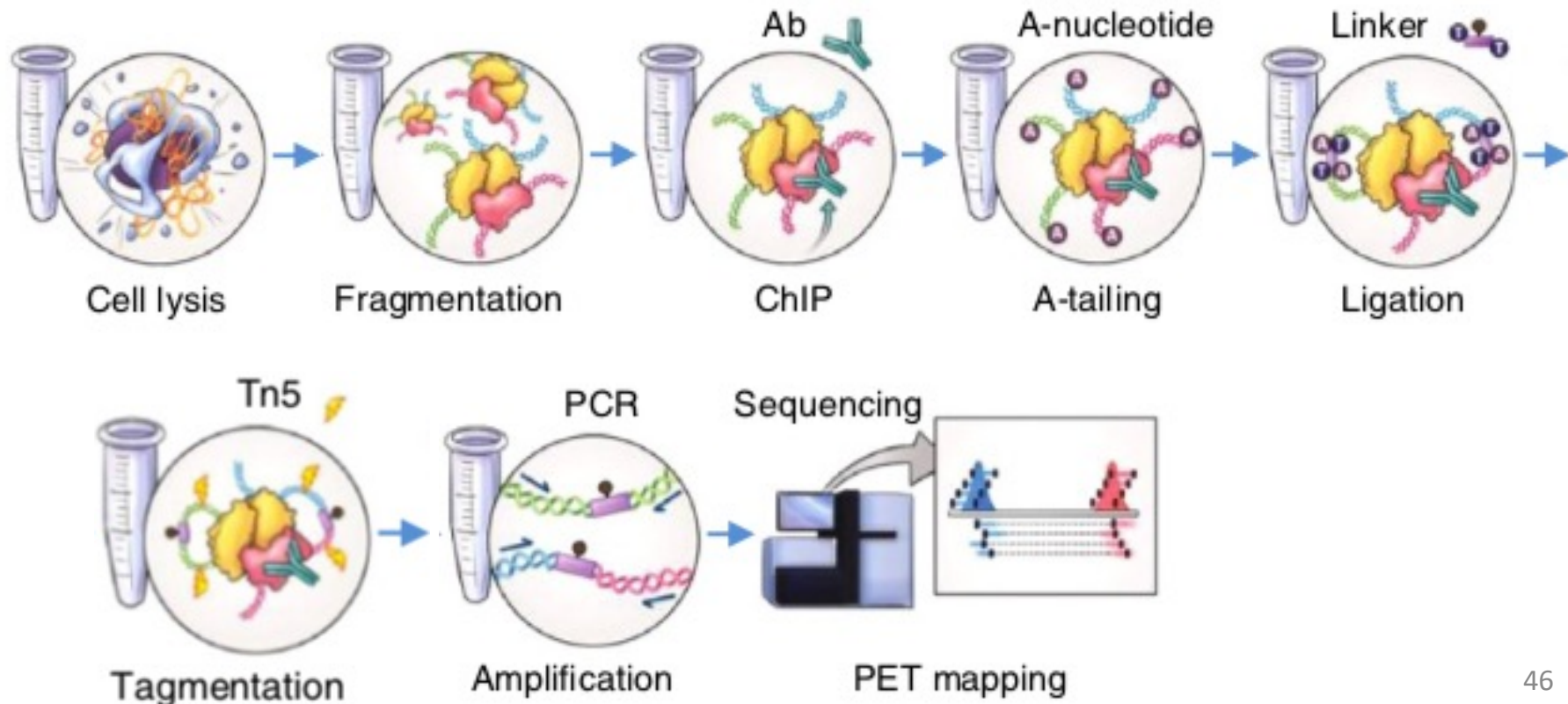
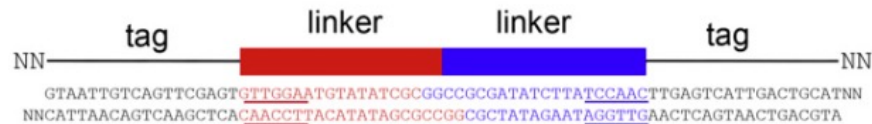
Kyong-Rim Kieffer-Kwon,^{1,12} Zhonghui Tang,^{2,12} Ewy Mathe,^{1,12} Jason Qian,^{1,12} Myong-Hee Sung,^{3,12} Guoliang Li,² Wolfgang Resch,¹ Songjoon Baek,³ Nathanael Pruett,¹ Lars Grøntved,³ Laura Vian,¹ Steevenson Nelson,¹ Hossein Zare,⁴ Ofir Hakim,⁵ Deepak Reyon,^{6,7} Arito Yamane,¹ Hirotaka Nakahashi,¹ Alexander L. Kovalchuk,⁸ Jizhong Zou,⁹ J. Keith Joung,^{6,7} Vittorio Sartorelli,⁴ Chia-Lin Wei,¹⁰ Xiaolan Ruan,² Gordon L. Hager,^{3,12} Yijun Ruan,^{2,12} and Rafael Casellas^{1,11,12,*}

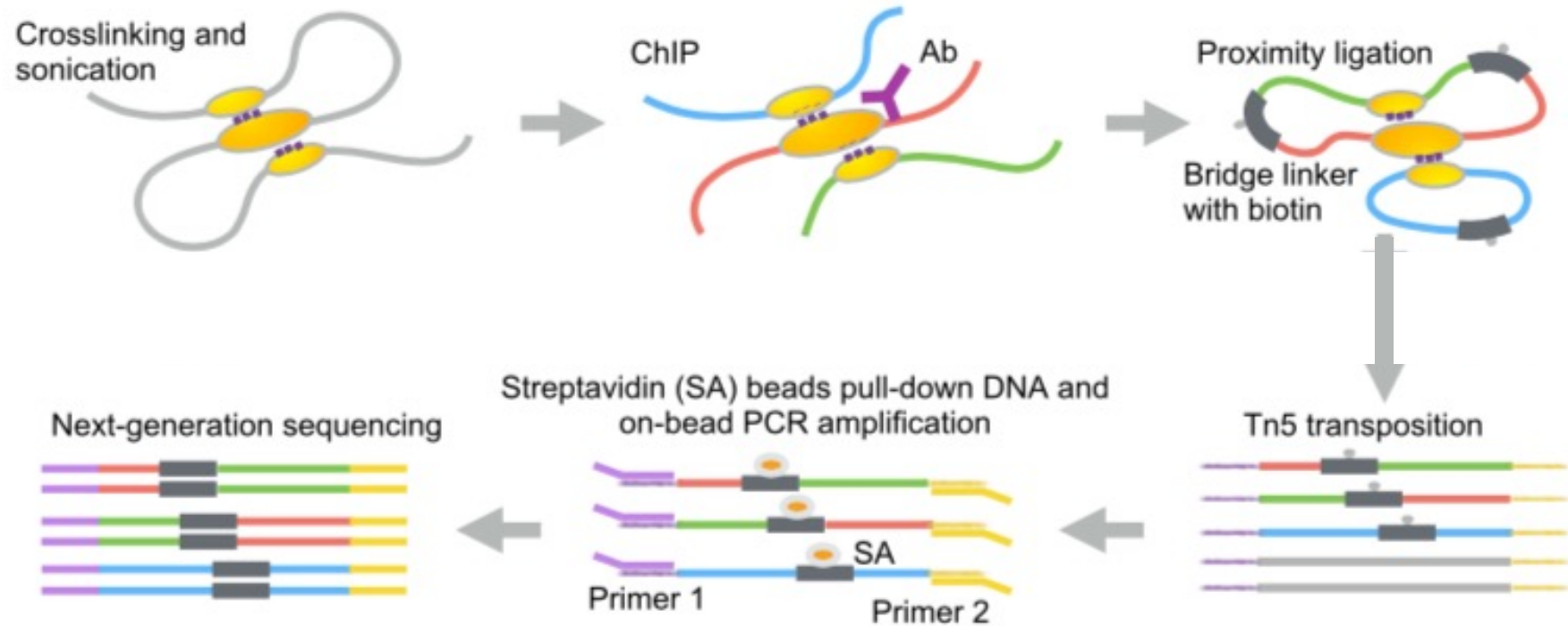


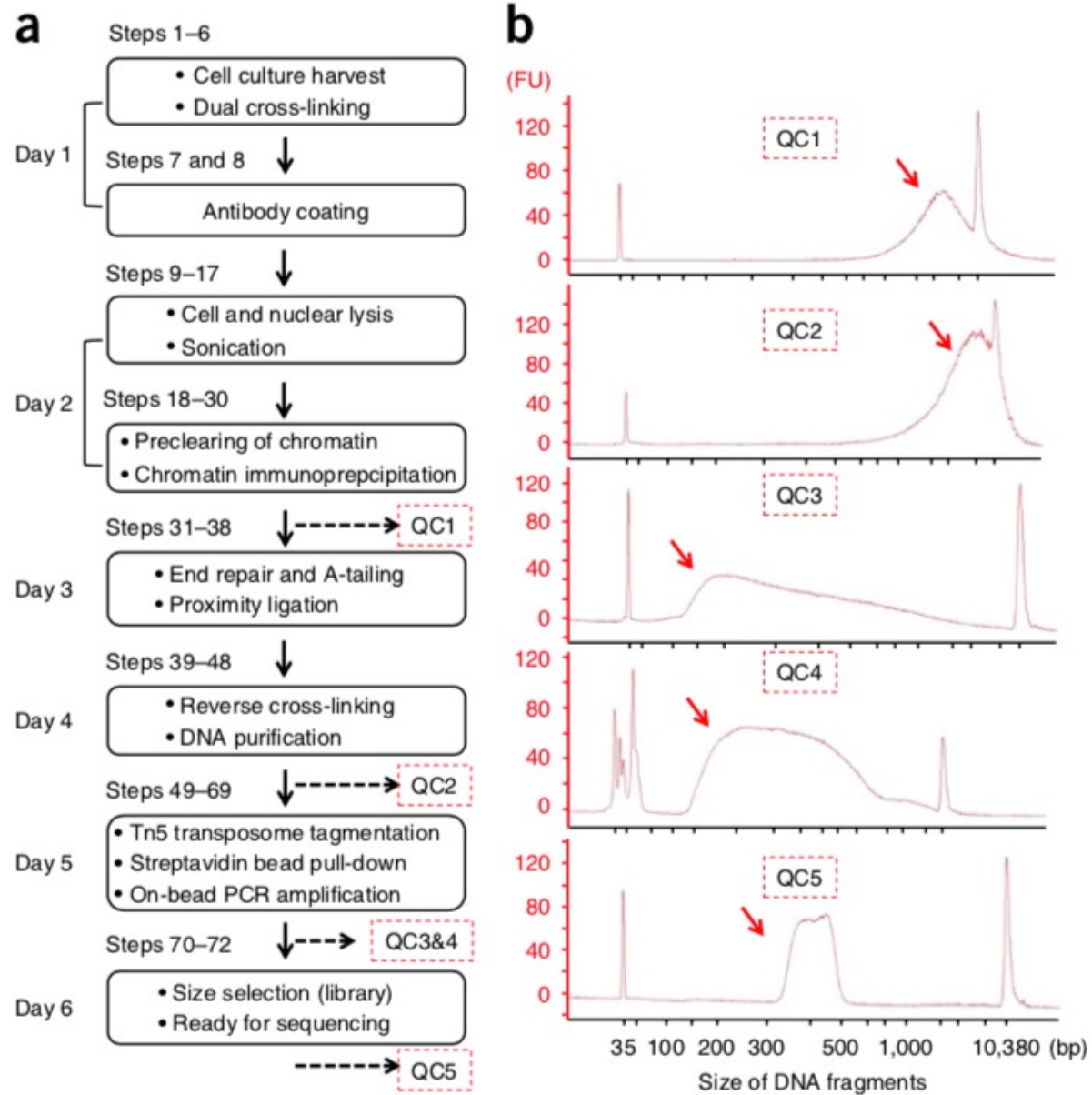
Long-read ChIA-PET for base-pair-resolution mapping of haplotype-specific chromatin interactions

Long-read ChIA-PET

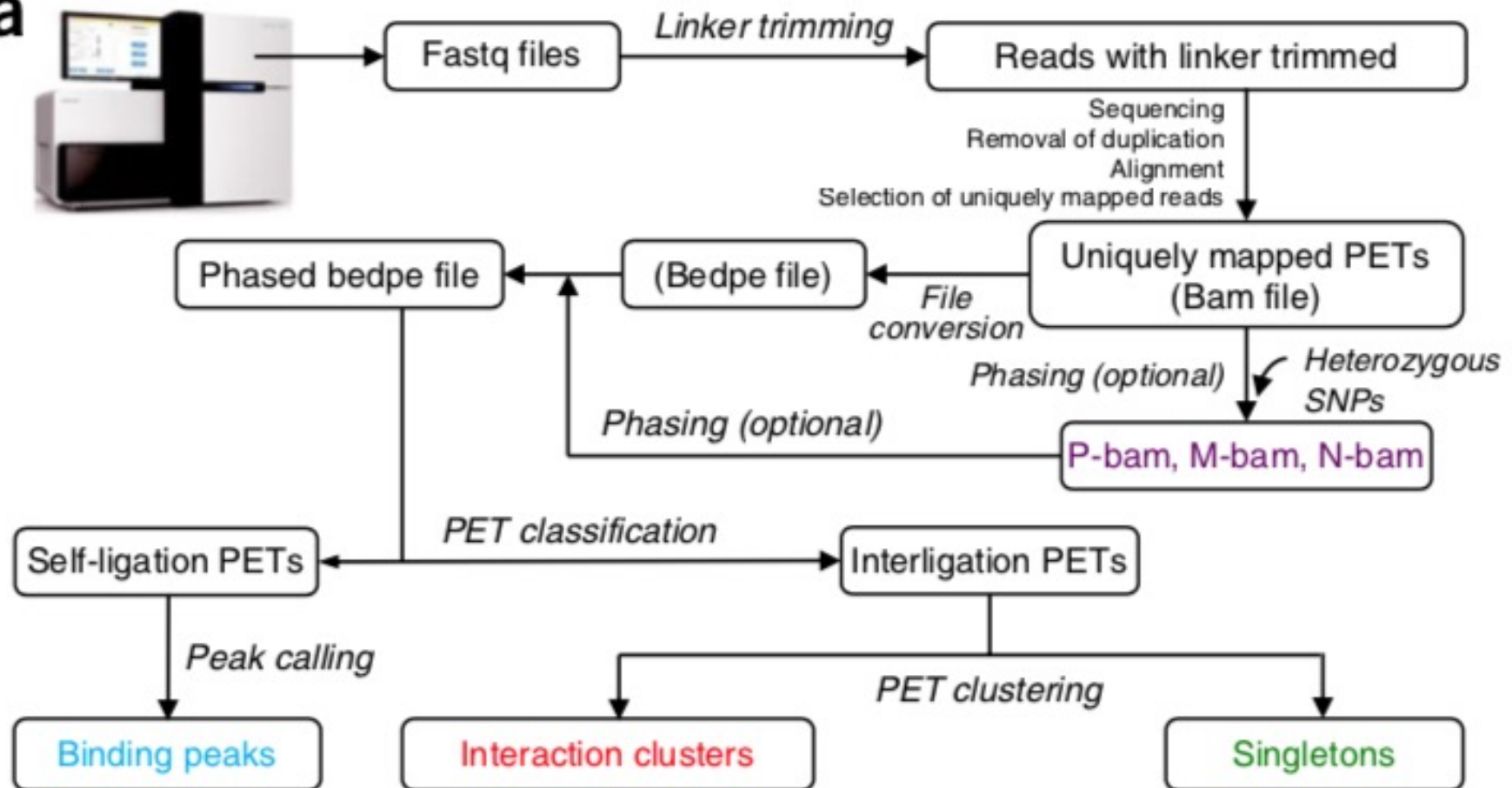
Xingwang Li^{1,2,5}, Oscar Junhong Luo^{1,4,5}, Ping Wang¹, Meizhen Zheng¹, Danjuan Wang¹, Emaly Piecuch^{1,3}, Jacqueline Jufen Zhu^{1,3}, Simon Zhongyuan Tian¹, Zhonghui Tang¹, Guoliang Li² & Yijun Ruan¹⁻³

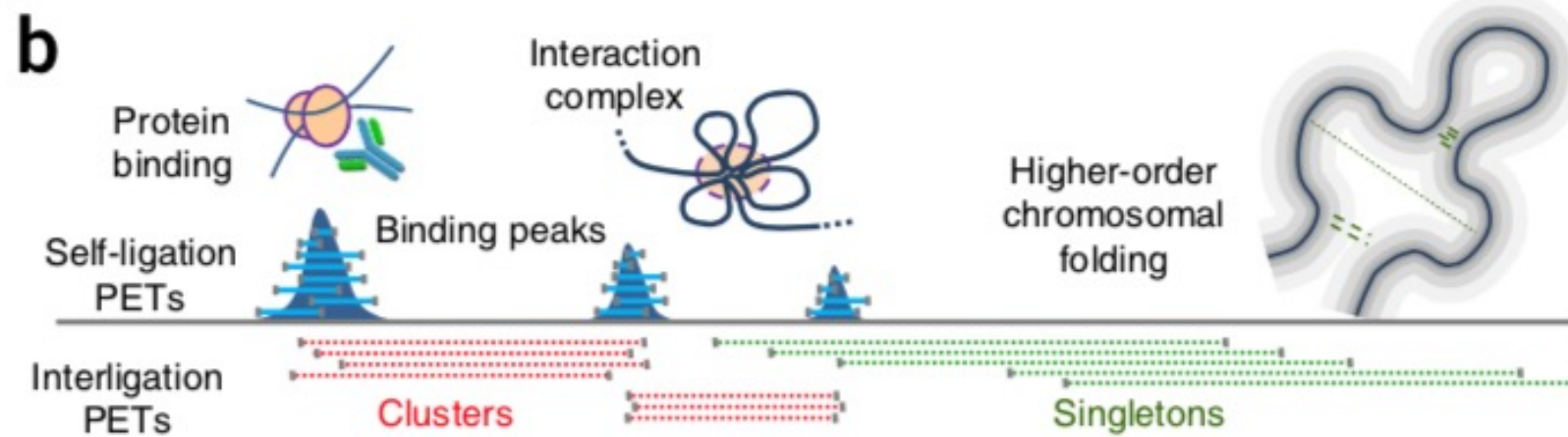


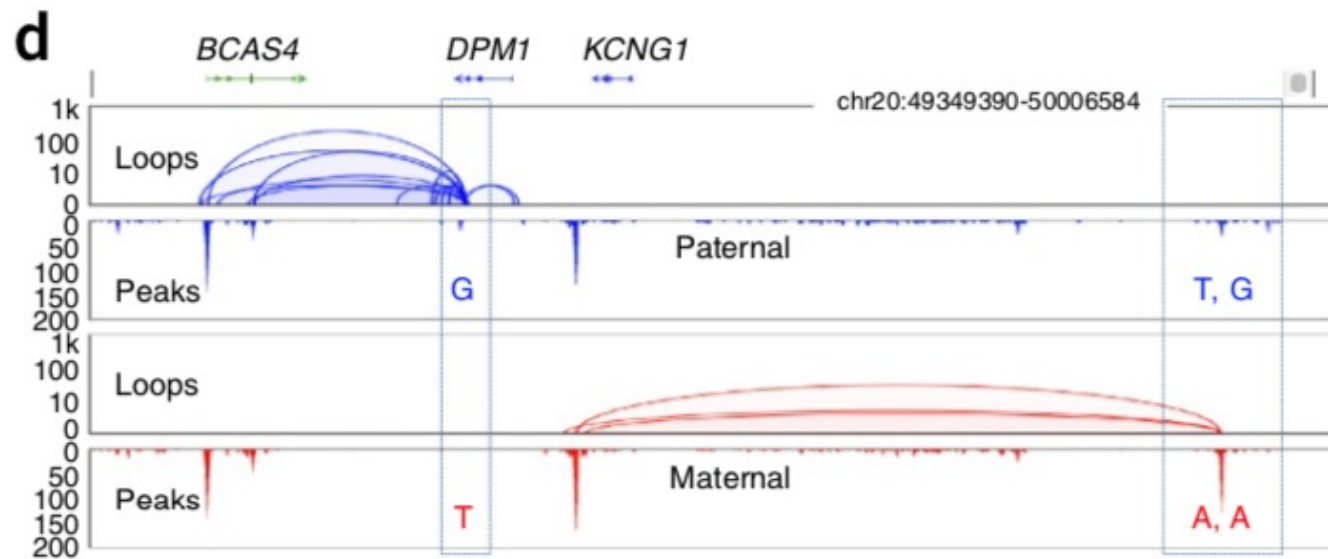
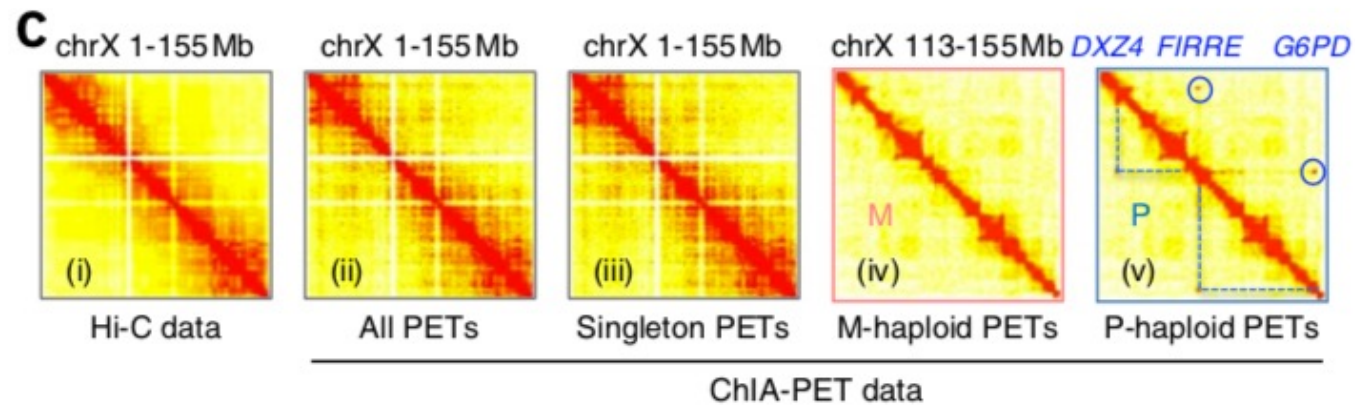




a



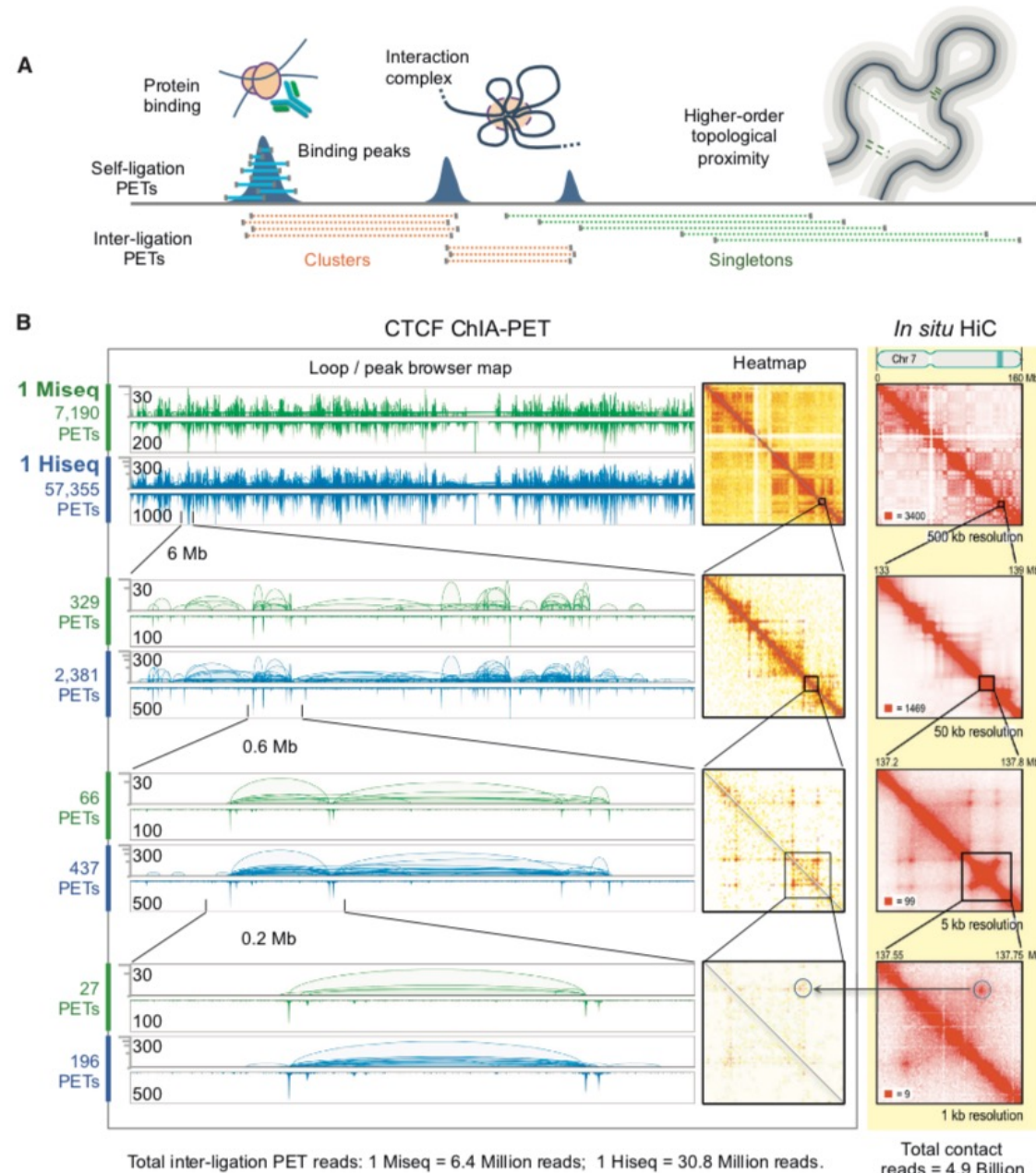


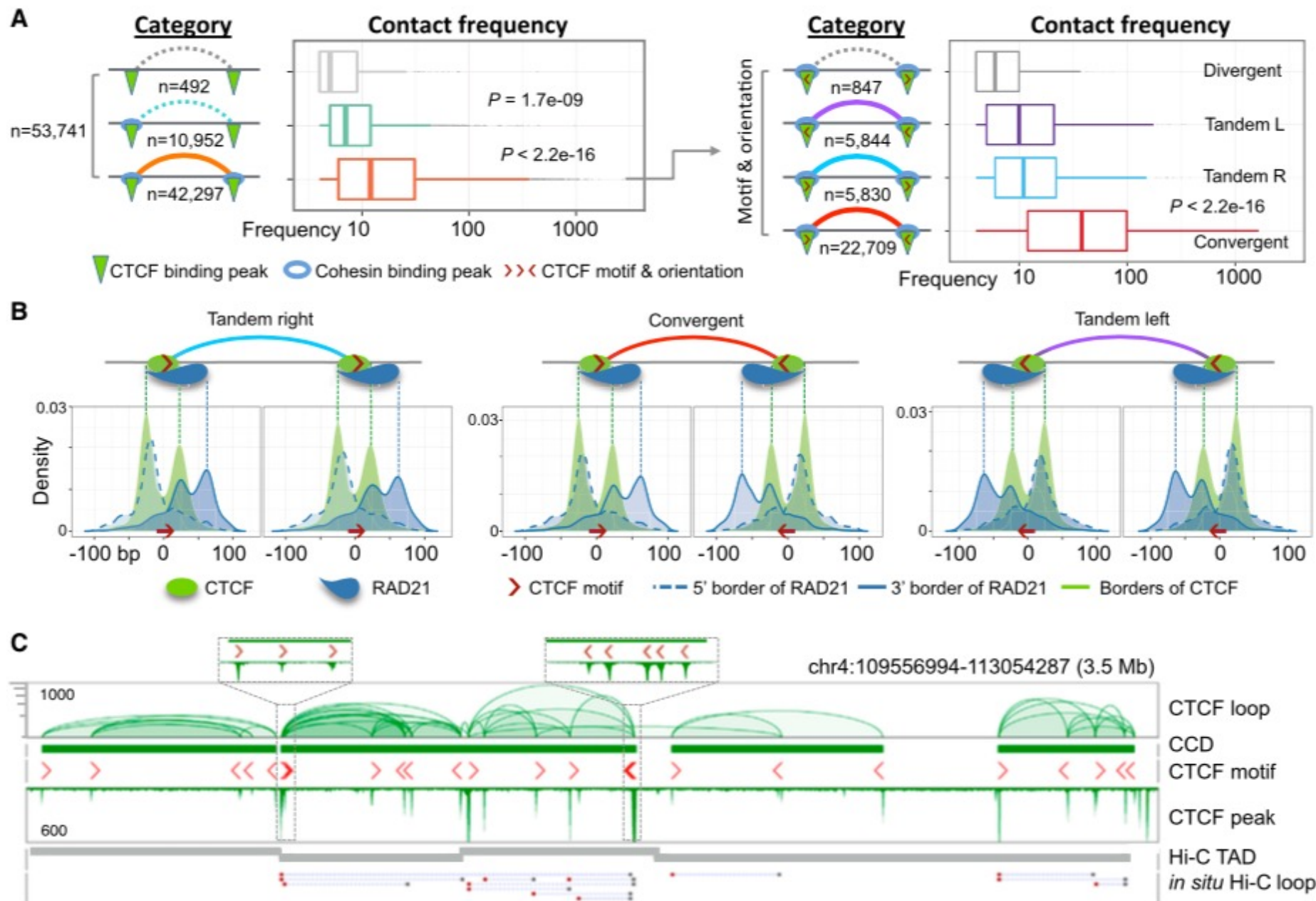


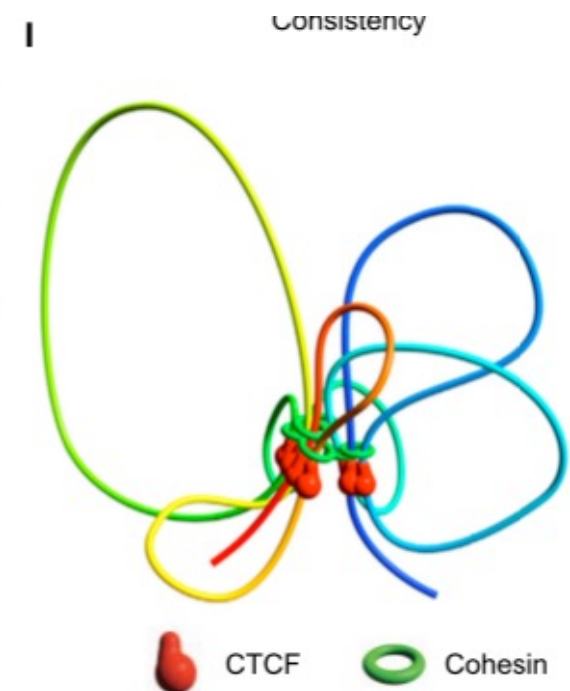
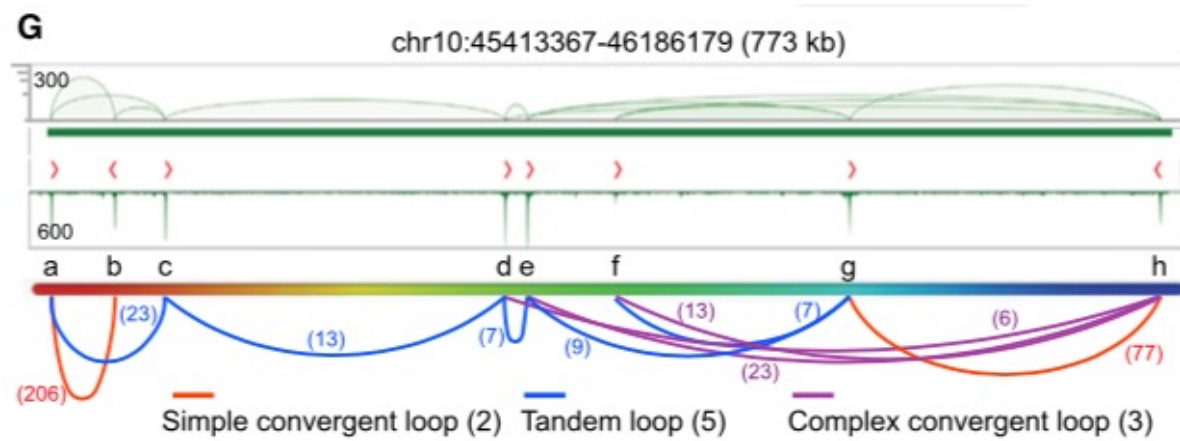
CTCF-Mediated Human 3D Genome Architecture Reveals Chromatin Topology for Transcription

Zhonghui Tang,^{1,12} Oscar Junhong Luo,^{1,12} Xingwang Li,^{1,2,12} Meizhen Zheng,¹ Jacqueline Jufen Zhu,^{1,3} Przemysław Szalaj,^{4,5,6} Paweł Trzaskoma,⁷ Adriana Magańska,⁸ Jakub Włodarczyk,⁹ Blazej Ruszczycki,⁹ Paul Michalski,¹ Emały Piecuch,^{1,3} Ping Wang,¹ Danjuan Wang,¹ Simon Zhongyuan Tian,¹ May Penrad-Mobayed,¹ Laurent M. Sachs,² Xiaolan Ruan,¹ Chia-Lin Wei,¹⁰ Edison T. Liu,¹ Grzegorz M. Wilczynski,⁷ Dariusz Plewczynski,⁶ Guoliang Li,^{2,11} and Yijun Ruan^{1,2,3,*}

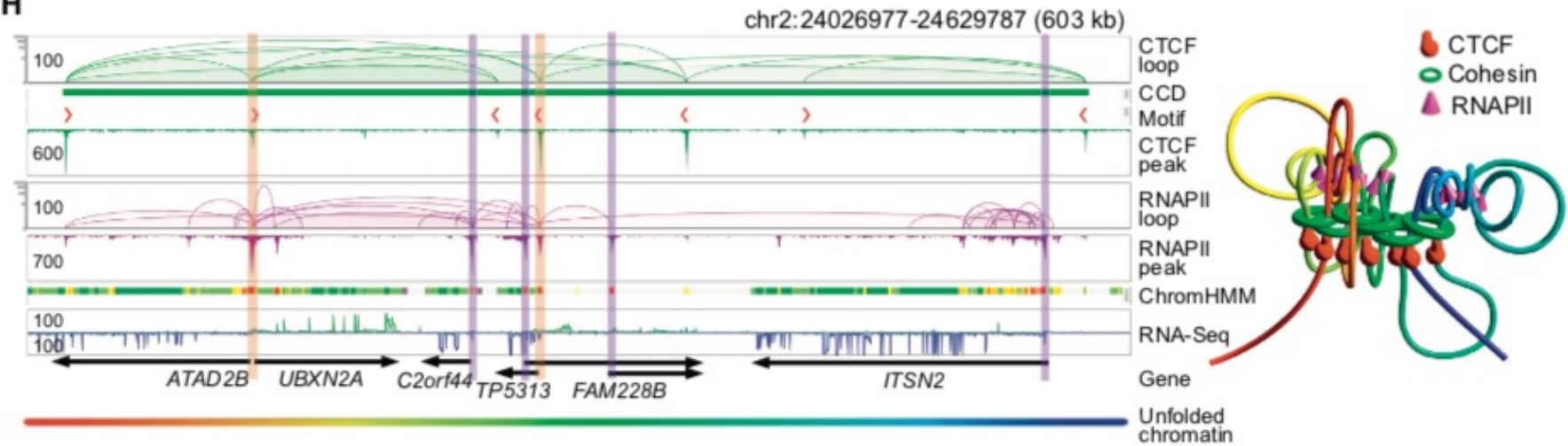
CTCF ChIA-PET Cell 2015

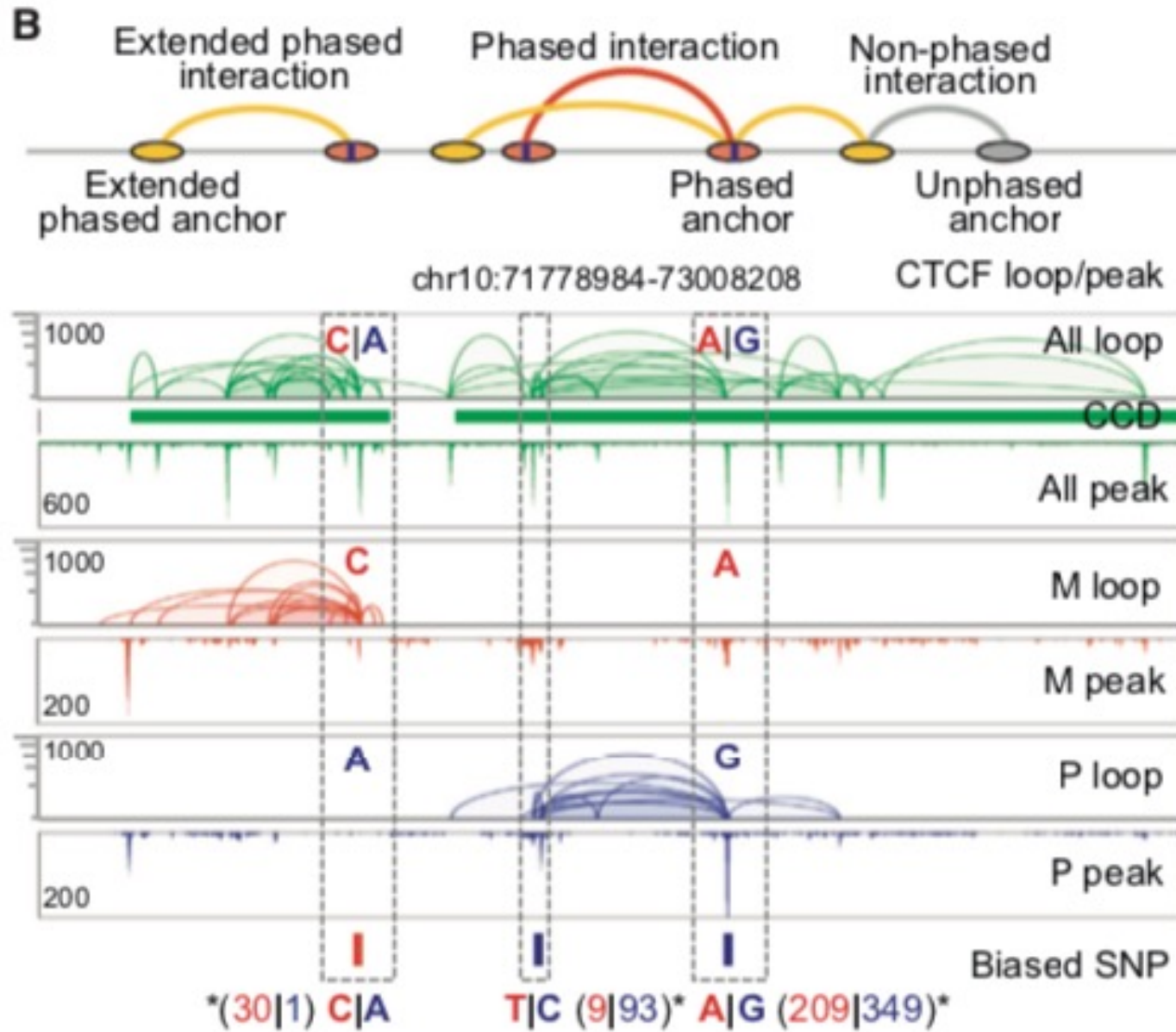


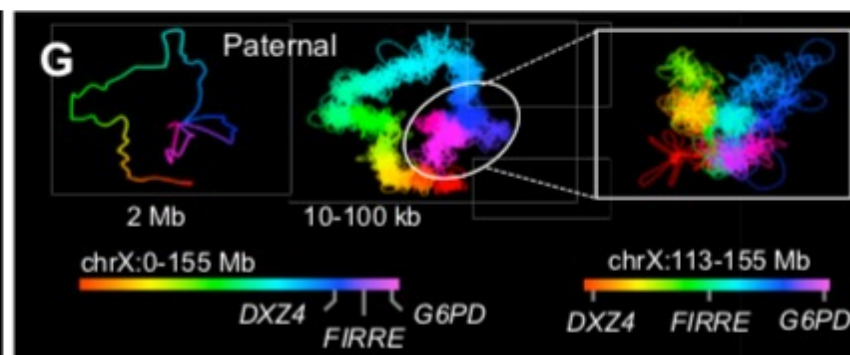
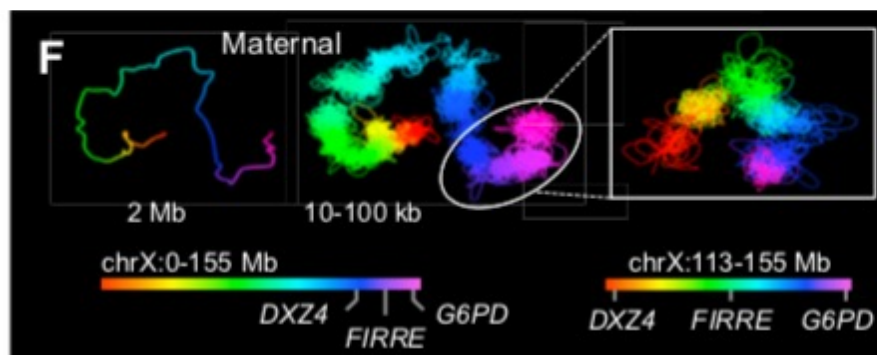
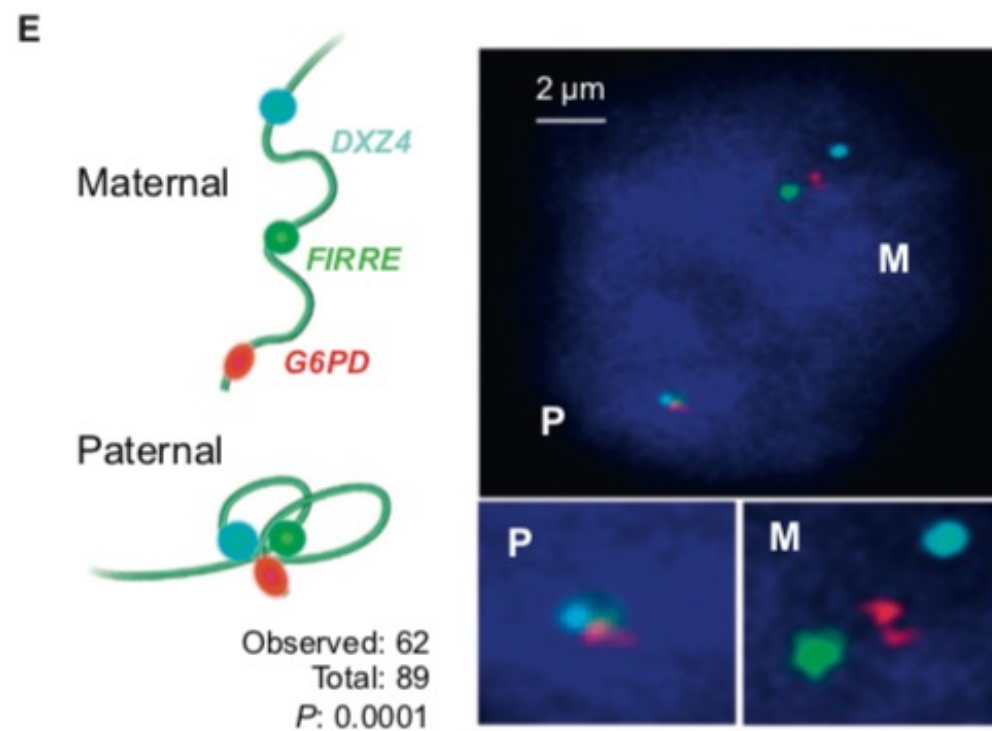
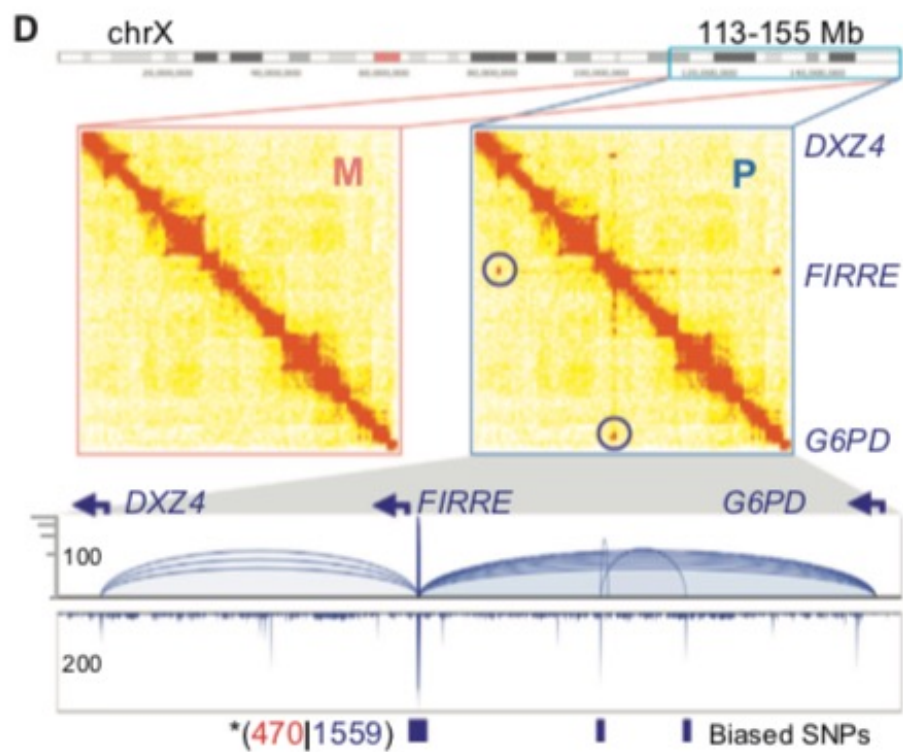




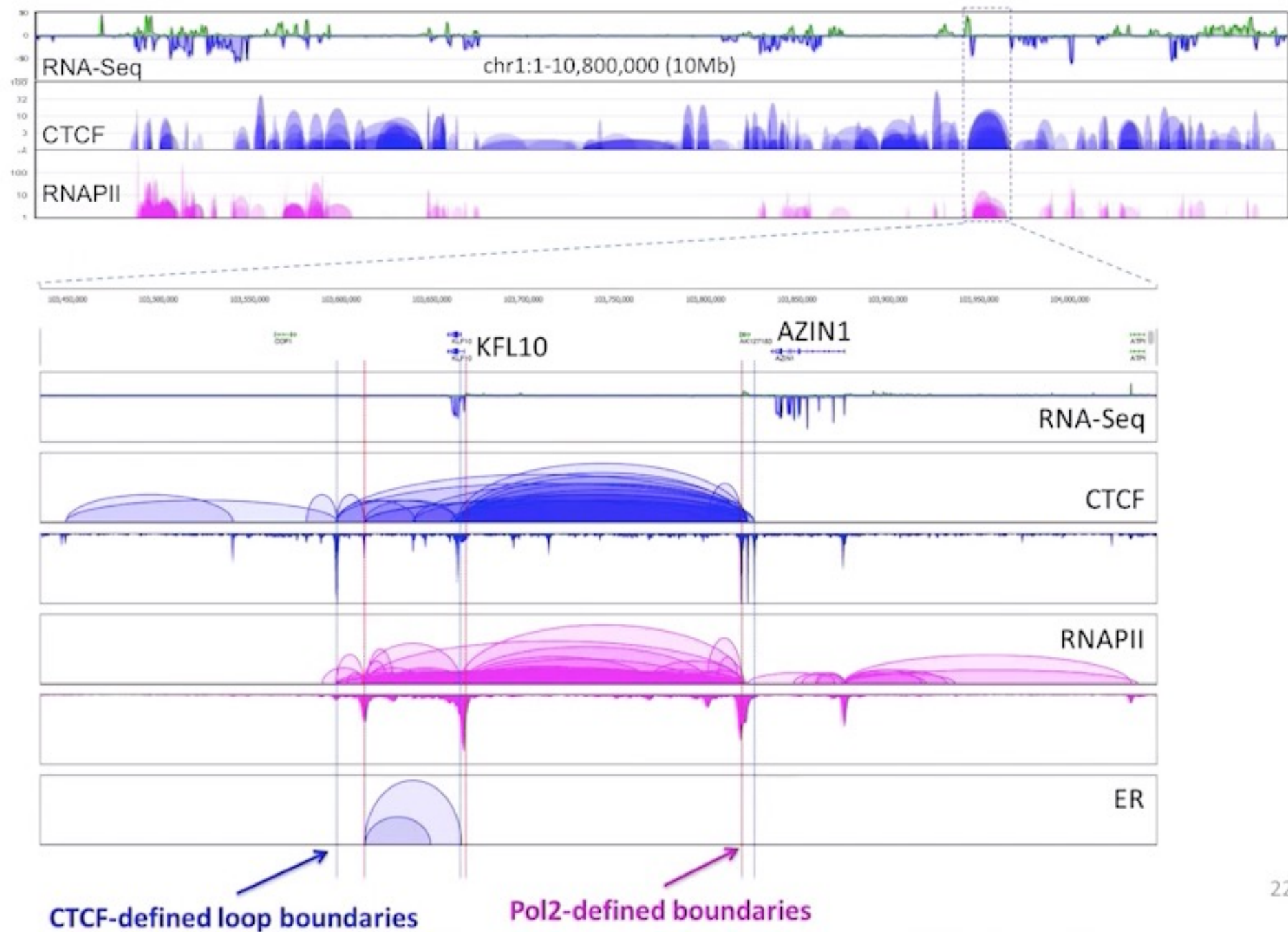
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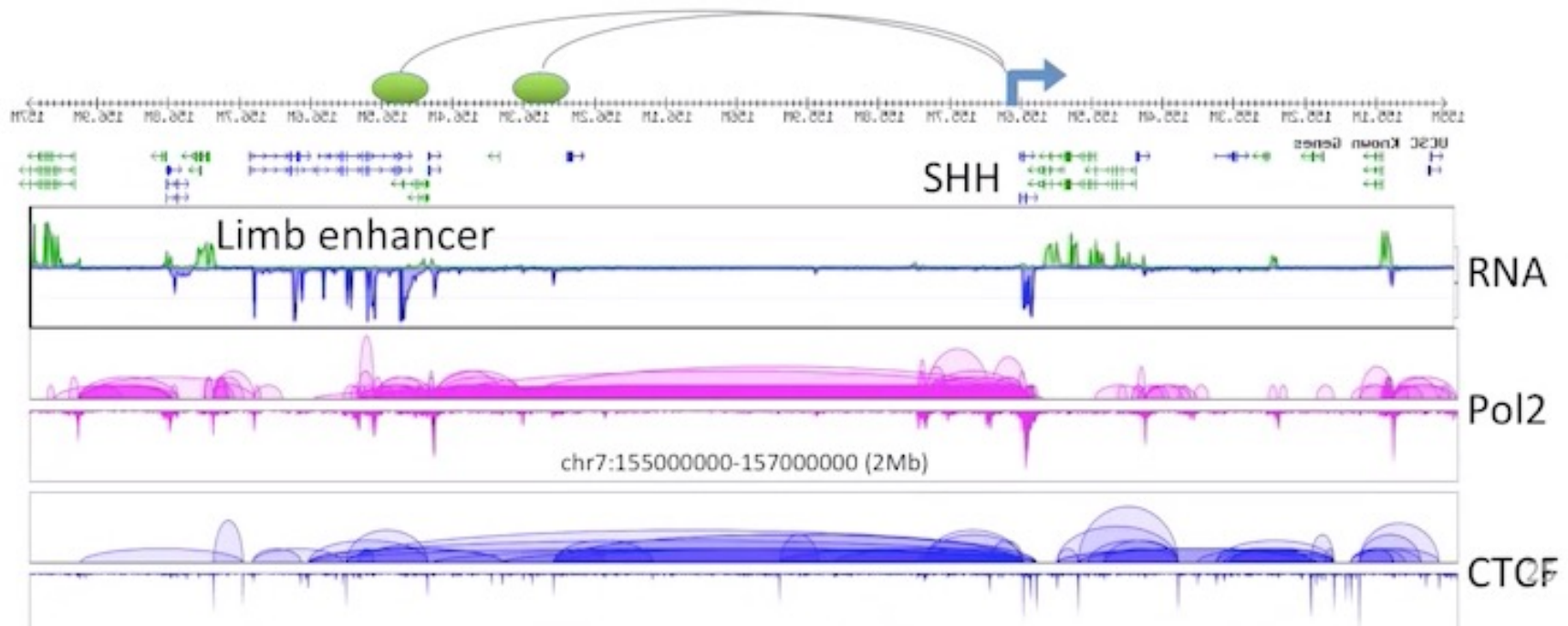
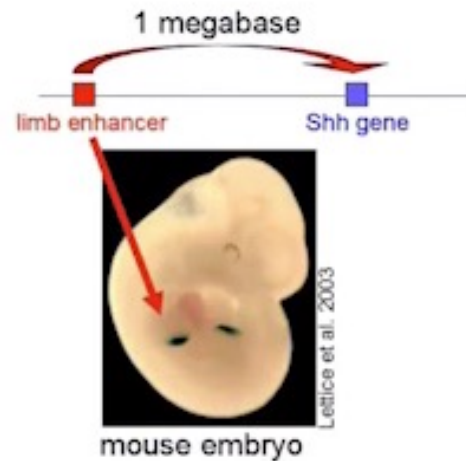


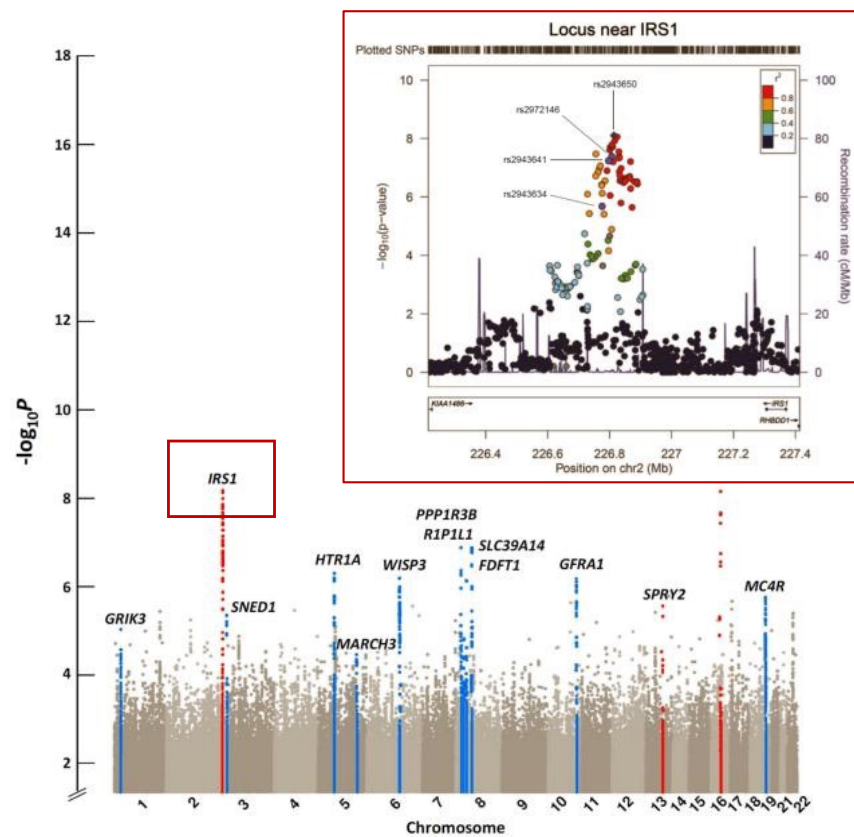


CTCF-defined chromatin looping space for transcription regulation



The importance of genome 3D structure to genome functions



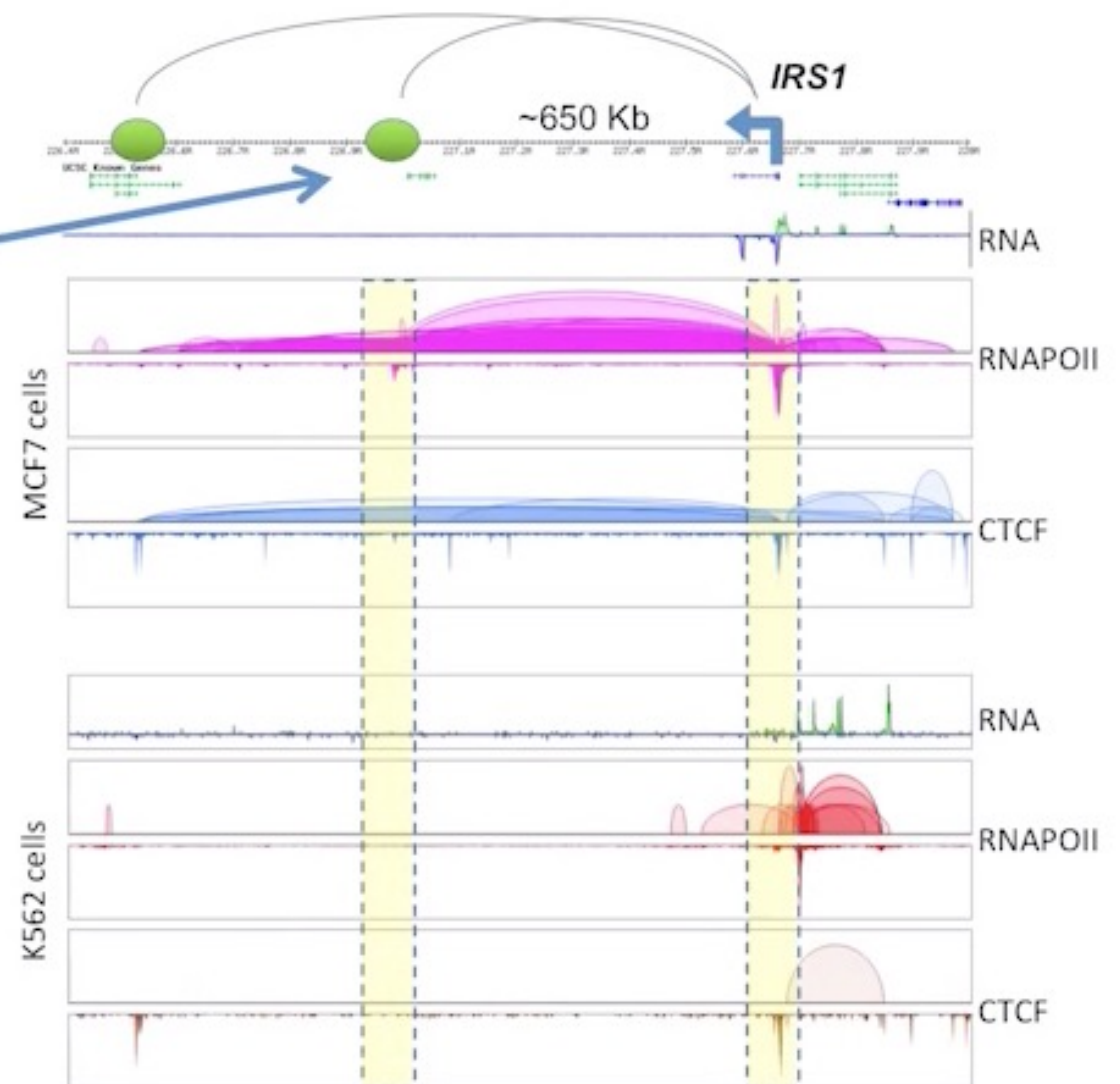


Published: 26 June 2011

Genetic variation near *IRS1* associates with reduced adiposity and an impaired metabolic profile

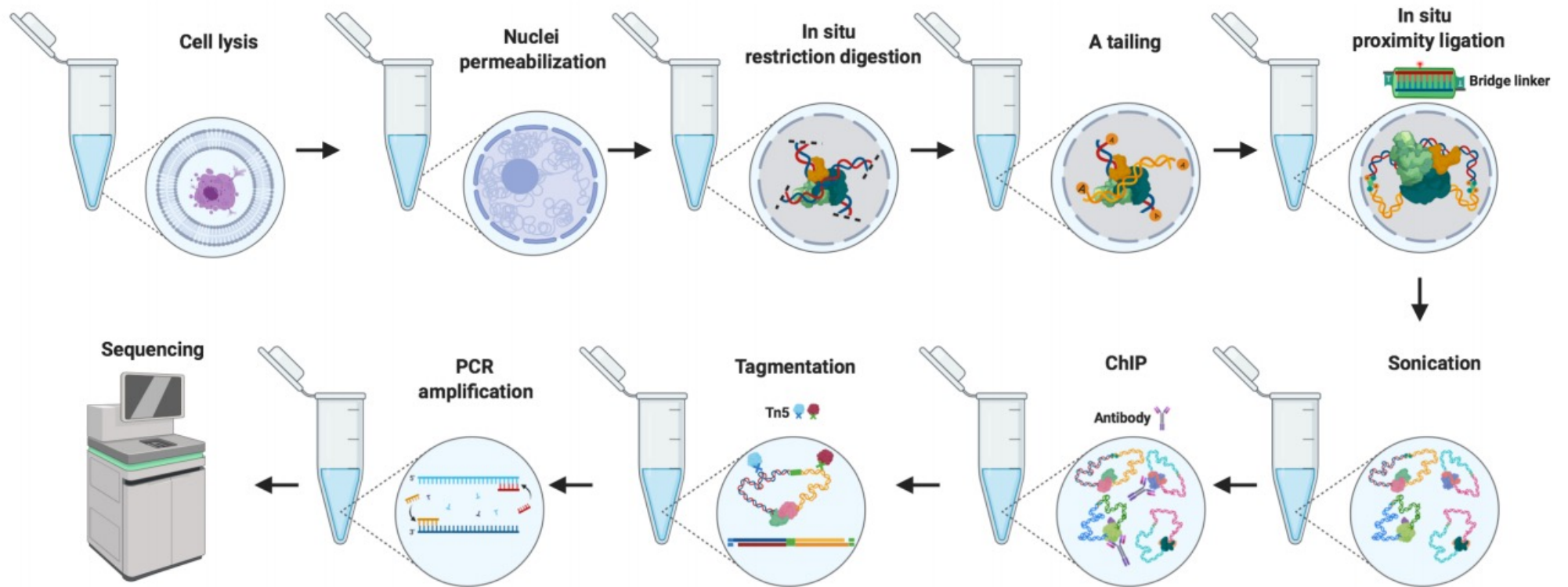
Tommaso O Kilpeläinen, M Carola Zillikens, [...] Ruth J F Loos

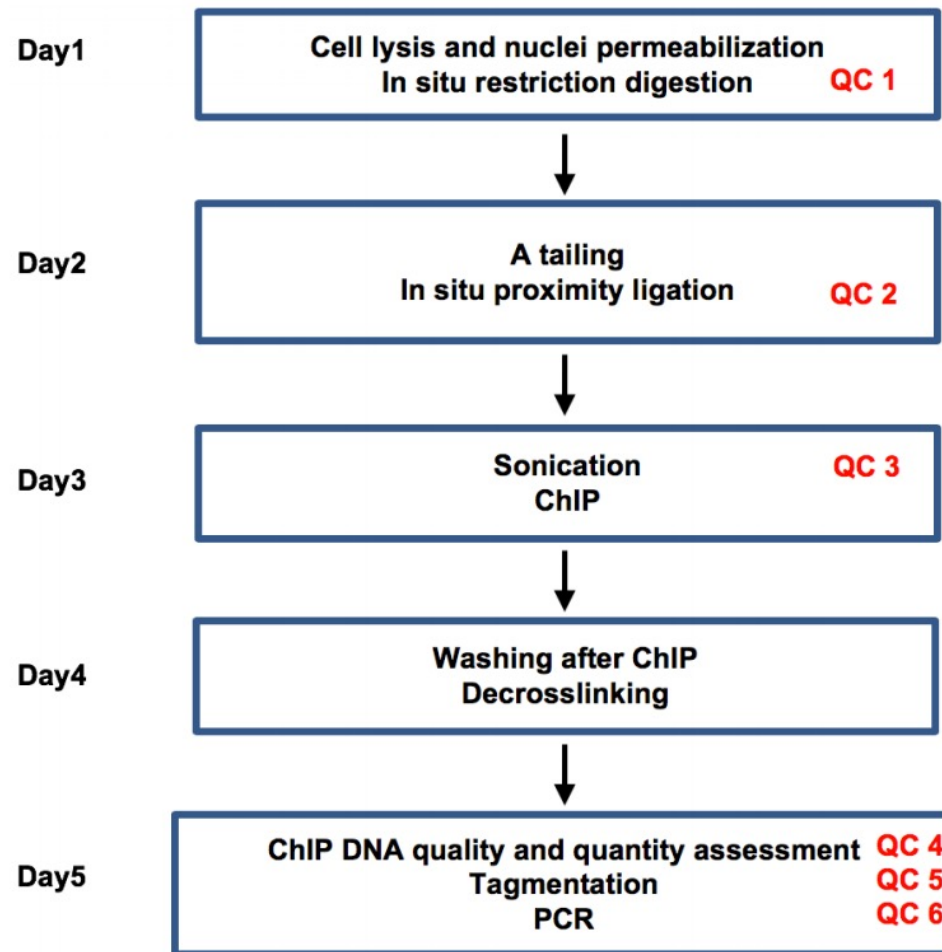
Nature Genetics 43, 753–760(2011) | [Cite this article](#)



In situ ChIA-PET

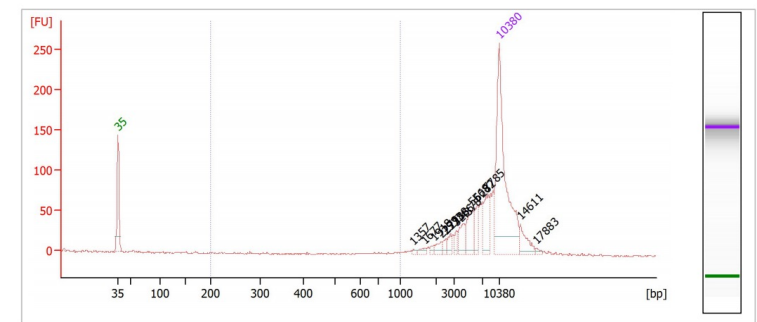
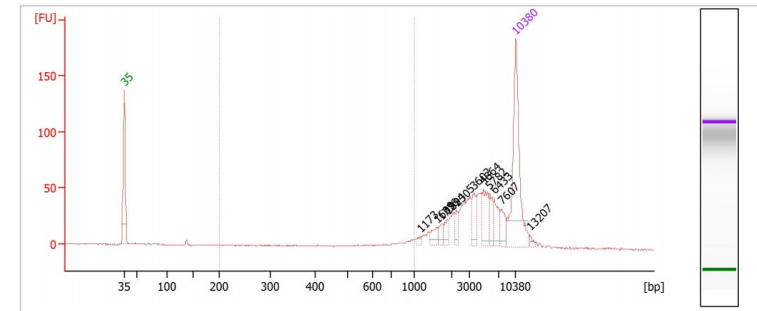
In situ ChIA-PET



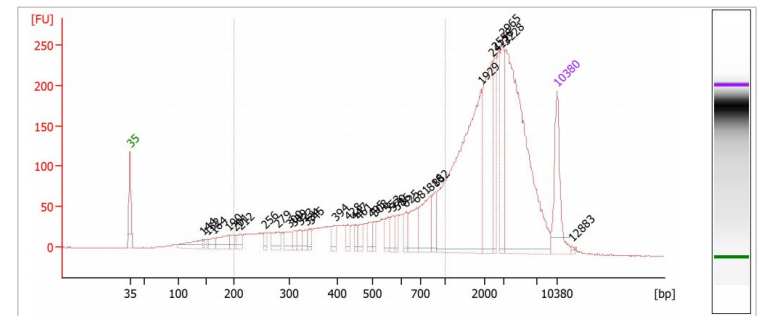


In situ ChIA-PET library ready for sequencing analysis

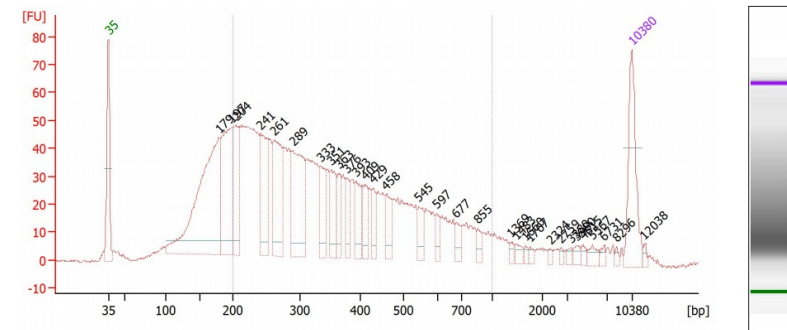
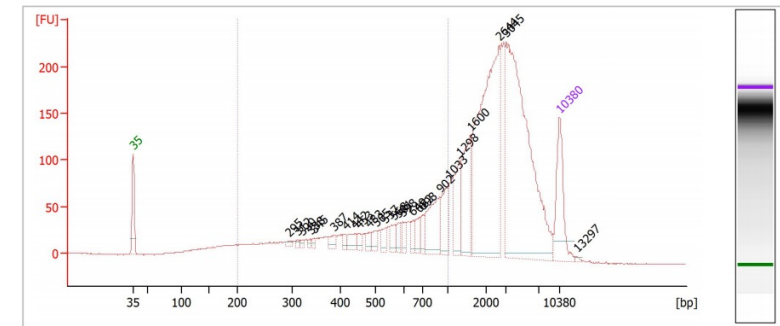
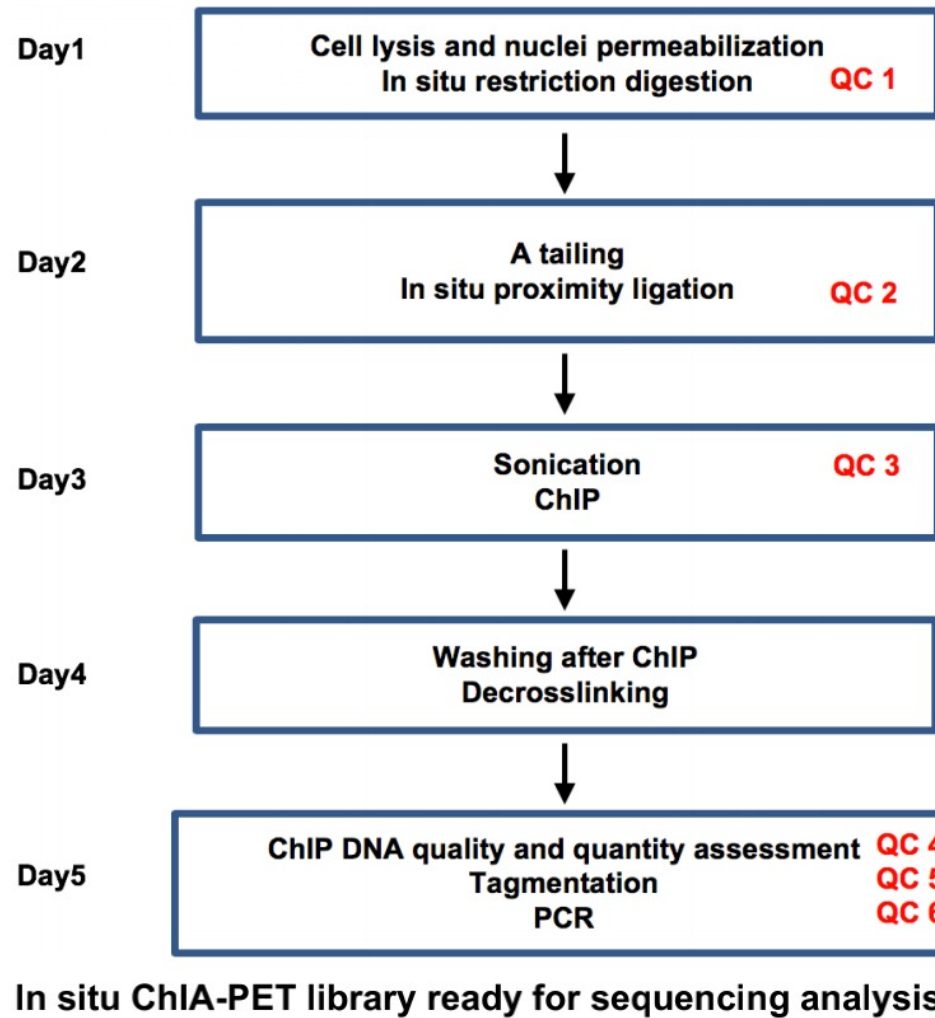
QC1: Profiles of the "In situ" chromatin DNA digestions (*Bioanalyzer, 2100-HS DNA Chip*)



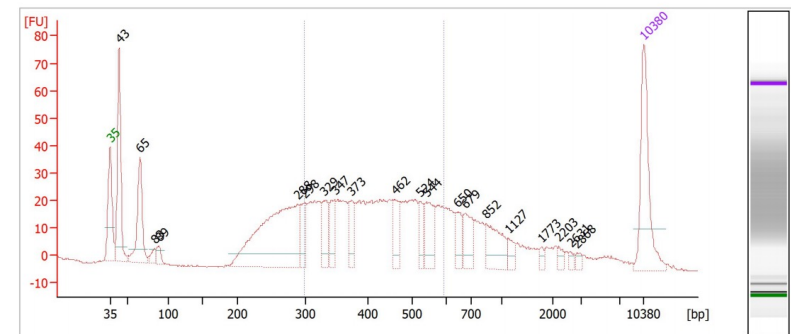
QC3: Sheared chromatin DNA profiling after the in situ proximity ligation step.

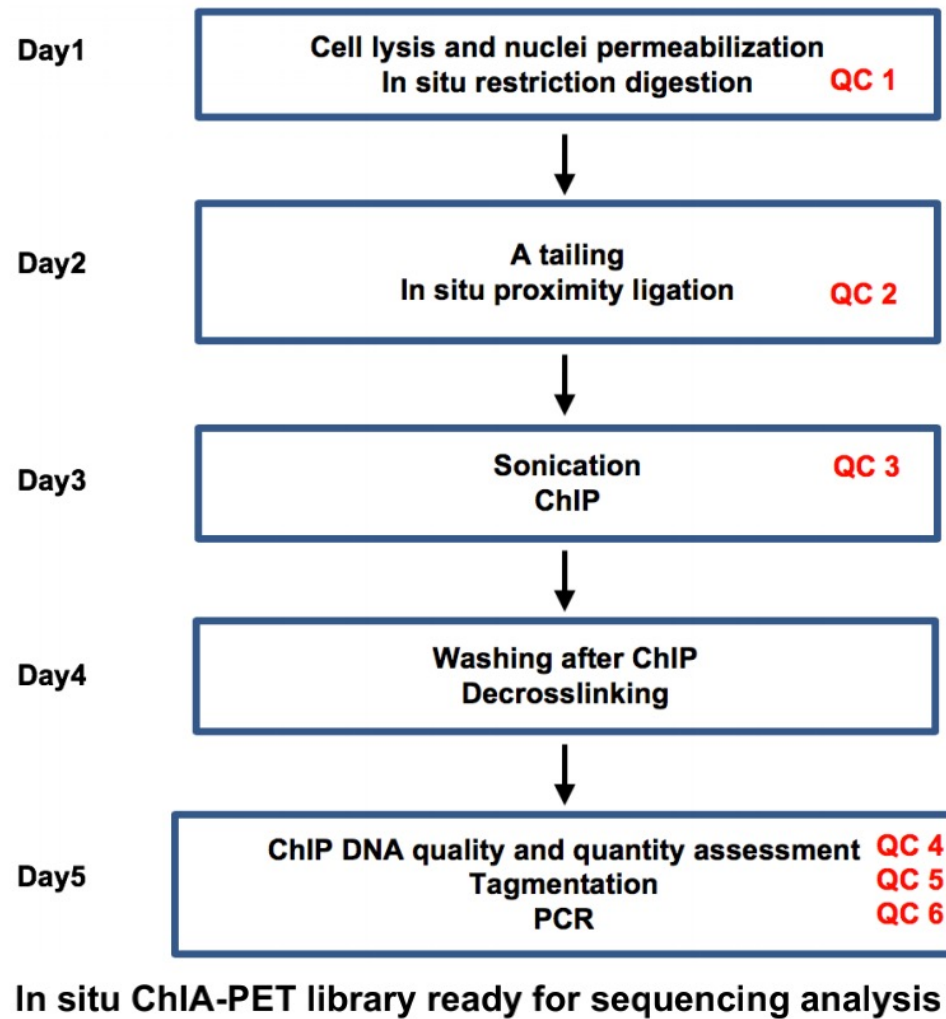


QC4-1: Quantify the purified DNA using Qubit and Agilent Bioanalyzer.

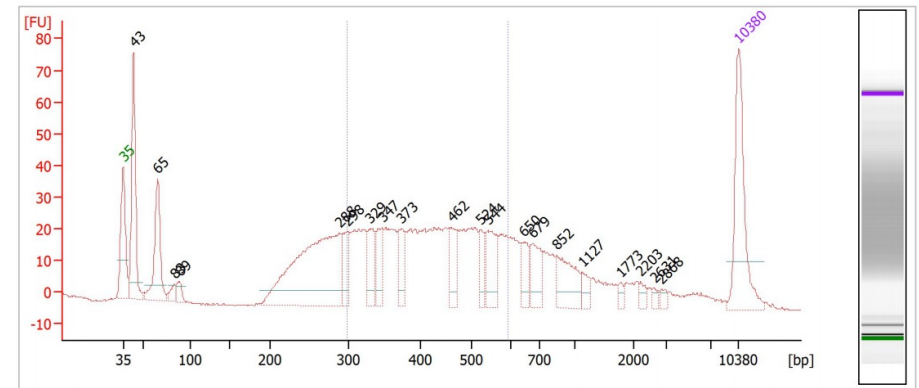


QC6-1: PCR product profiling.

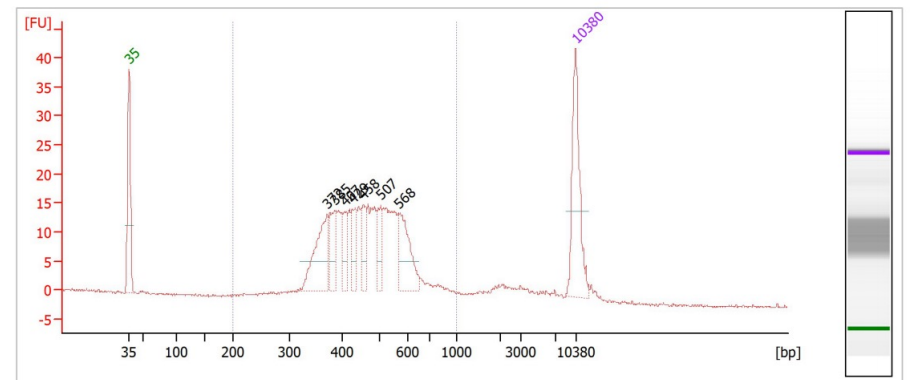


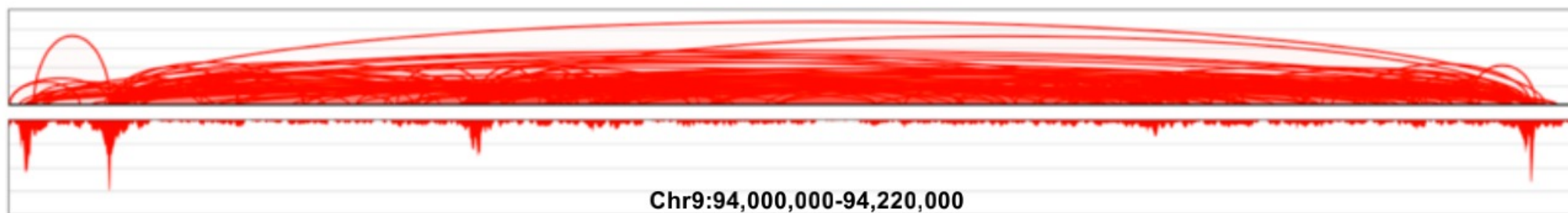
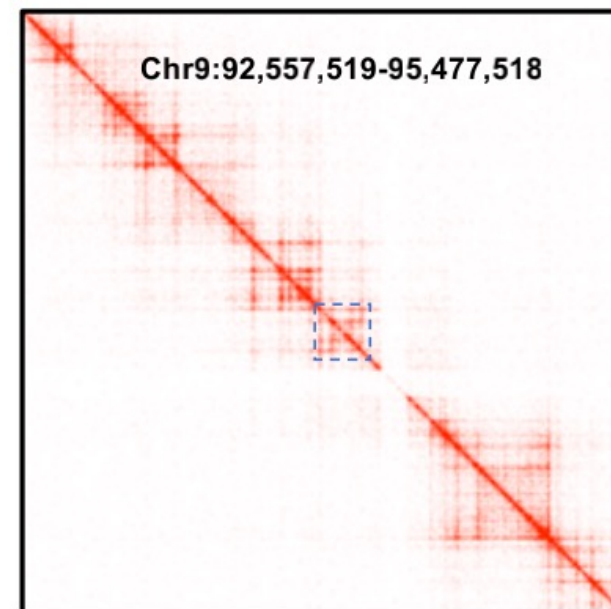
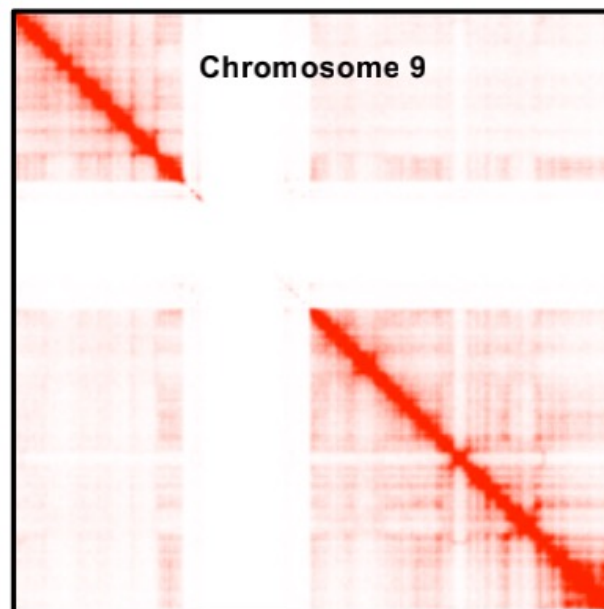
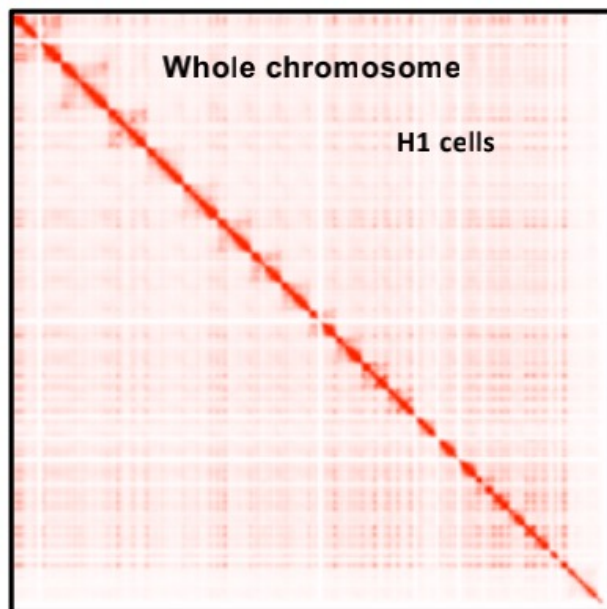


QC6-1: PCR product profiling.



QC6-2: Library profiling after size selection.





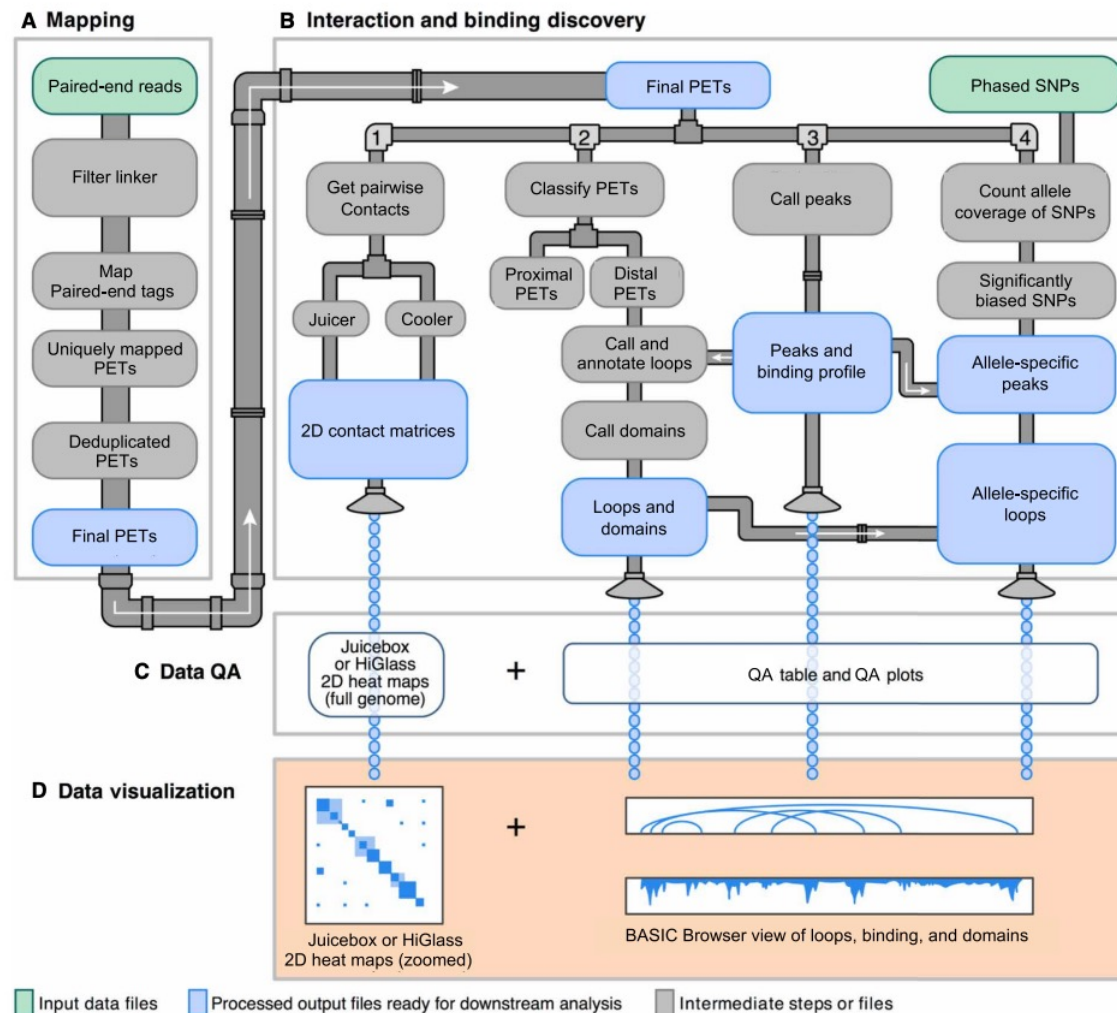
SYSTEMS BIOLOGY

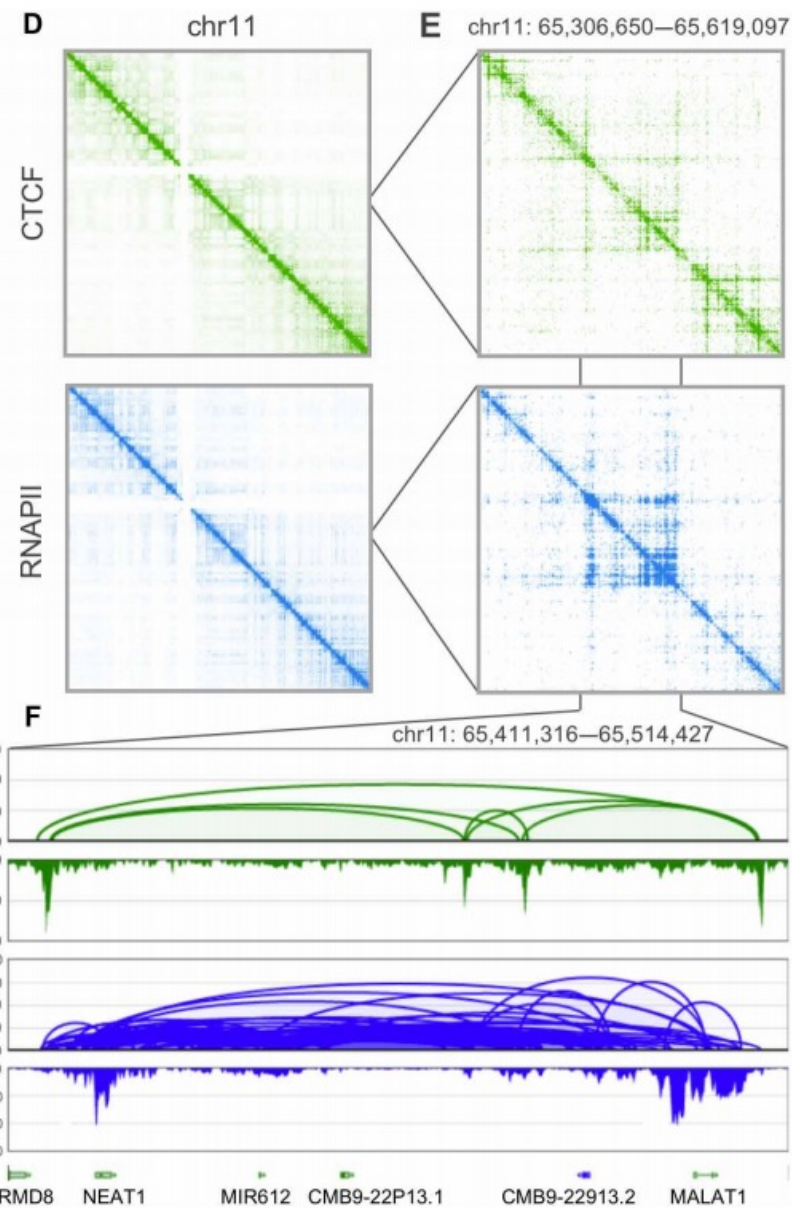
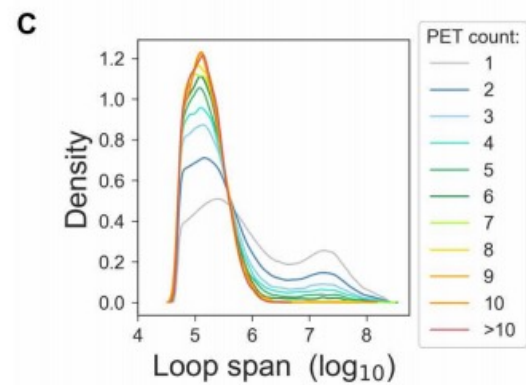
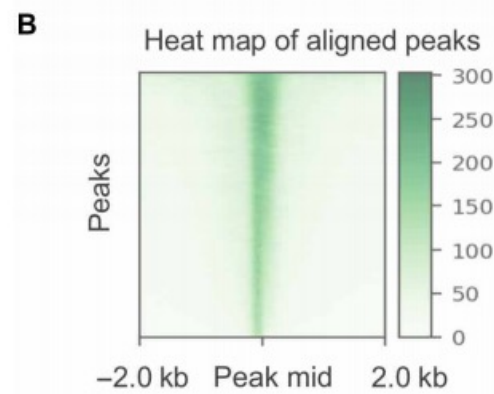
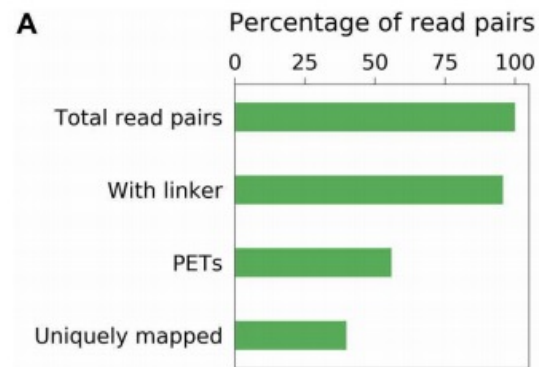
ChIA-PIPE: A fully automated pipeline for comprehensive ChIA-PET data analysis and visualization

Byoungkoo Lee¹, Jiahui Wang¹, Liuyang Cai¹, Minji Kim¹, Sandeep Namburi¹, Harianto Tjong¹, Yuliang Feng¹, Ping Wang¹, Zhonghui Tang¹, Ahmed Abbas¹, Chia-Lin Wei^{1,2*}, Yijun Ruan^{1,2,3*}, Sheng Li^{1,2,3,4*}

SCIENCE ADVANCES | RESEARCH ARTICLE

ChIA-PIPE

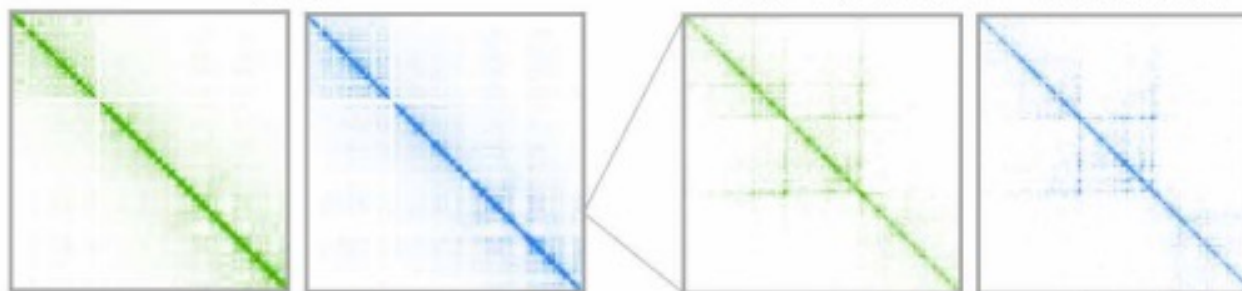




AHFFc6 ChIA-PET
CTCFHFFc6 ChIA-PET
RNAPII

chr8

chr8: 102,436,381—103,028,987

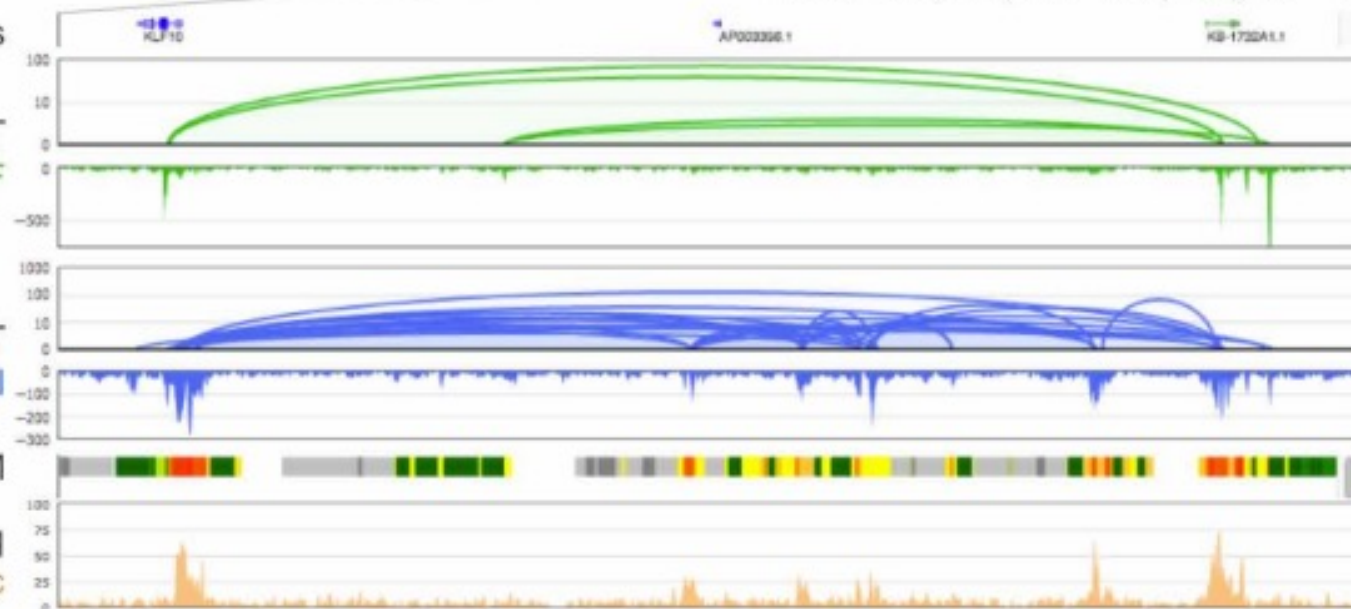
**B**

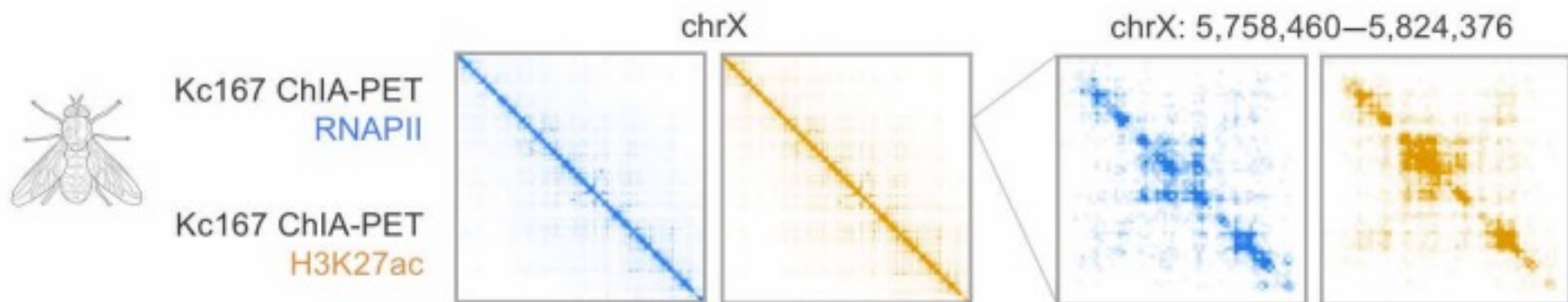
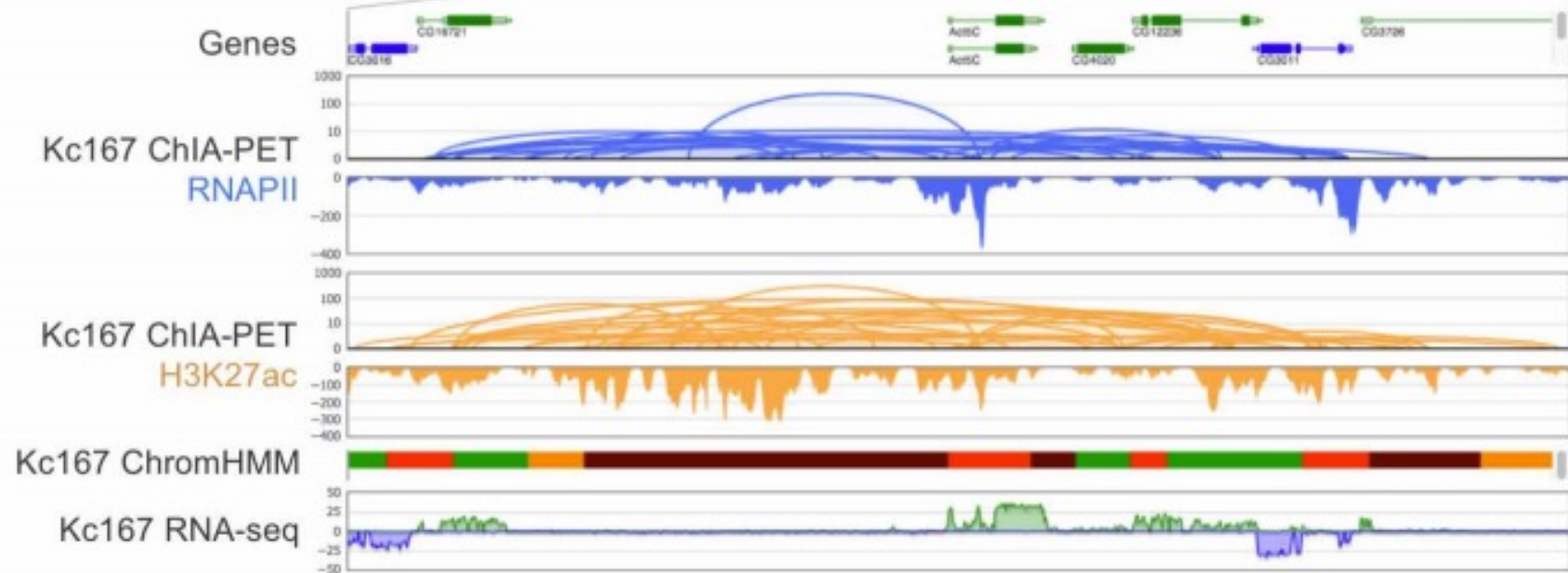
Genes

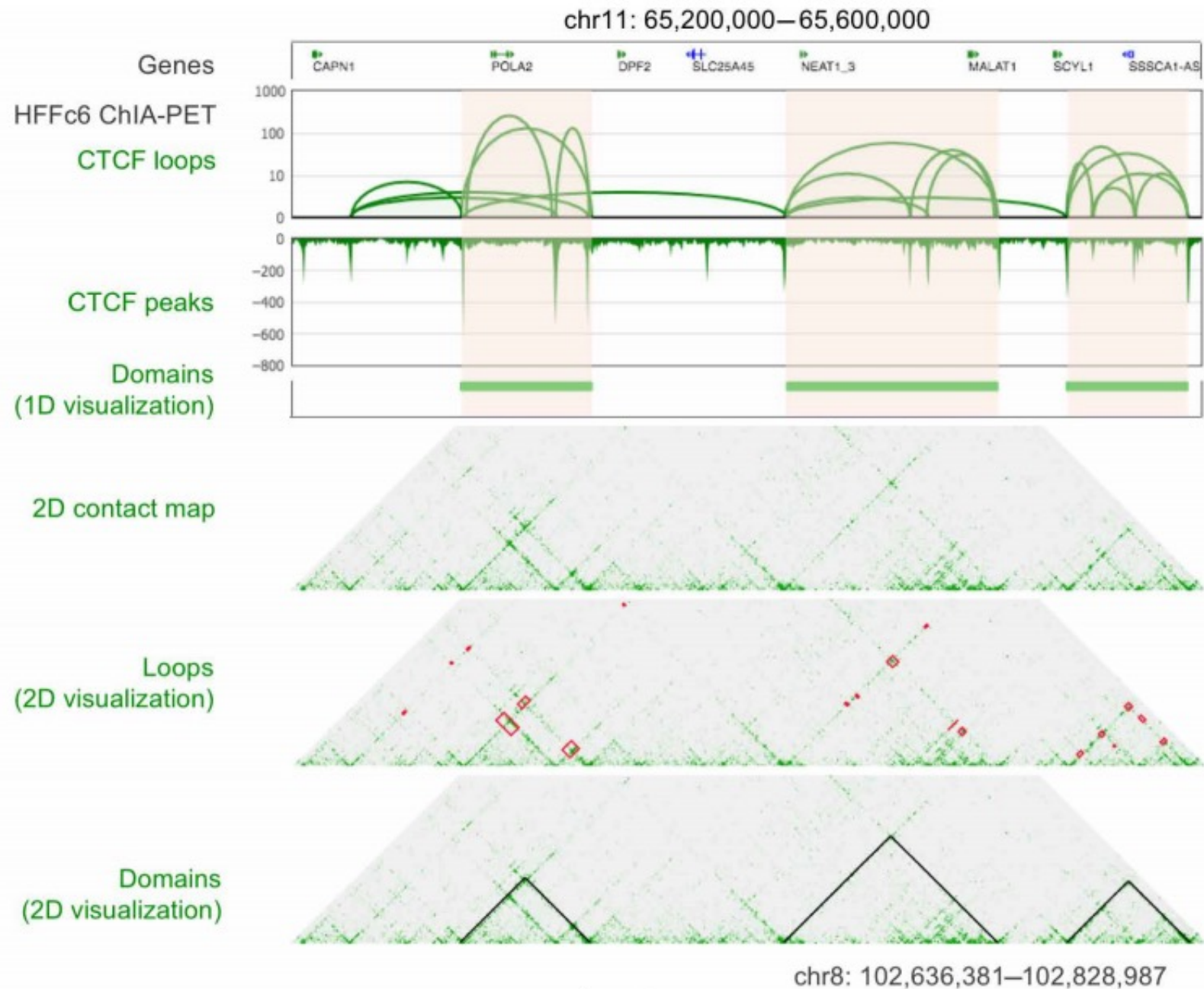
chr8: 102,636,381—102,828,987

HFFc6 ChIA-PET
CTCFHFFc6 ChIA-PET
RNAPII

HFF ChromHMM

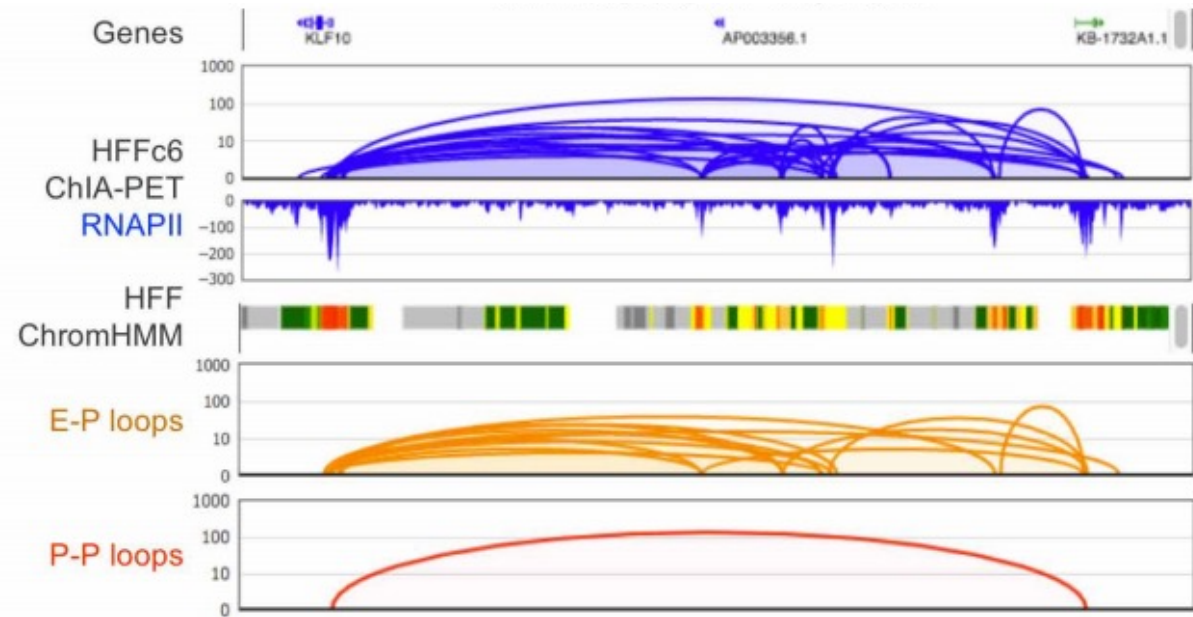
HFFc6 ChIP-seq
H3K27ac

C**D**

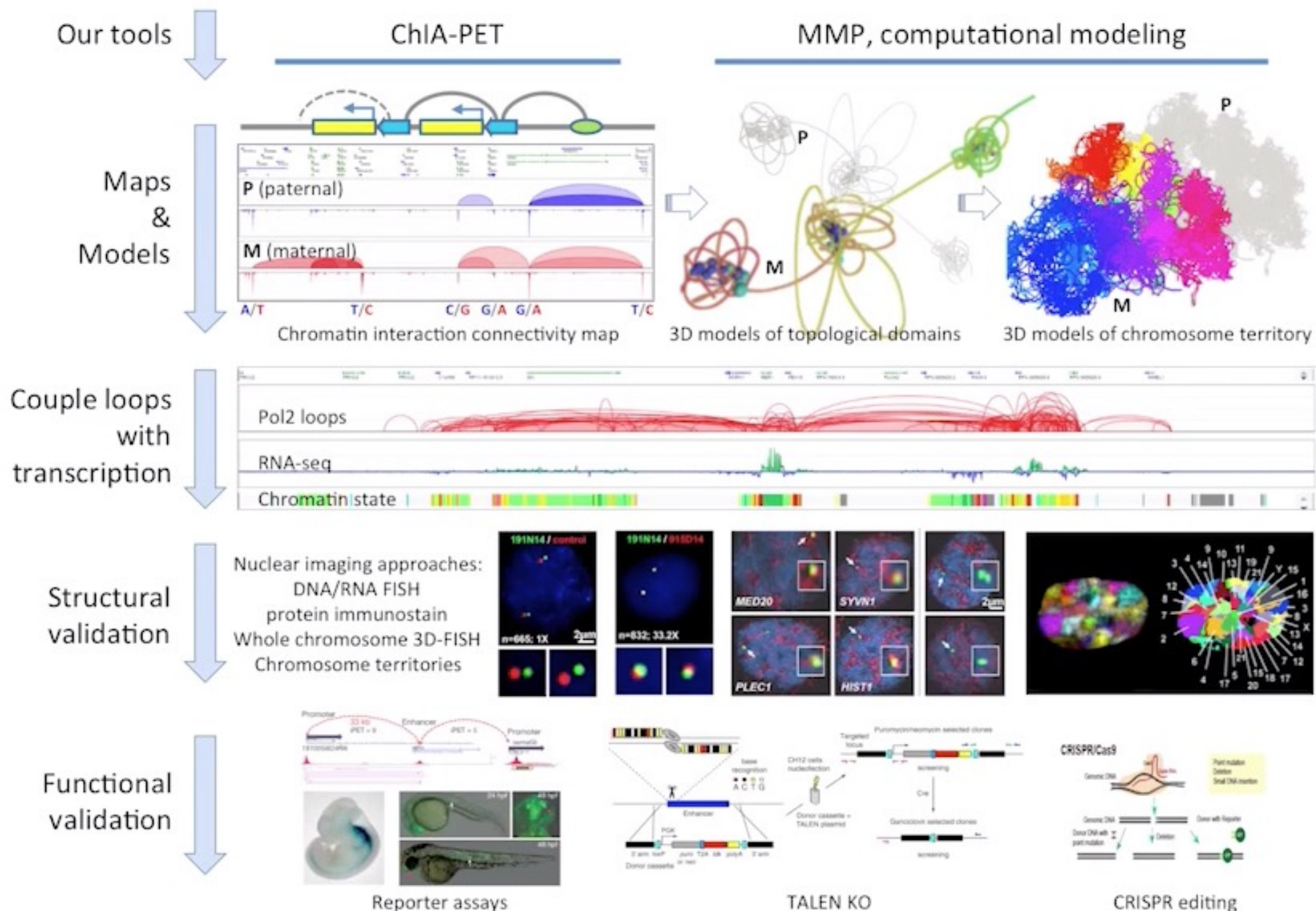


B

HFFc6 RNAPII ChIA-PET loops	45,214
Enhancer-promoter (E-P)	12,660
Promoter-promoter (P-P)	7,445



High-resolution and haplotype-specific 3D models of the human genomes



Question

- The difference between Hi-C and ChIA-PET ?

Hi-C

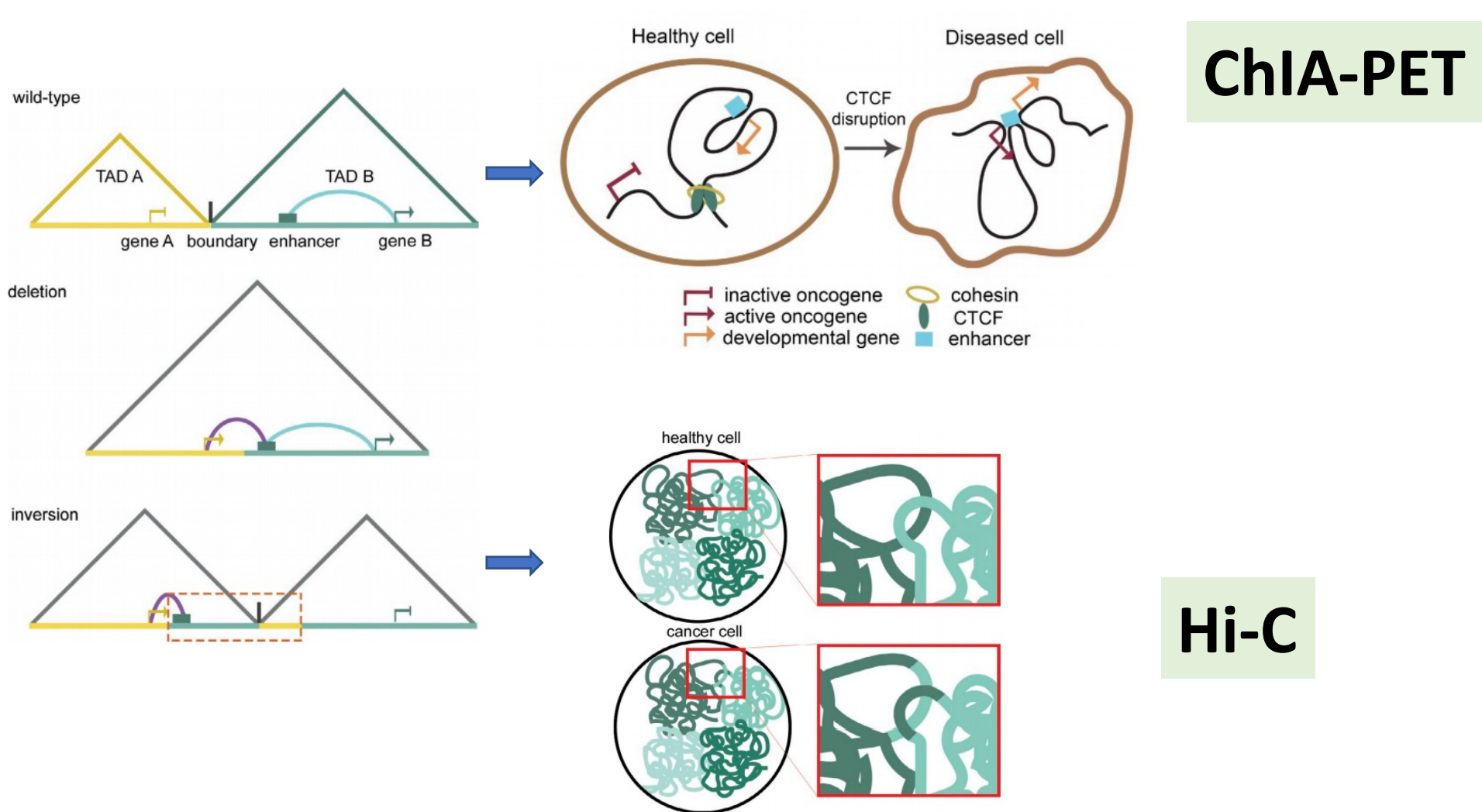


ChIA-PET



Question

- Use Hi-C or ChIA-PET ?



ChIA-PET visualization tools

BASIC browser